

MOSFET

Transistor

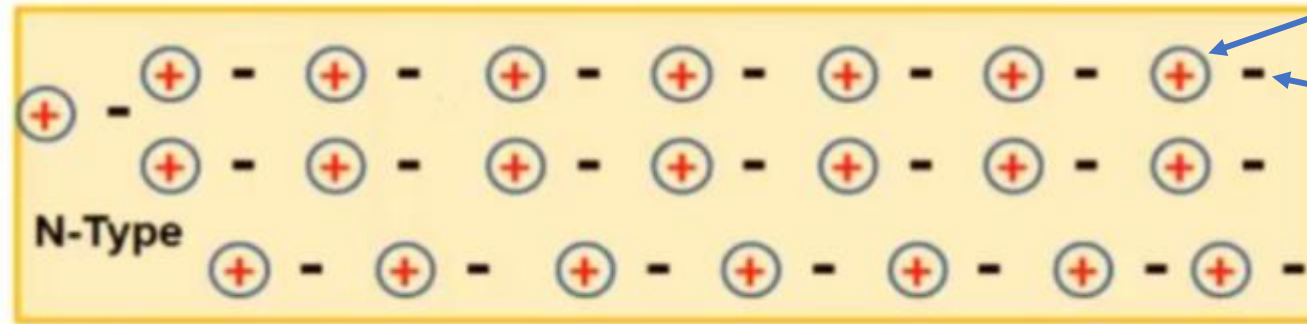


Current I_O is much more sensitive to V_{IN} than V_O

$$\frac{\partial I_o}{\partial V_{in}} \gg \frac{\partial I_o}{\partial V_o}$$

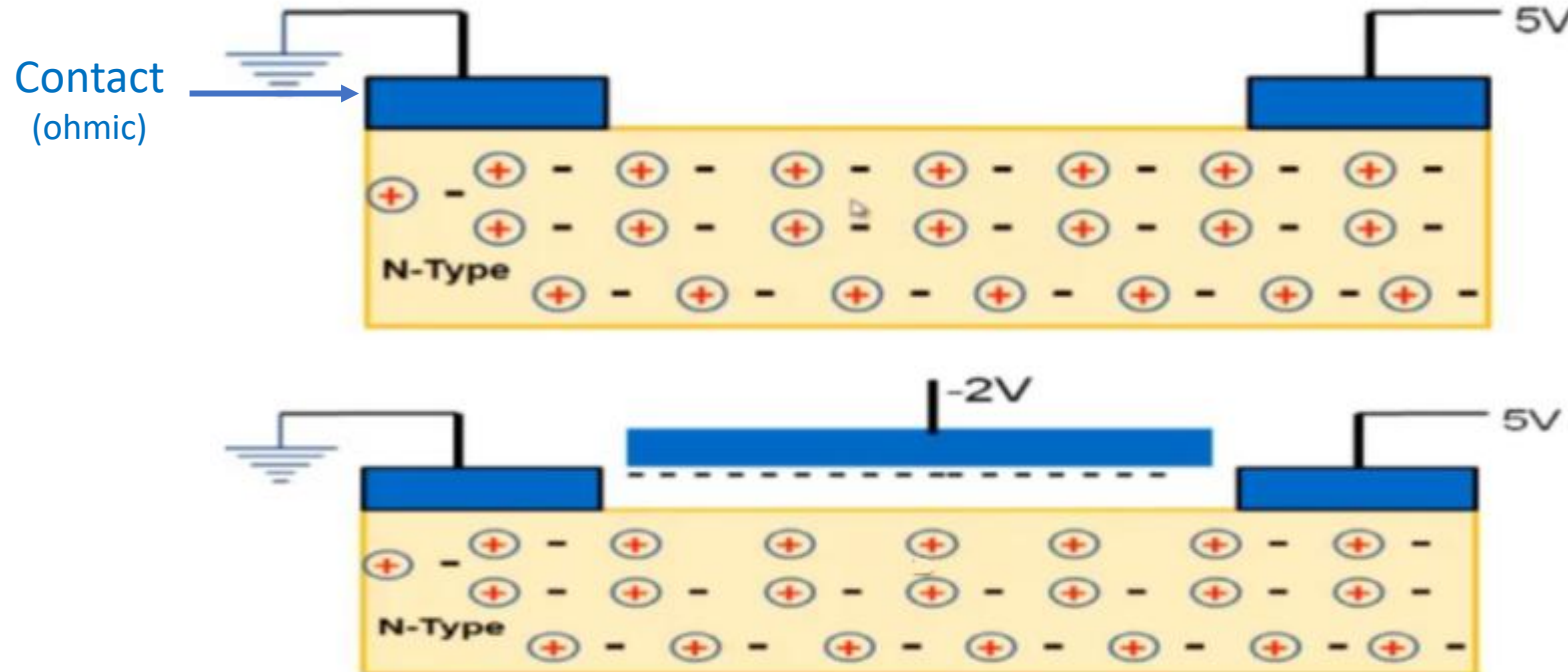
Field Effect Principle

$$\frac{\partial I_o}{\partial V_{in}} \gg \frac{\partial I_o}{\partial V_o}$$

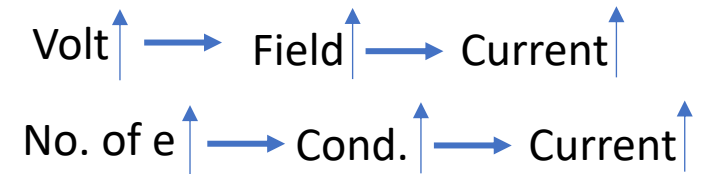


Donor atoms

Free electron



- Amount of current flow depend on applied voltage



Behaving as a Resistor

doping thickness area

- Take another plate, place very close (by some means) to n-type semiconductor.
- Apply a negative voltage.
- Push the e away from the surface.

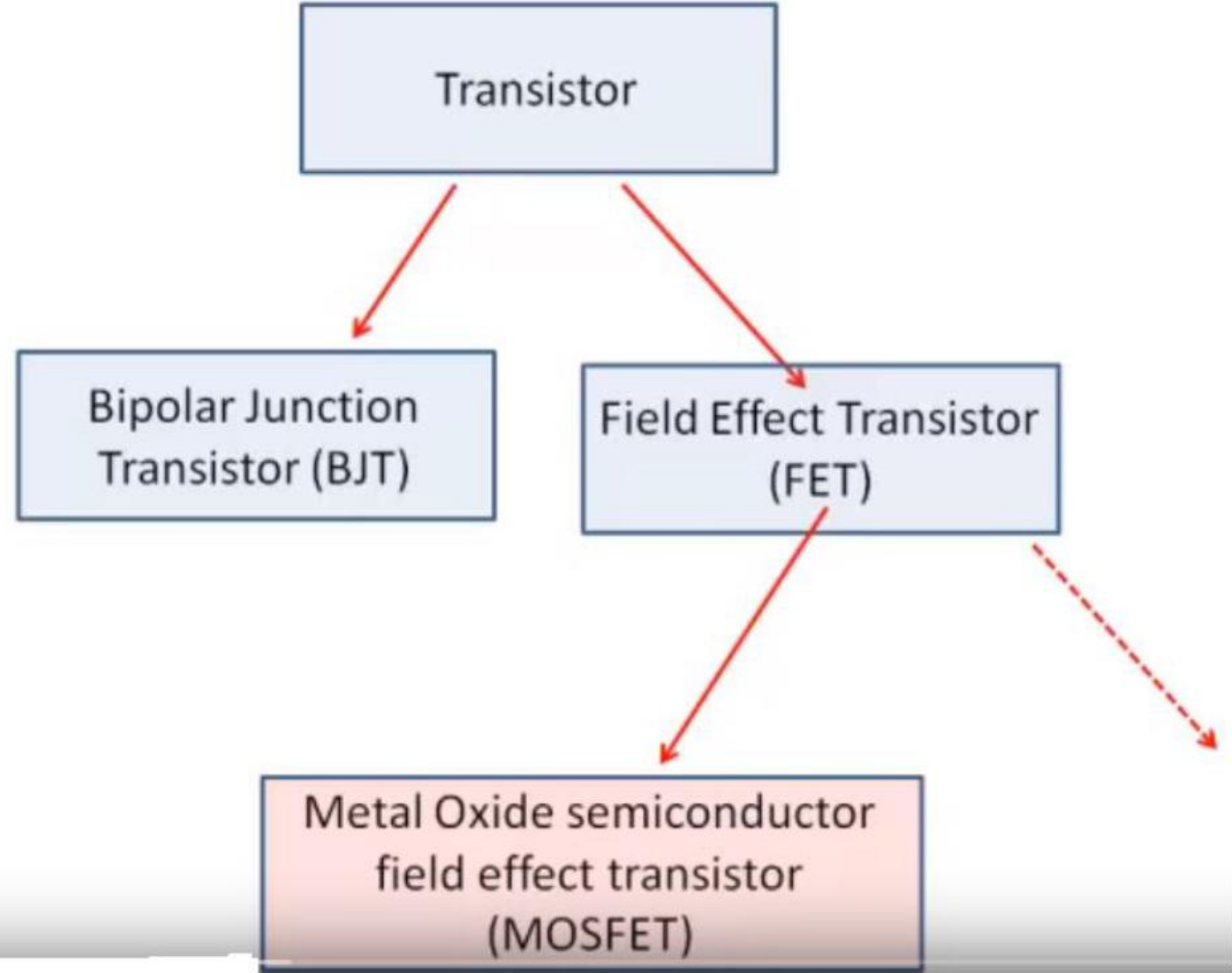
Capacitor

Modulation of conductivity using electric field

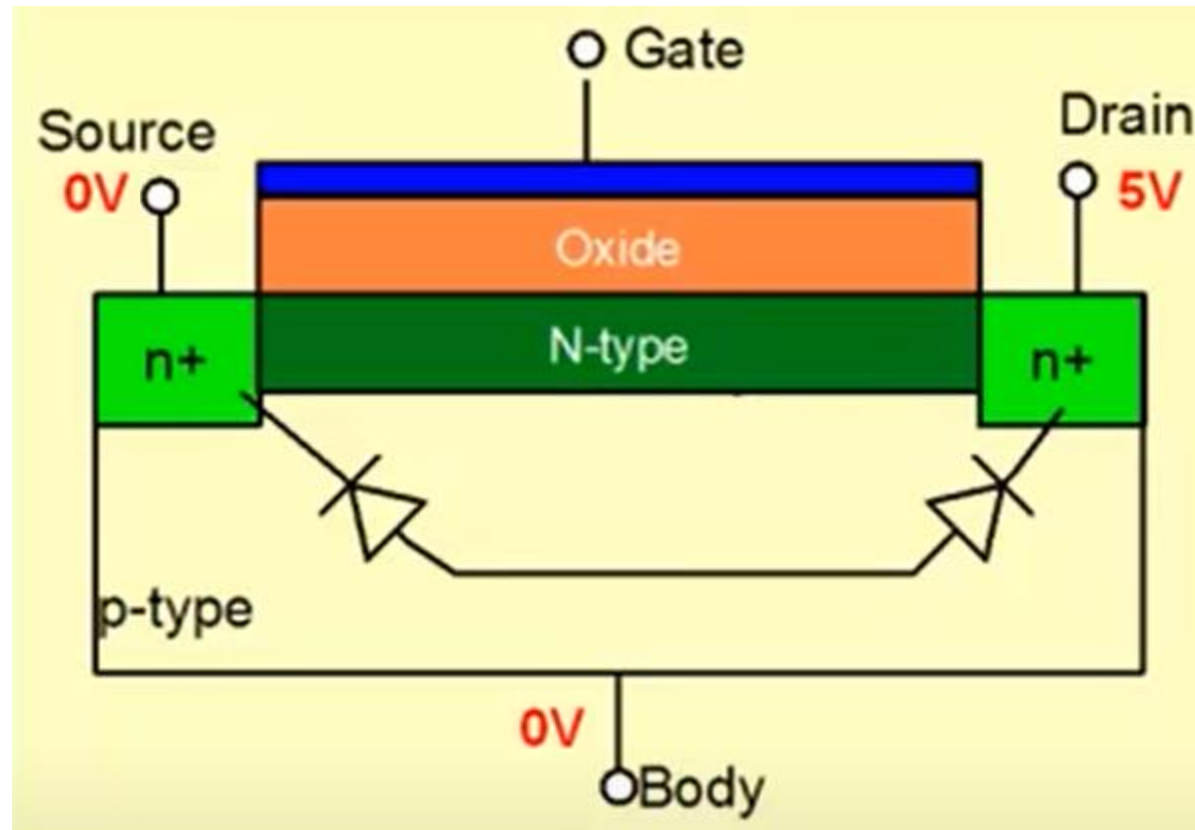
Transconductance

Field Effect Principle



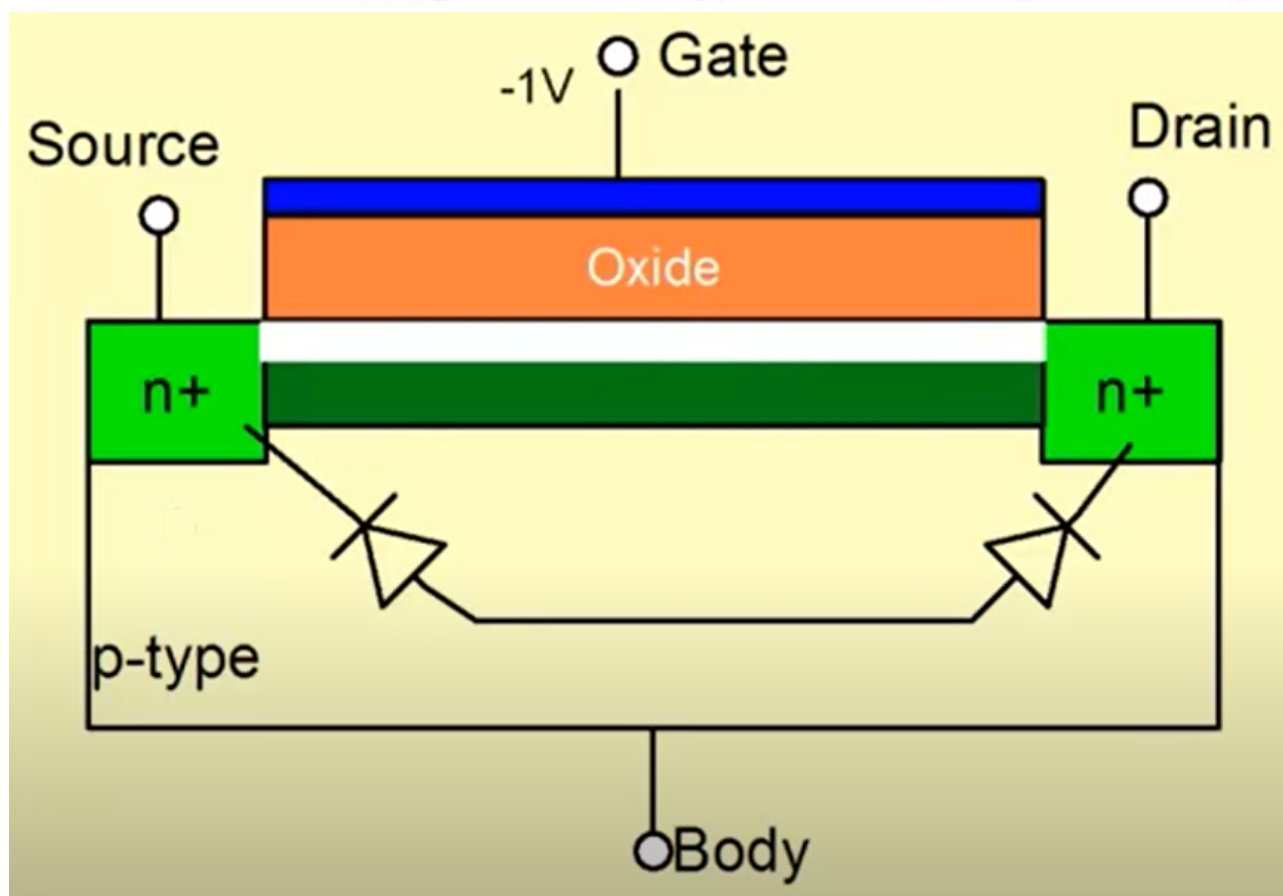


Depletion-Mode Transistor



Depletion-Mode Transistor

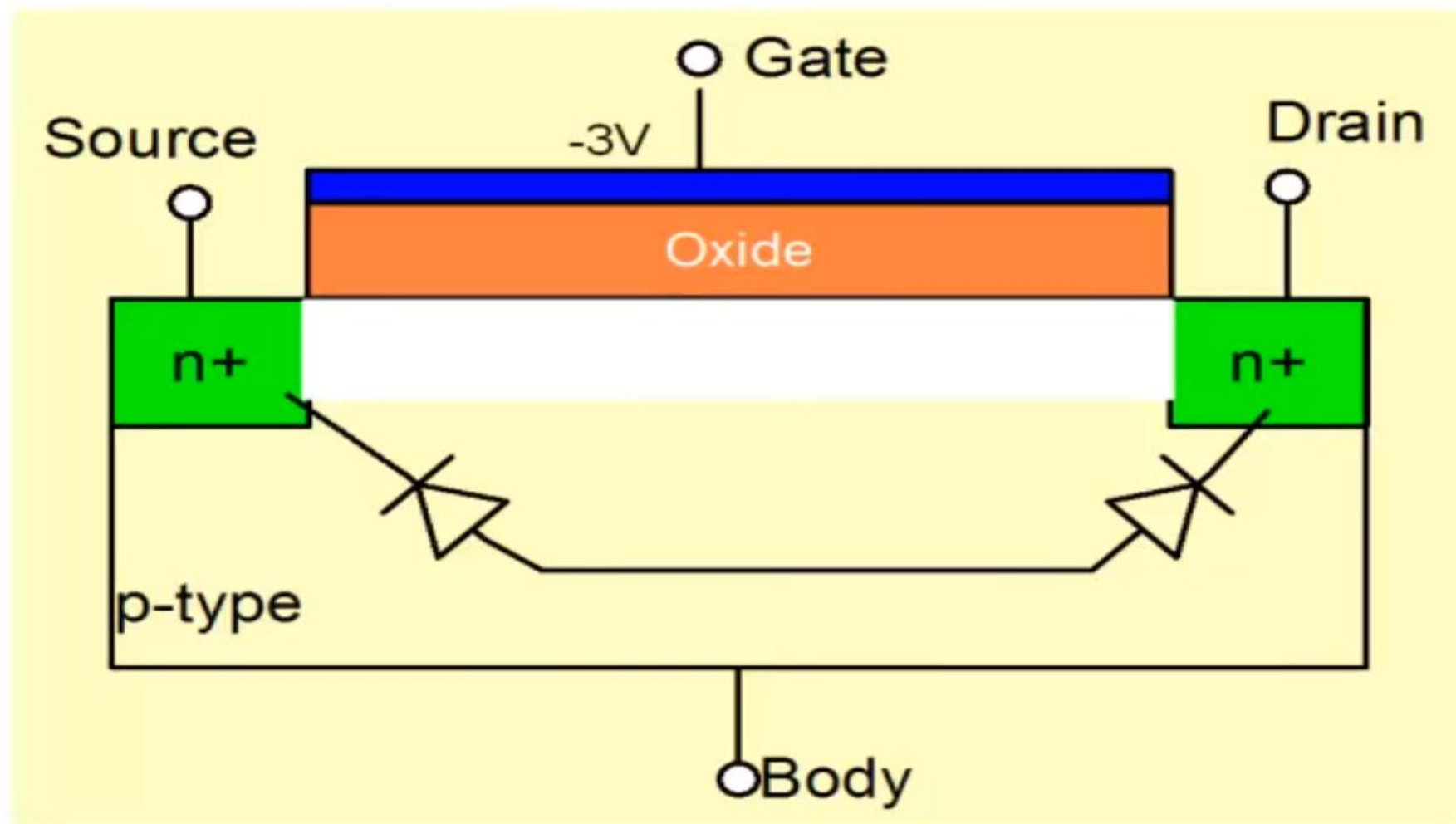
Channel exists at zero gate voltage and is depleted by gate voltage



In a depletion-mode transistor, a channel exists without any gate voltage being applied and current flows when drain voltage is applied.

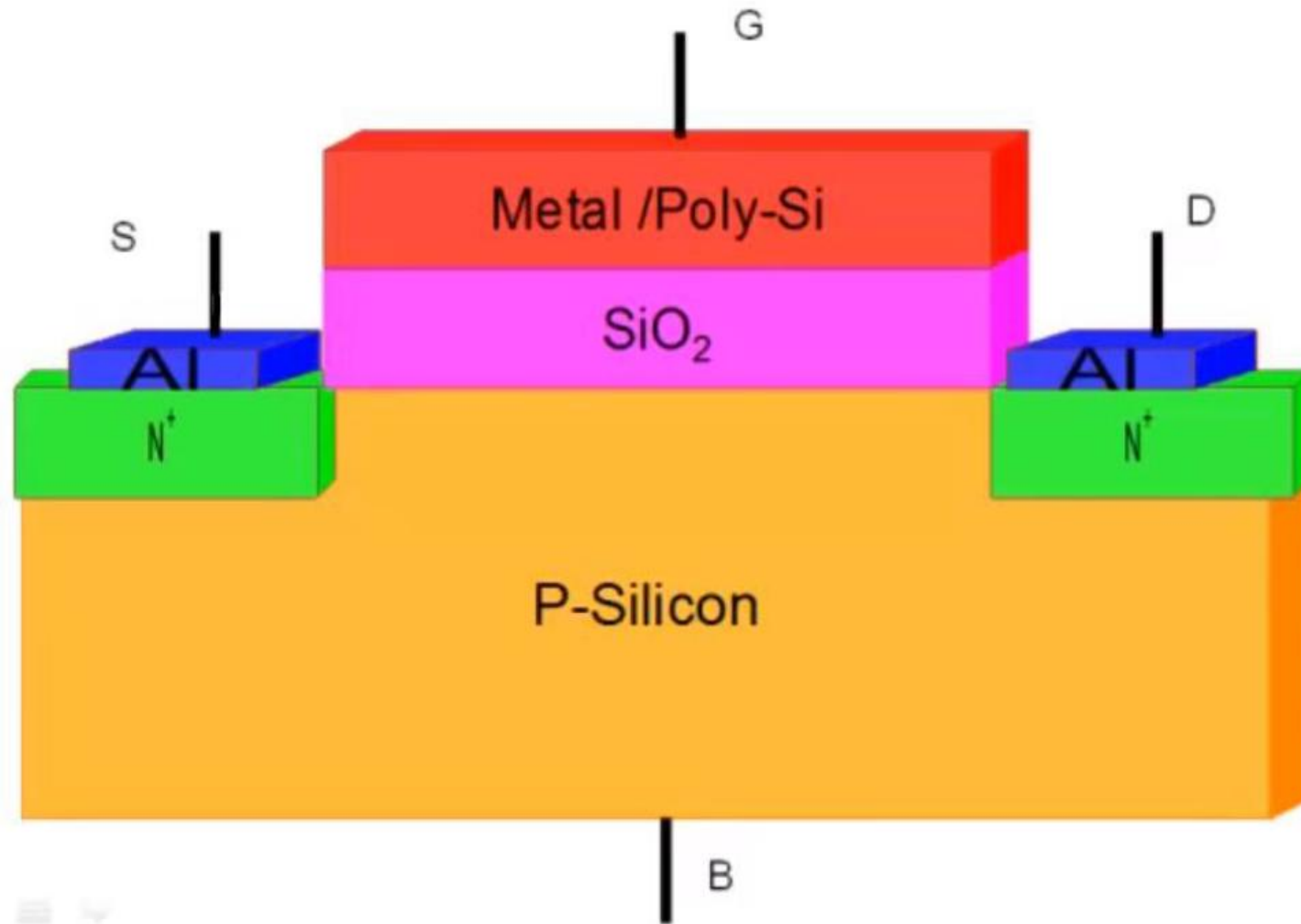
Negative gate voltage is applied to deplete the channel of carriers and cause current to reduce.

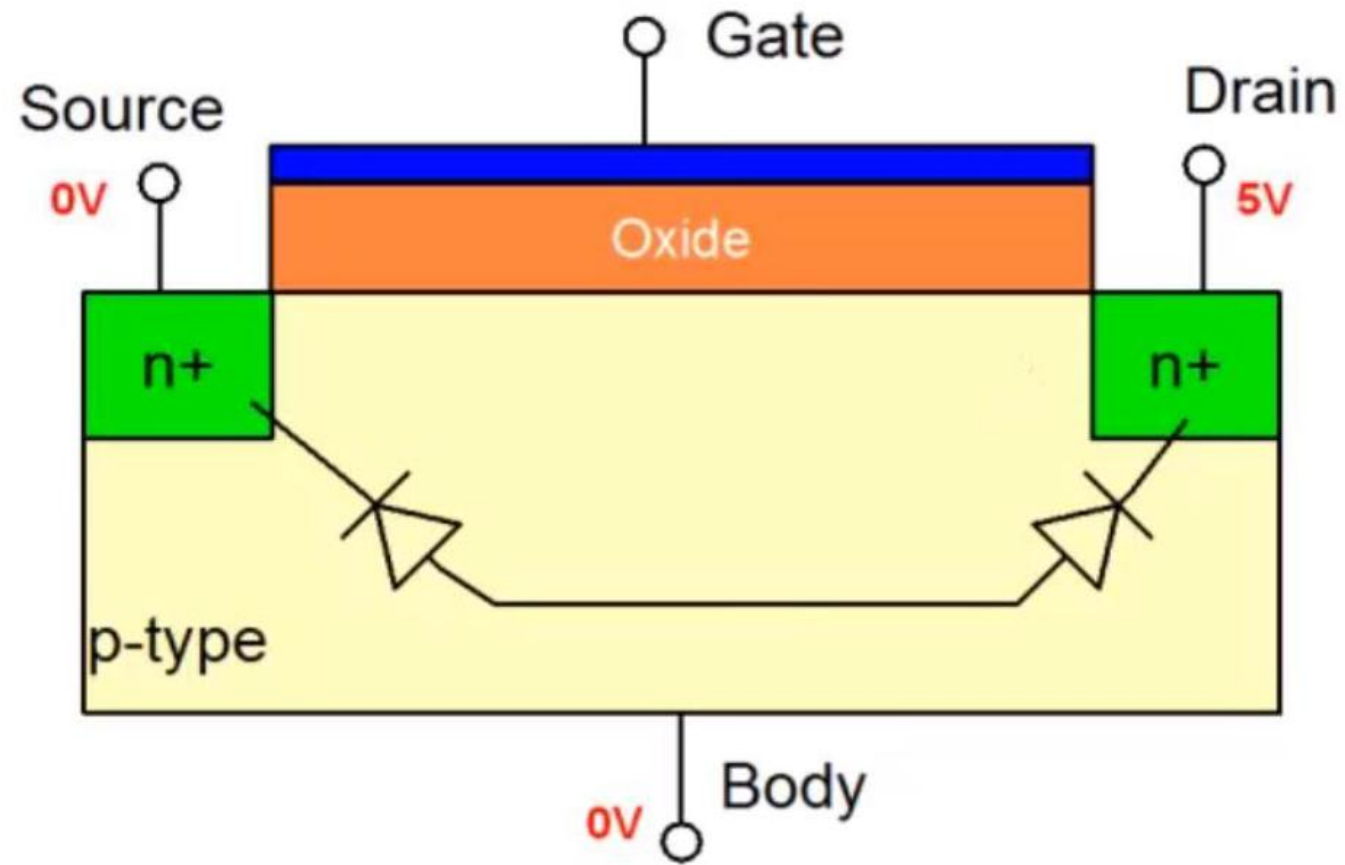
Channel exists at zero gate voltage and is depleted by gate voltage



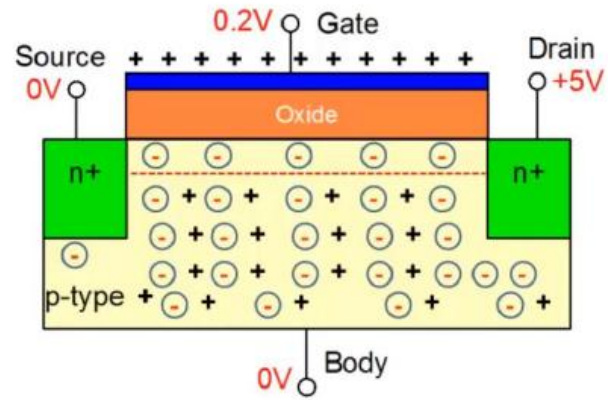
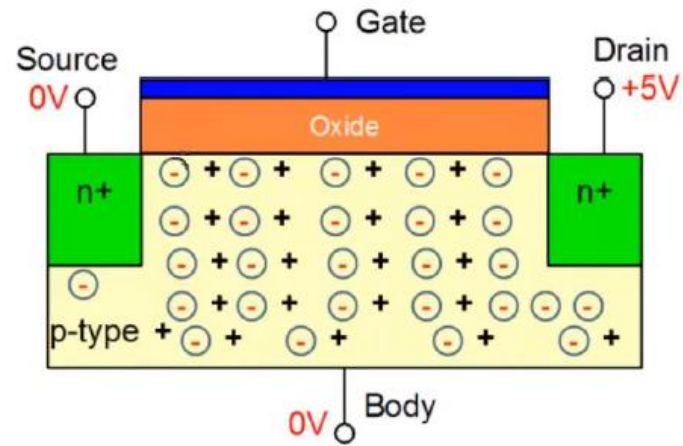
Channel is completely pinched off and current ~zero

NMOS Enhancement mode transistor: Inversion Mode Transistor

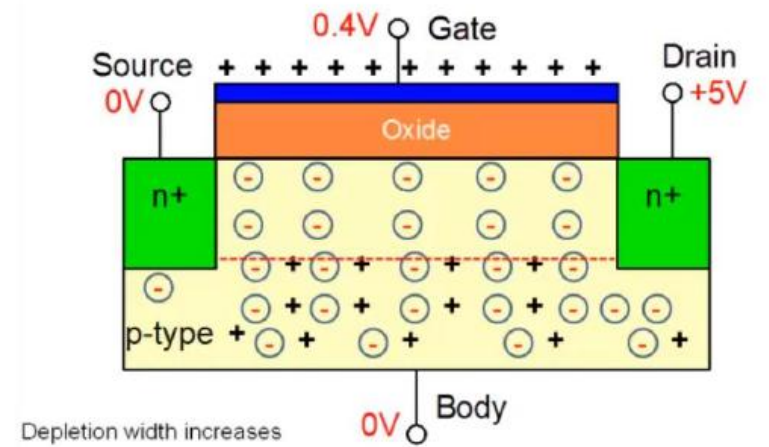




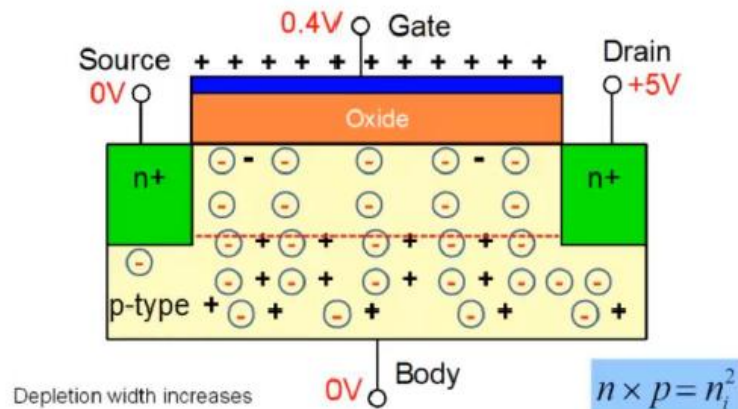
No channel exists when gate voltage is zero and current is zero as well.



Depletion Region is formed near the Si/SiO₂ interface



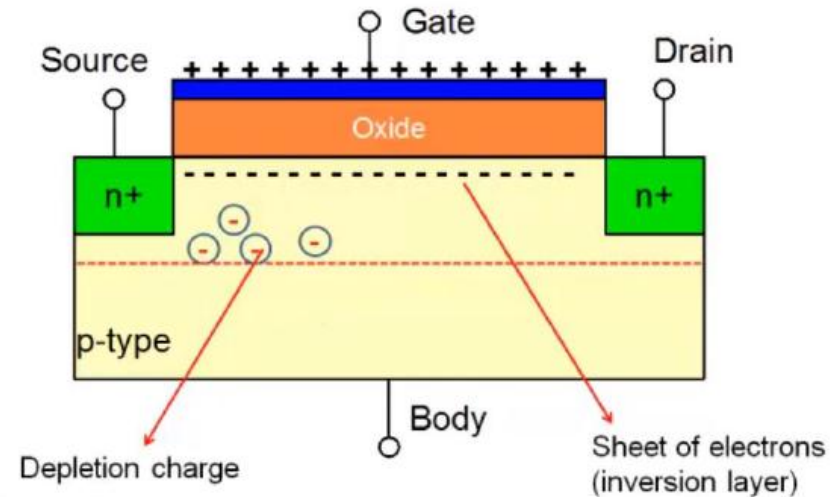
Depletion width increases



Depletion width increases

But something interesting happens: electron density at the surface also increases

At a sufficiently large voltage ($>V_{THN}$) a channel of electrons forms at the Si/SiO₂ interface.

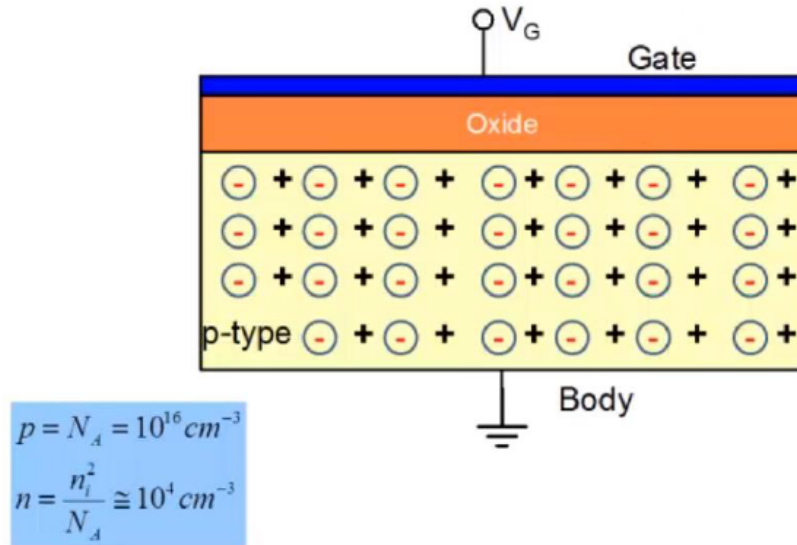


Depletion charge

Sheet of electrons
(inversion layer)

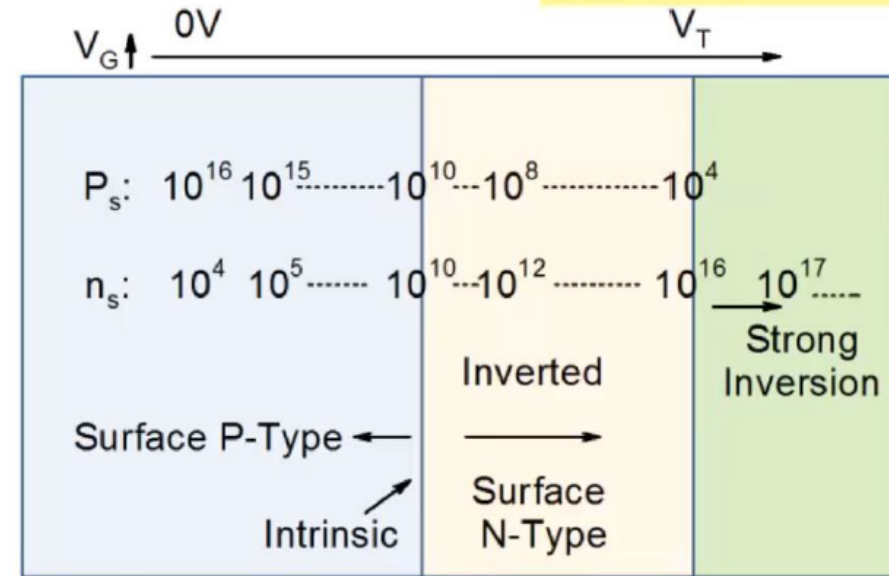
Conductivity modulation at the surface?

MOS capacitor constitutes the heart of a MOSFET



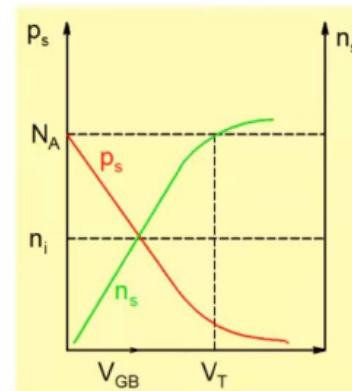
Field Effect...

$$n_s \times p_s = n_i^2 \approx 10^{20} \text{ cm}^{-3}$$



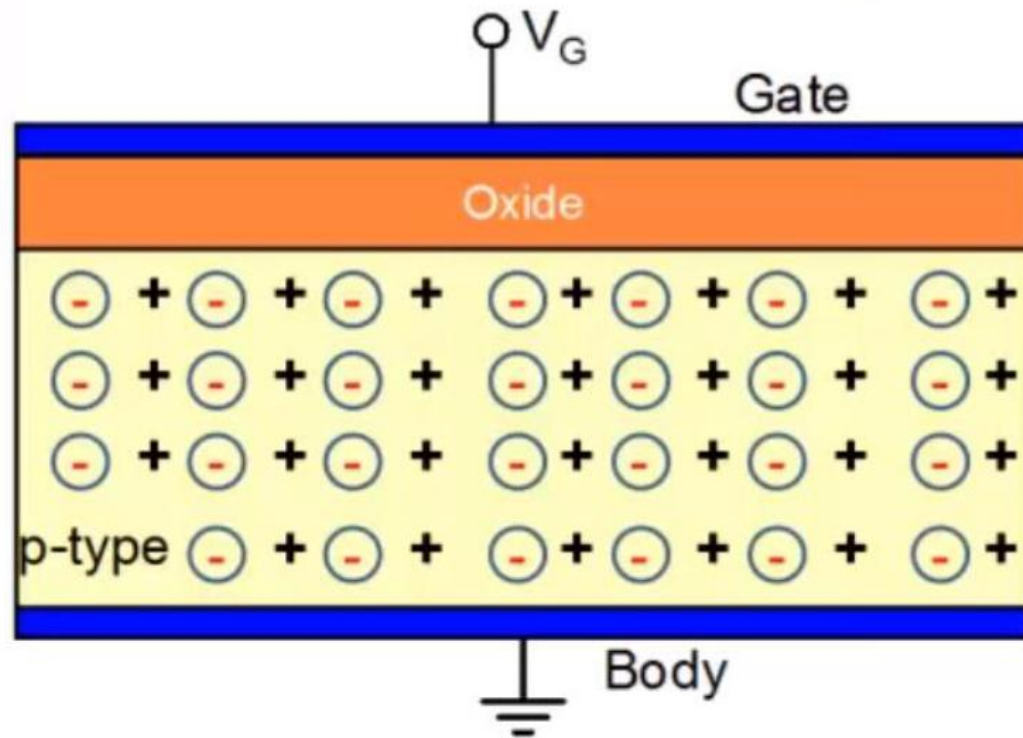
Surface carrier density can be changed from P-type to N-type

Surface Carrier Density



Flat band condition

$$V_G = V_{FB}$$

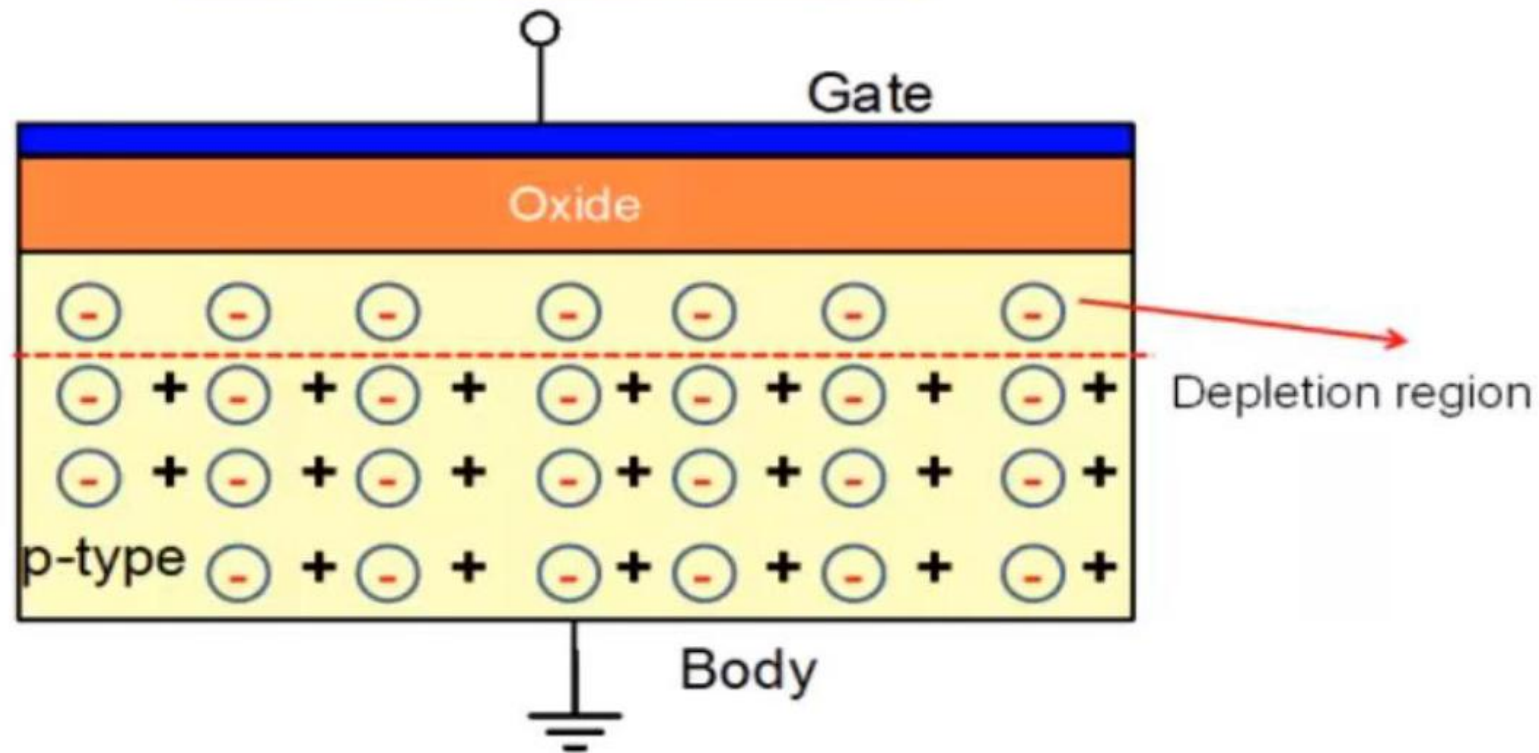


Whenever two different material are brought into contact, an internal potential difference develops like in a pn junction. Thus even when no gate voltage is applied, there is a voltage across the mos capacitor.

$V_G = V_{FB}$; Flat-band condition meaning no NET voltage across the capacitor.
Uniform hole density everywhere

Depletion

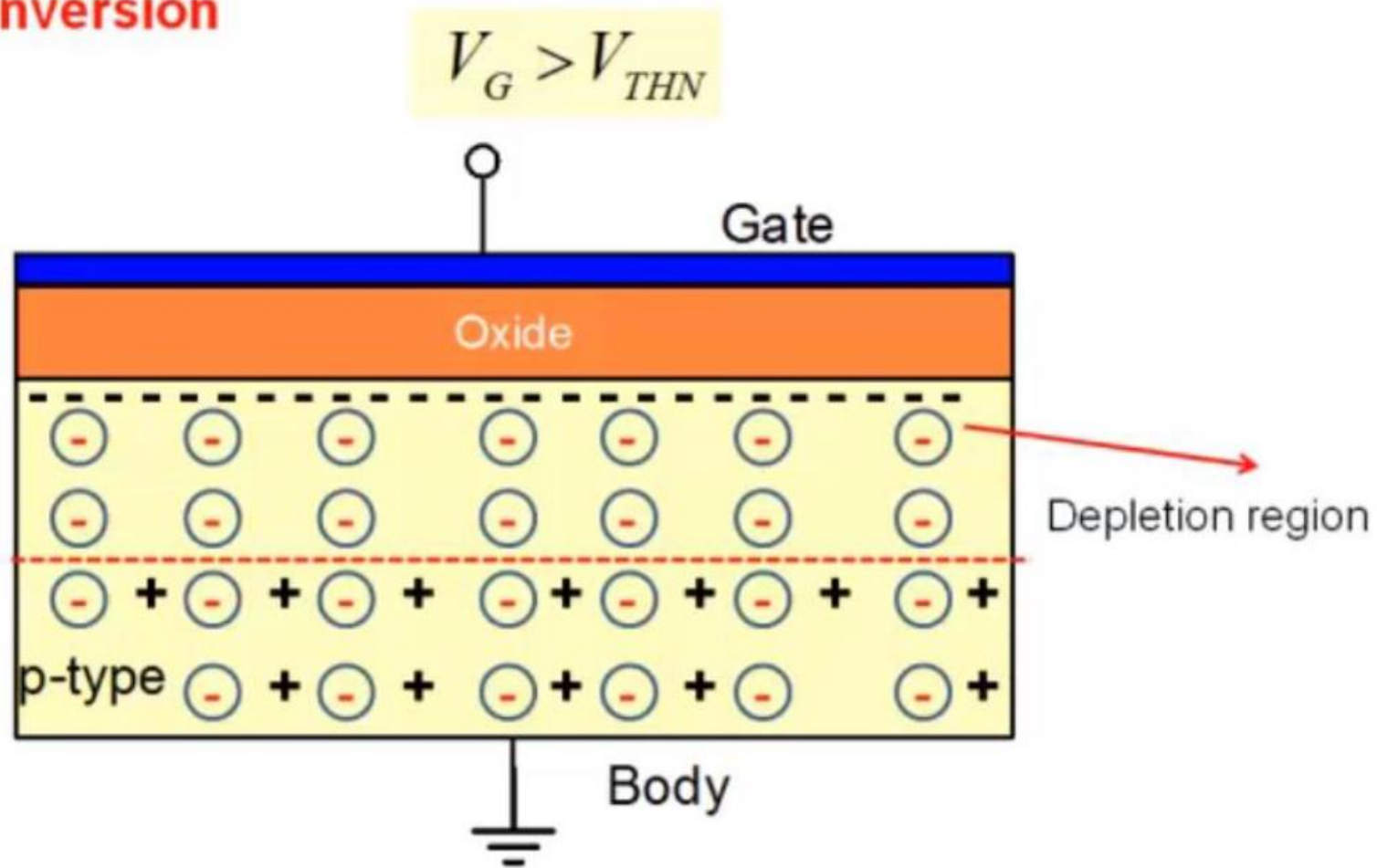
$$V_G > V_{FB} \text{ but } V_G < V_{THN}$$



Holes are depleted from the surface $p_S < p_B$

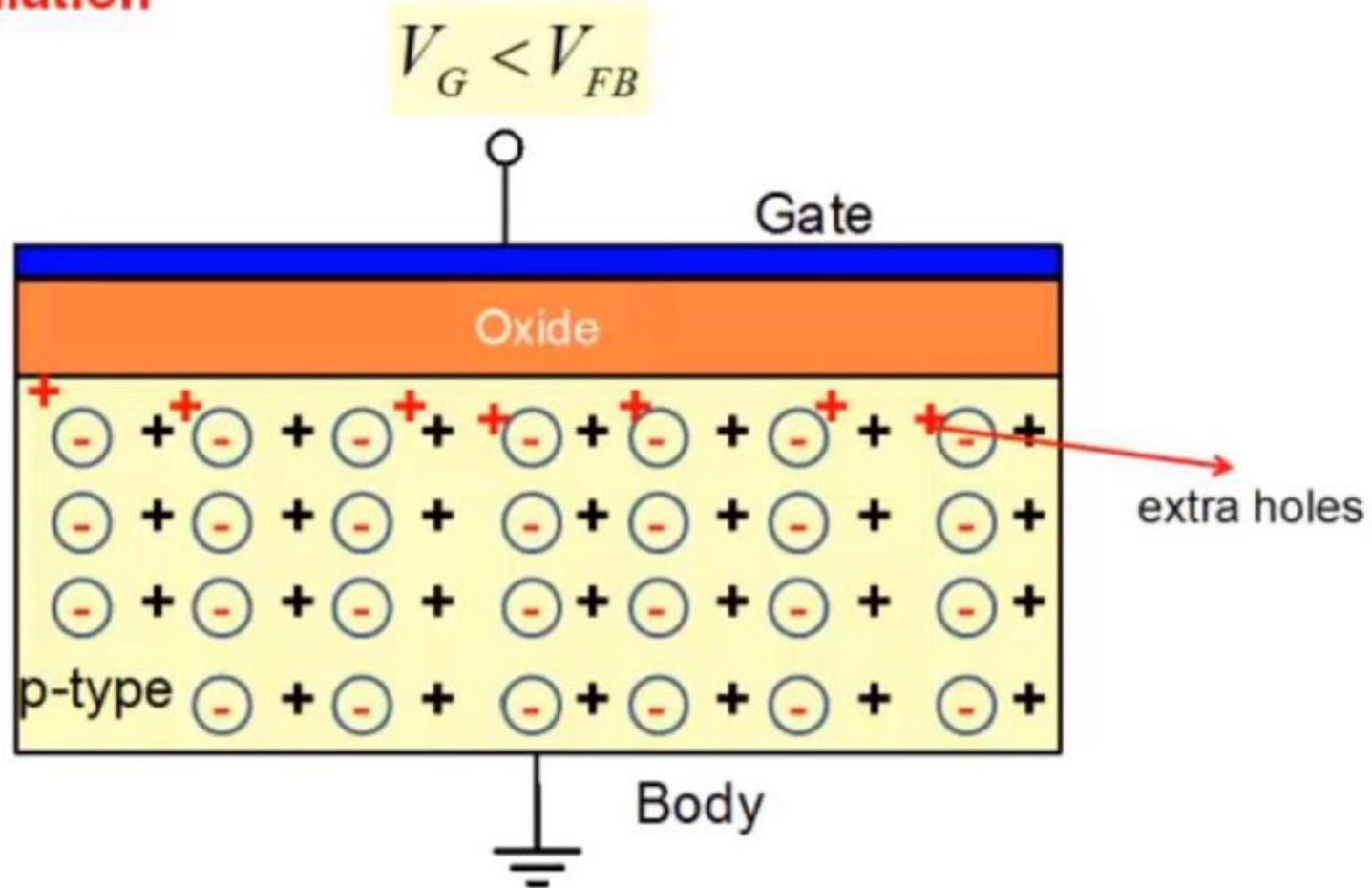
Although $n_S > n_B$ electron density is also very small

Strong Inversion



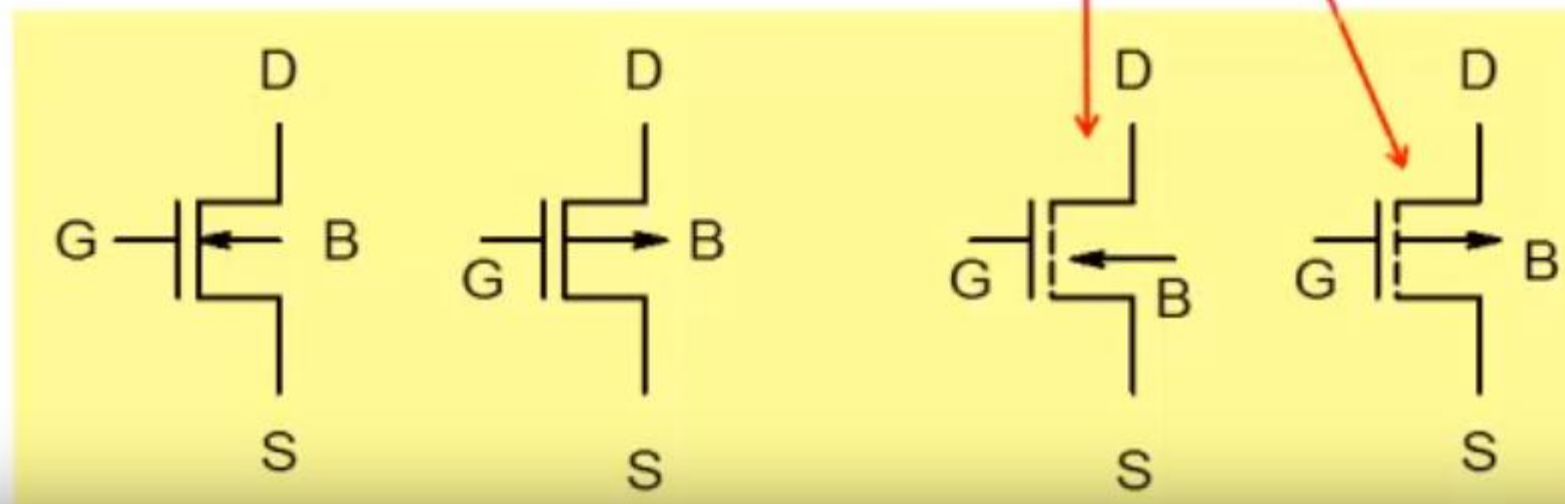
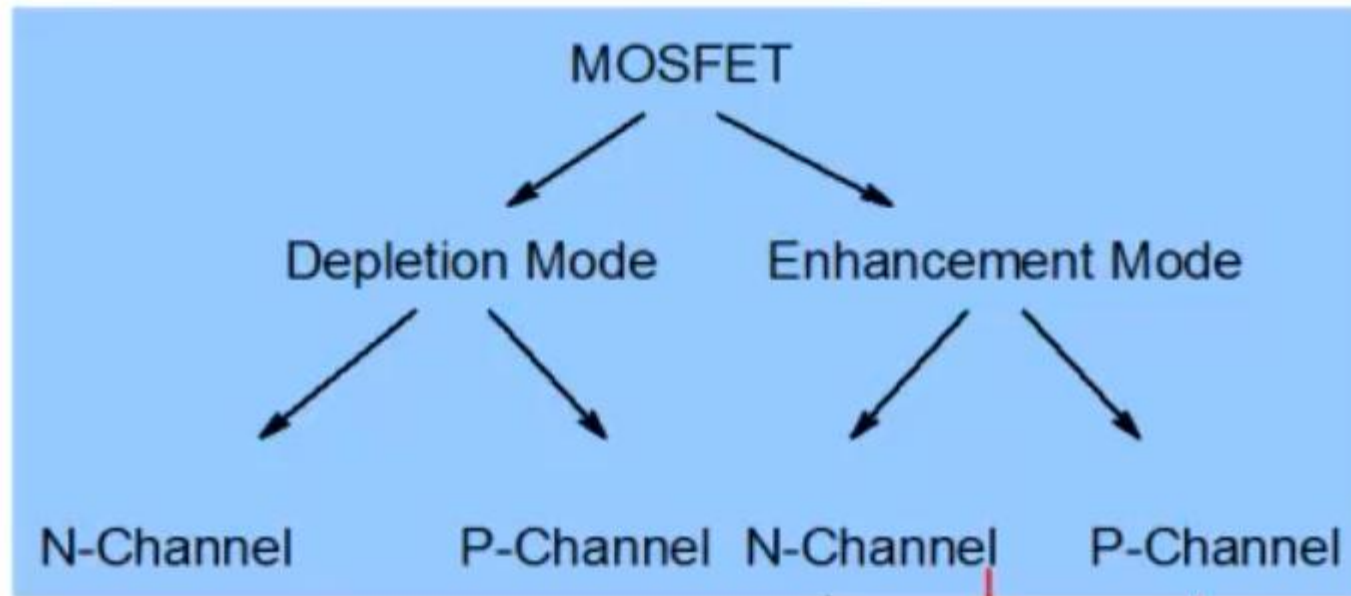
Electrons are accumulated at the surface $n_S \gg N_A$

Accumulation

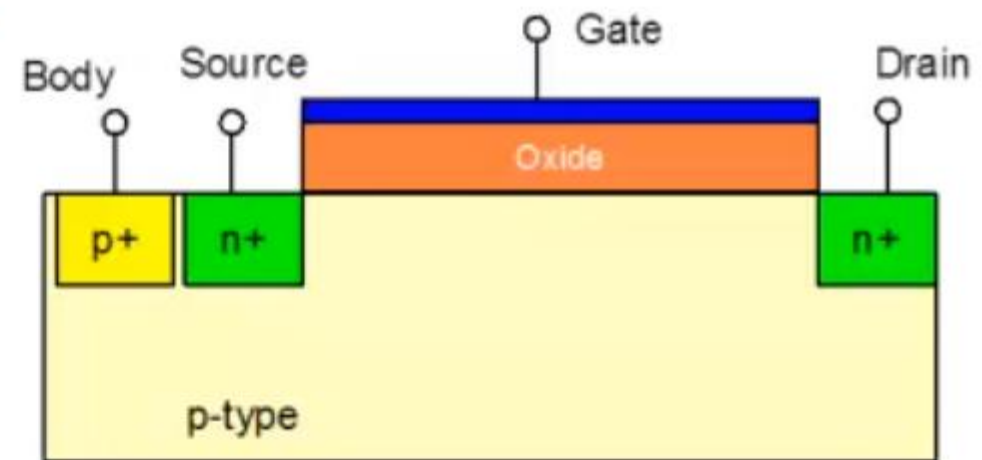
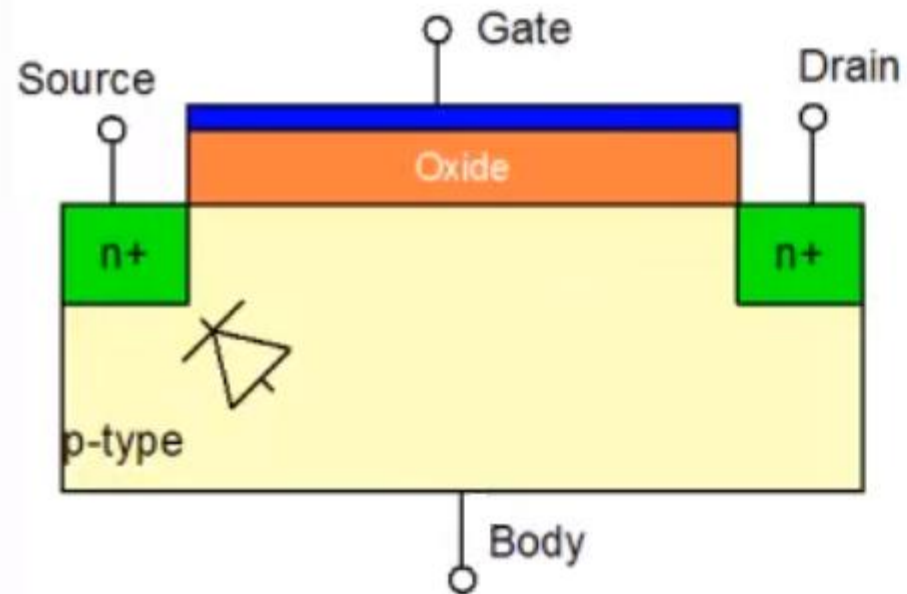
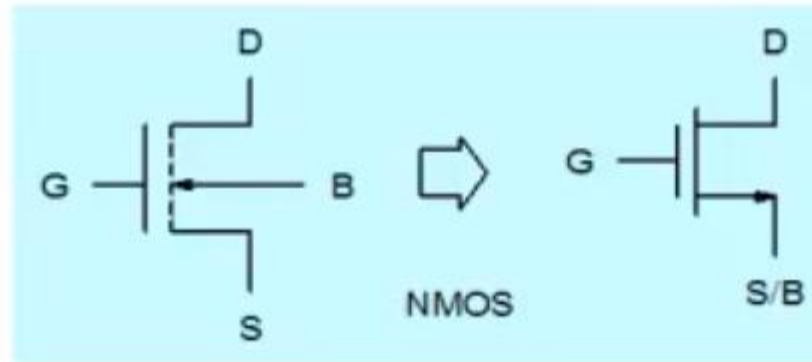


Holes are accumulated at the surface $p_S > p_B$

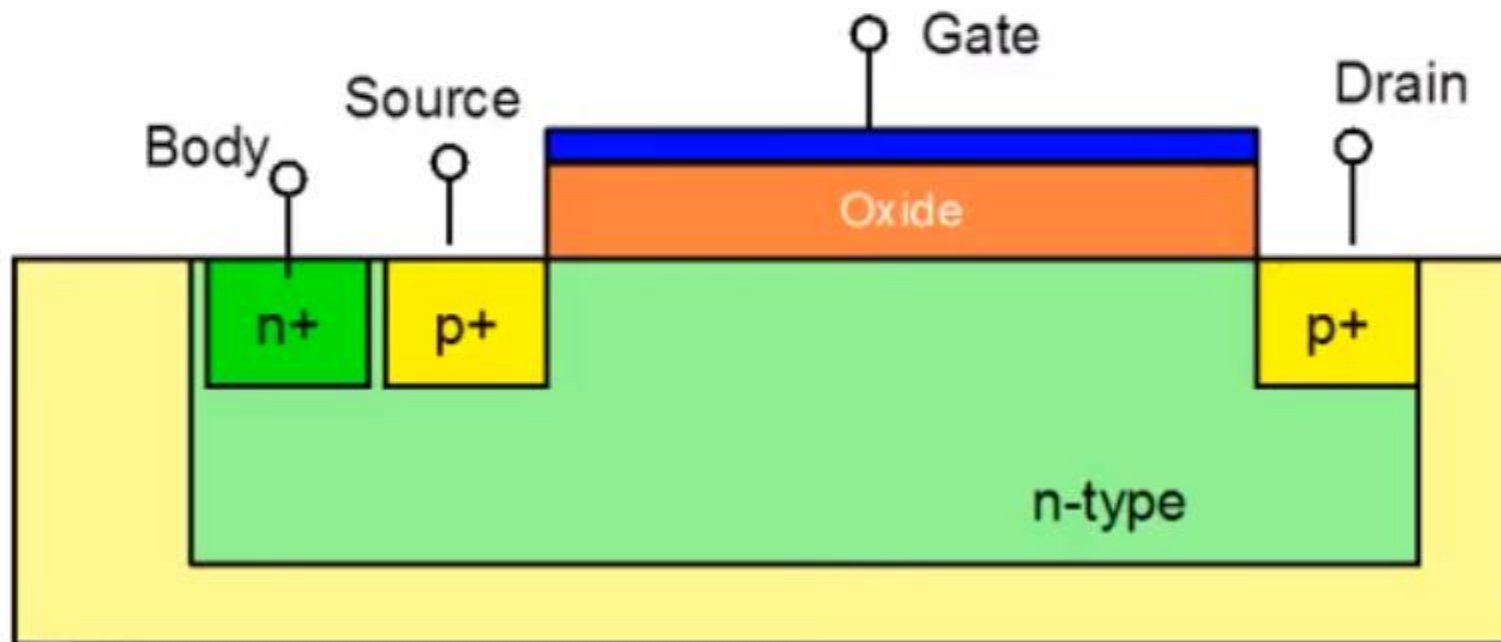
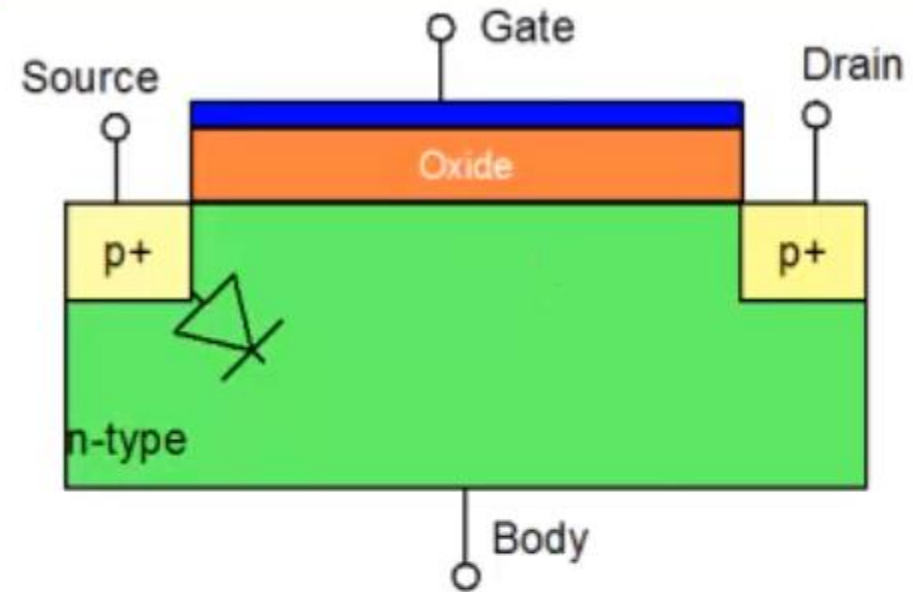
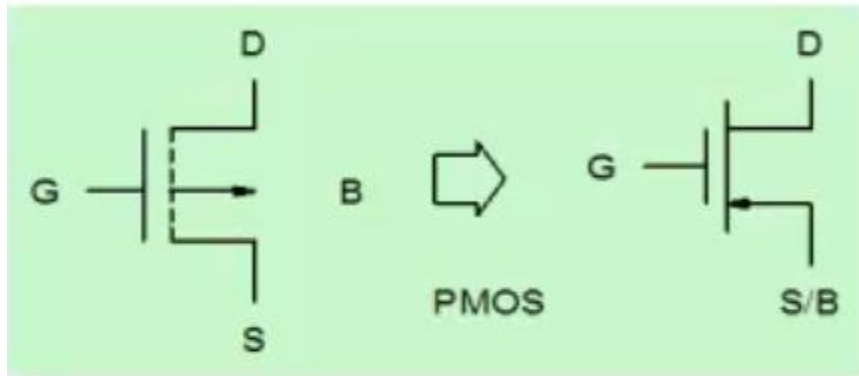
Metal Oxide Semiconductor Field Effect Transistor:

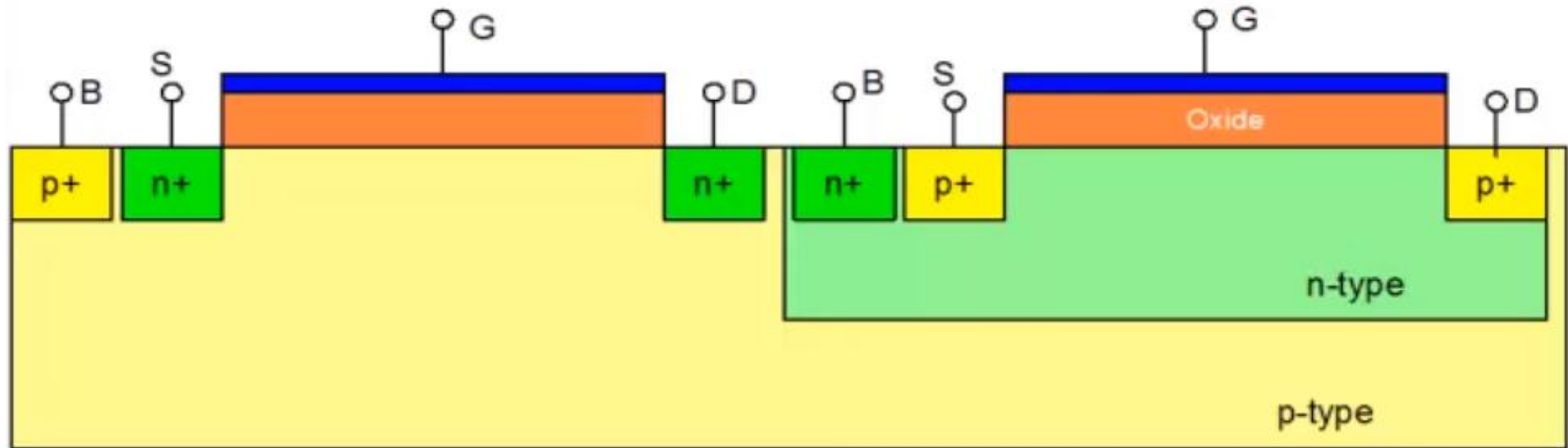


Simplified Symbols and structure

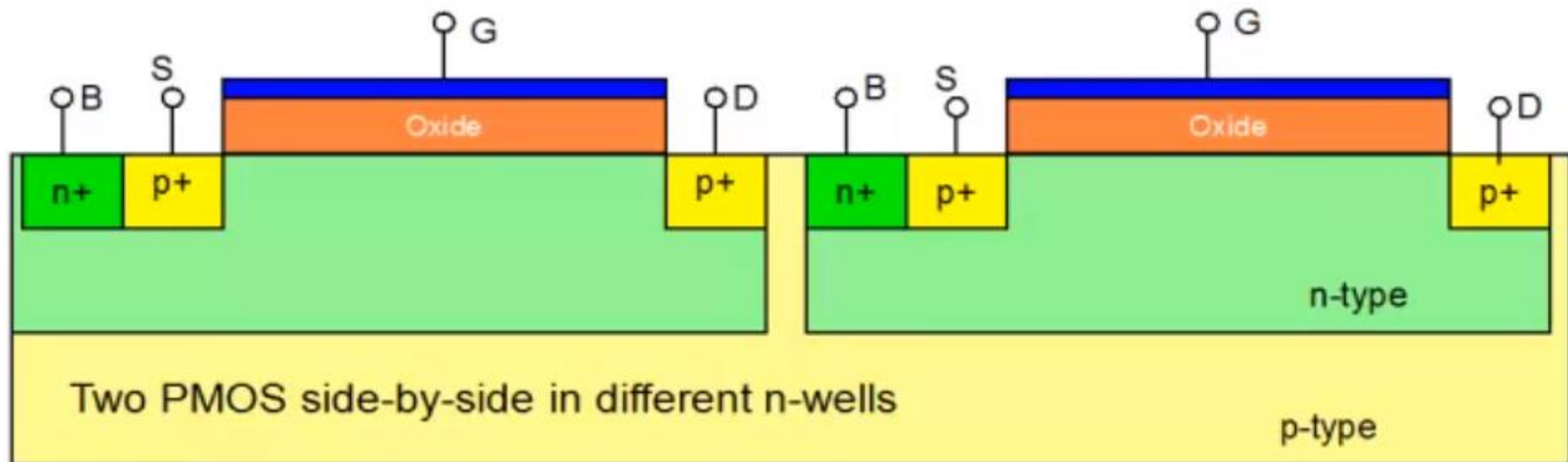
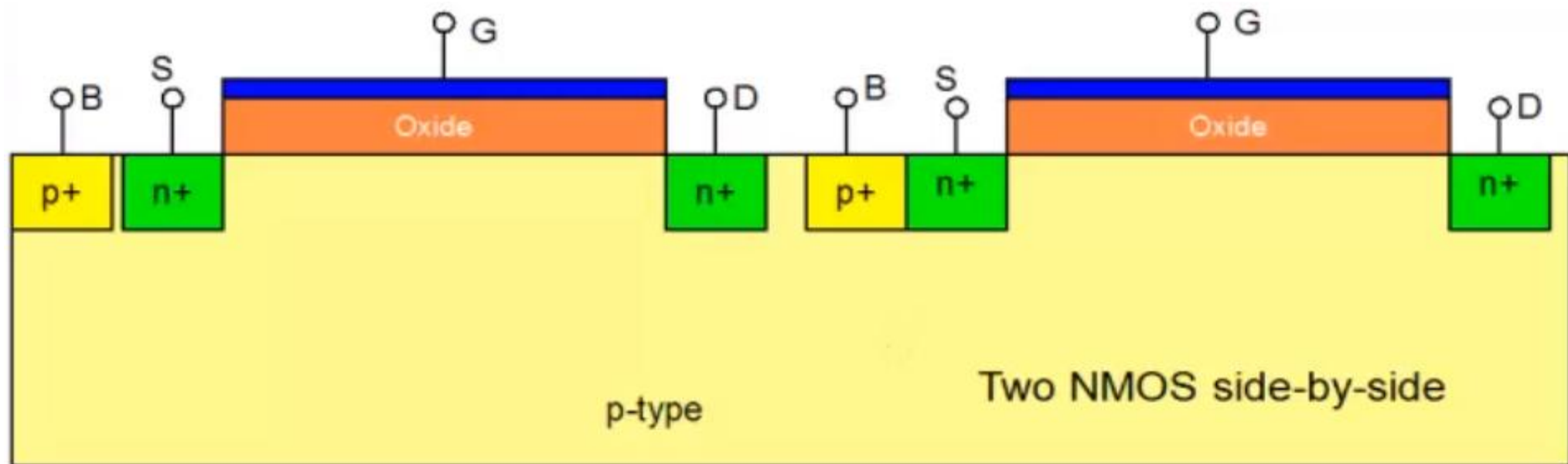


Simplified Symbols and structure

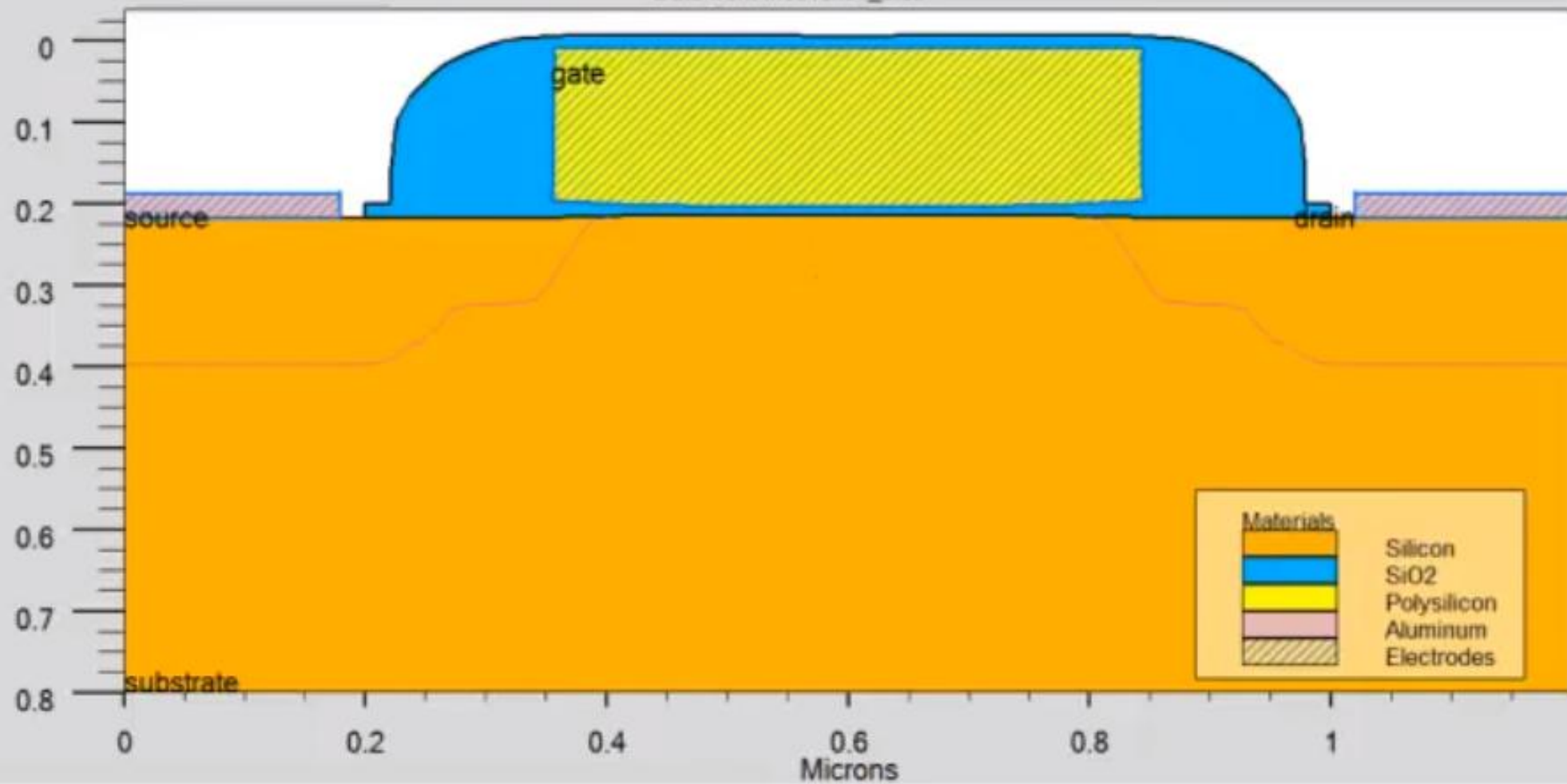


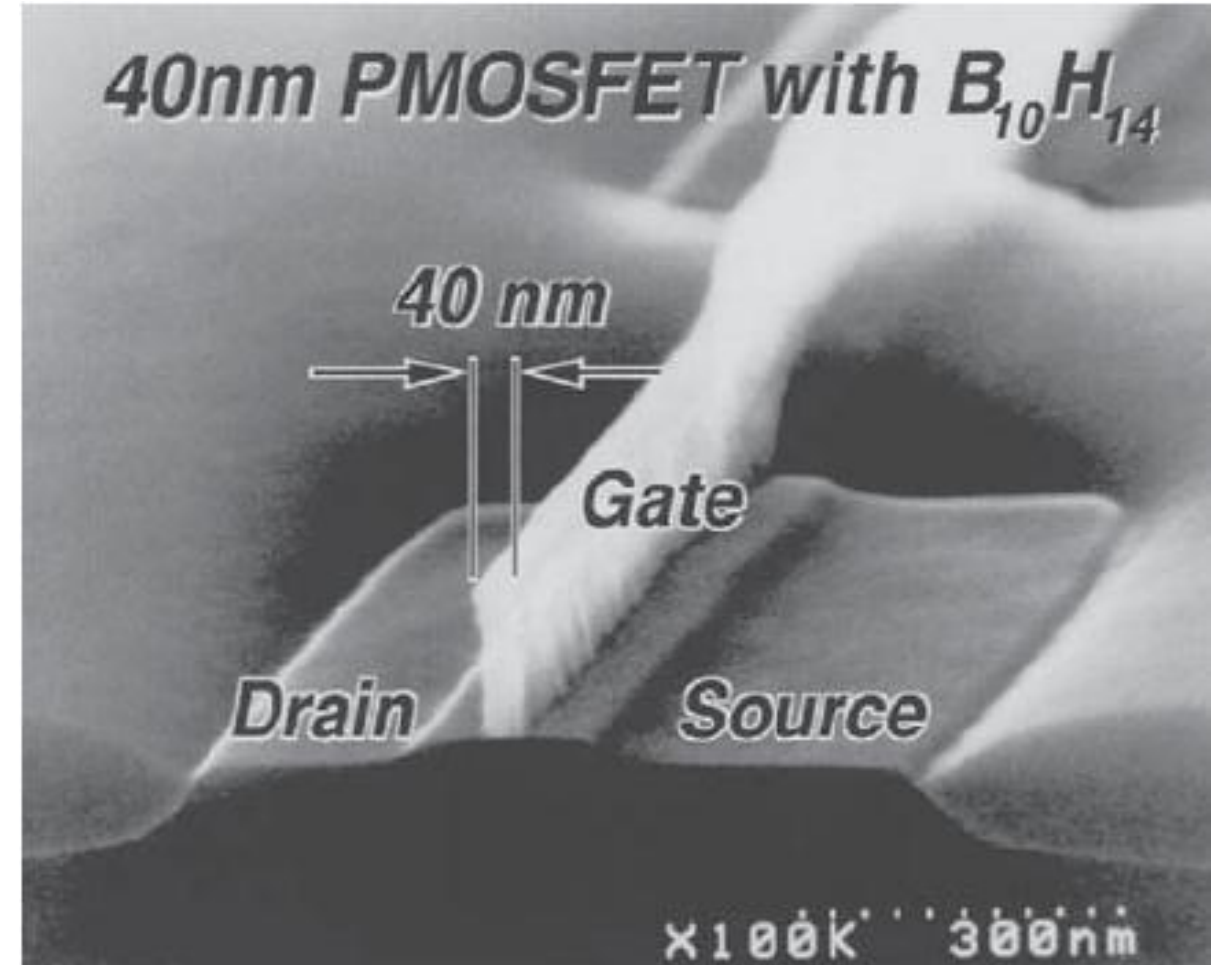
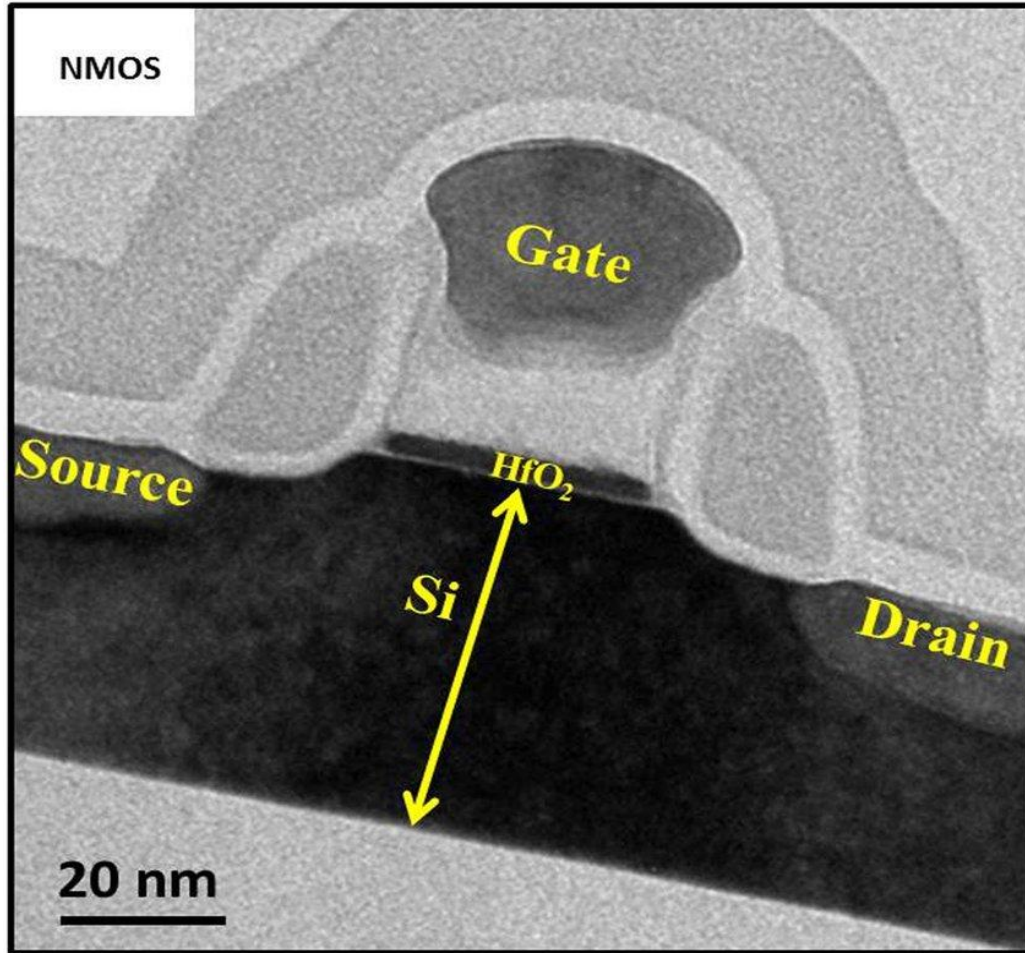


- Body potential of NMOS transistors is same and is normally connected to the most negative voltage in the circuit to ensure that $V_{BS} < 0$ and thus body-source PN junction is reverse biased.
- Each PMOS can be fabricated in a separate individual N-well and thus each pmos body terminal can have a distinct voltage. Normally body and source terminals of PMOS are shorted together.

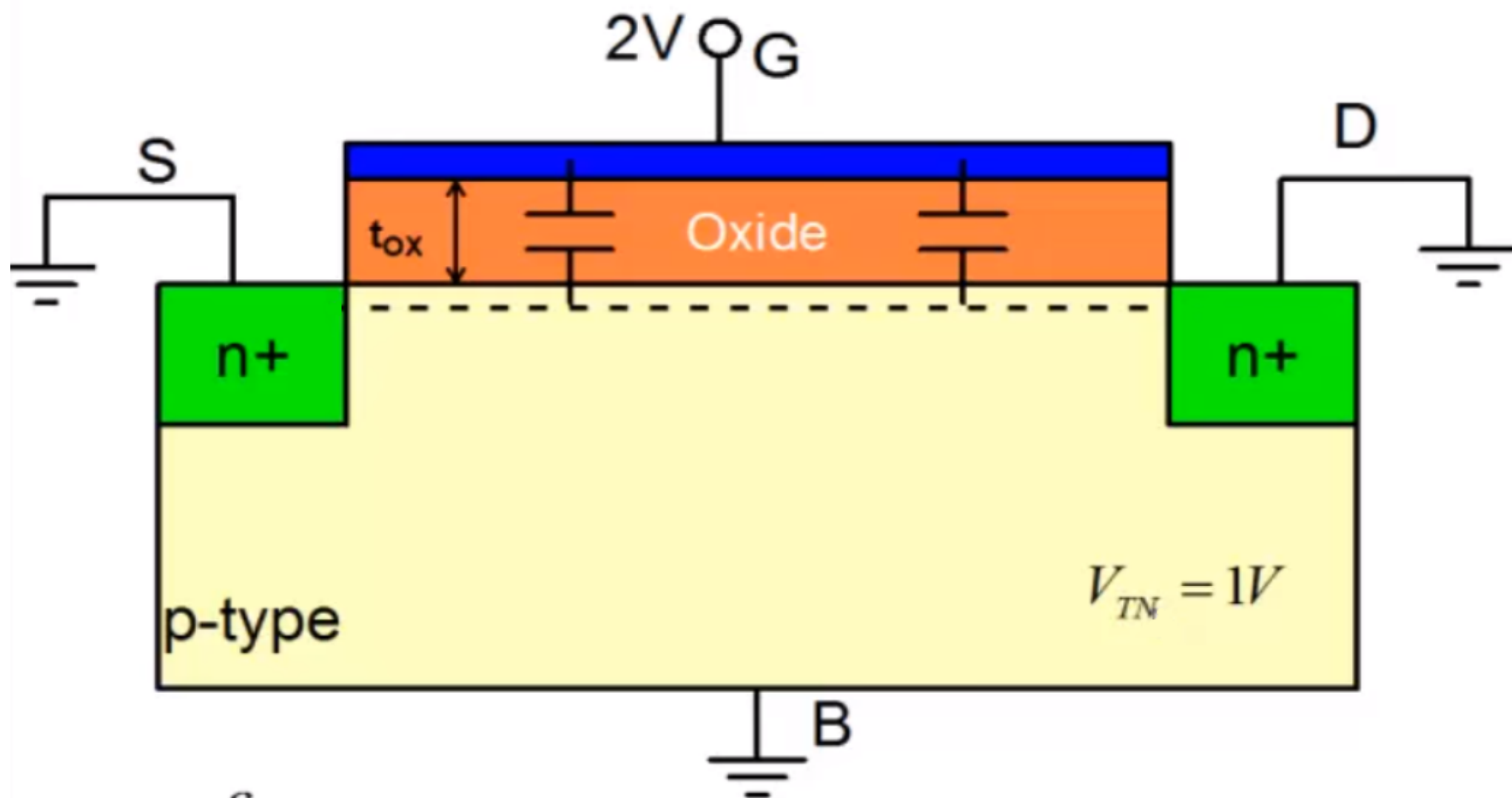


ATHENA
Data from mos1ex01_0.str



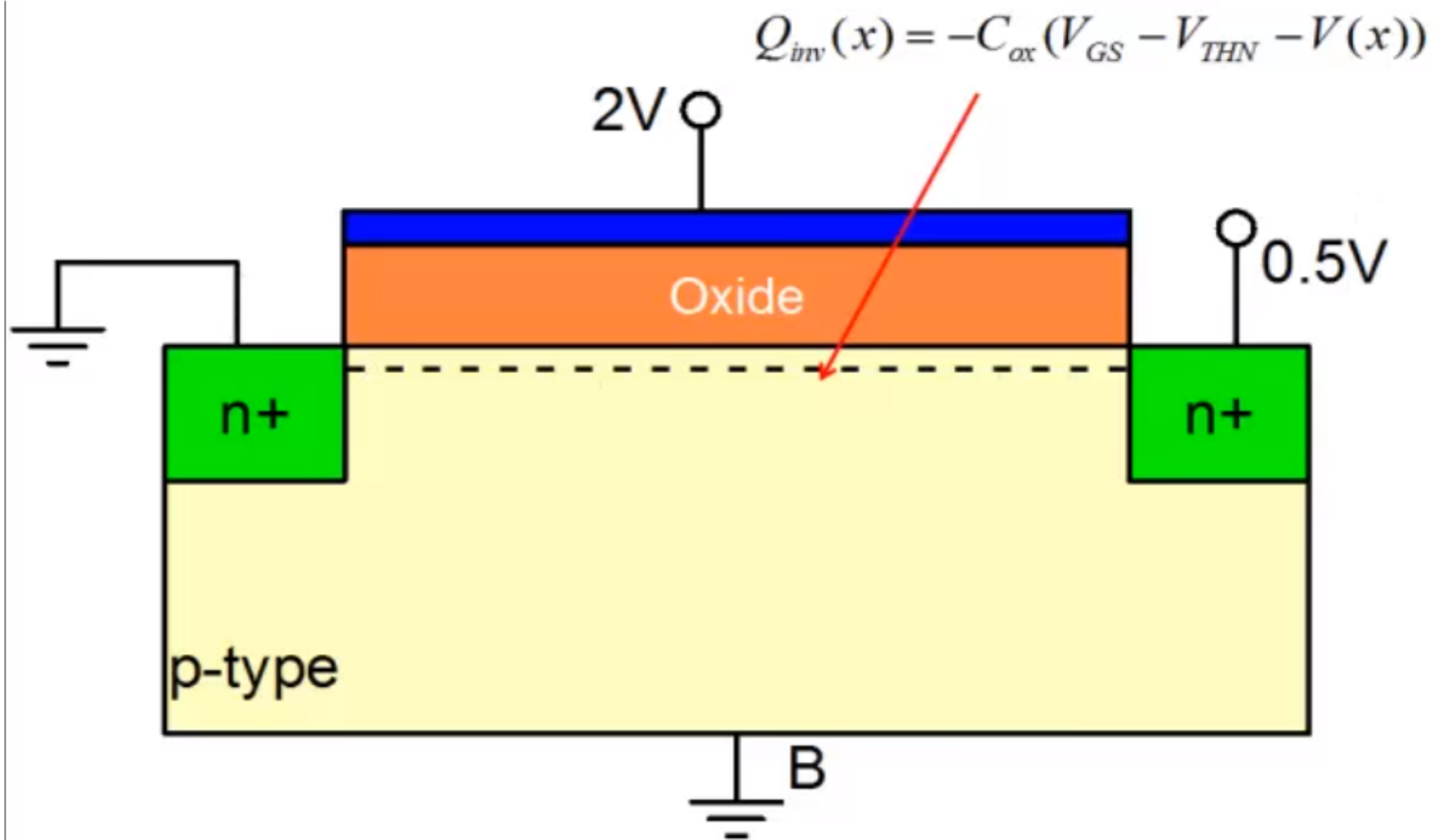


Operation of the MOSFET

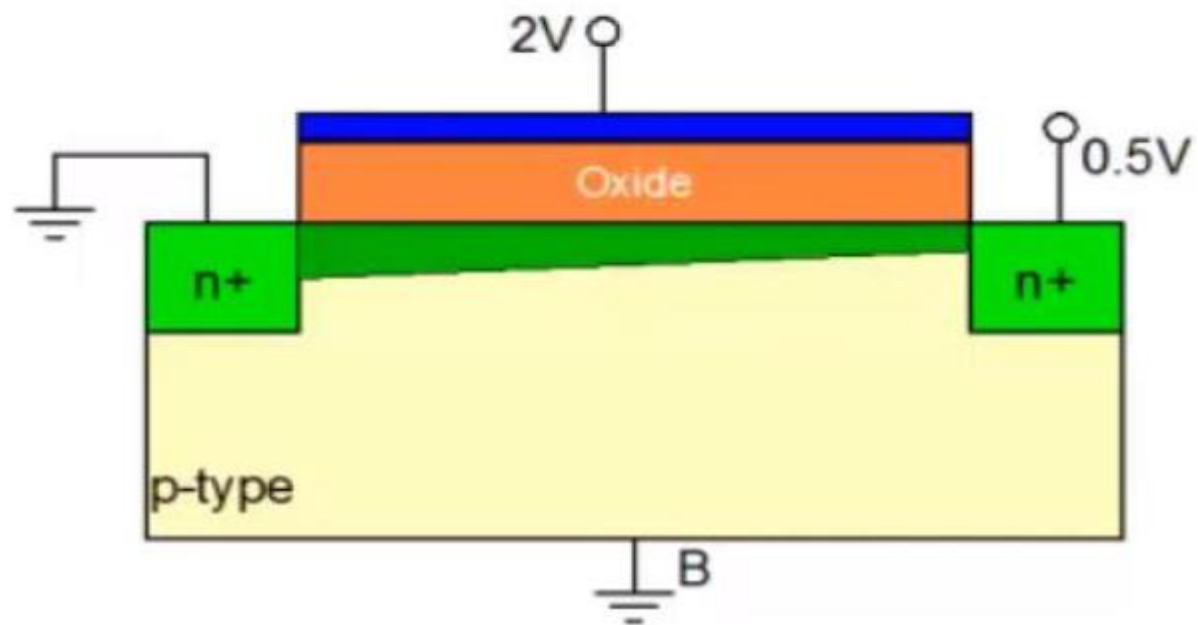
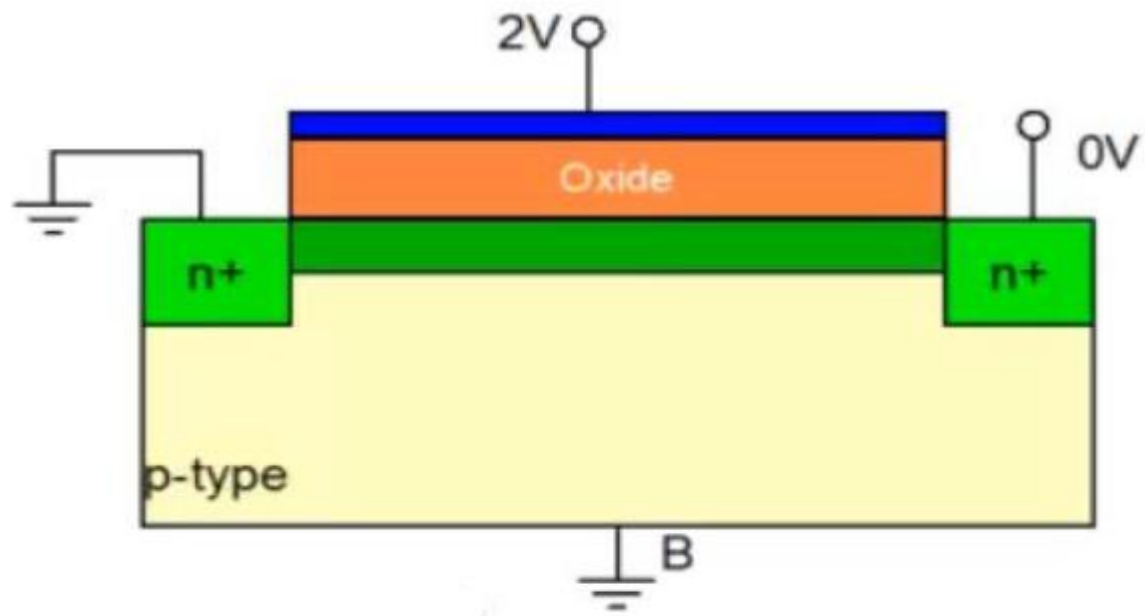


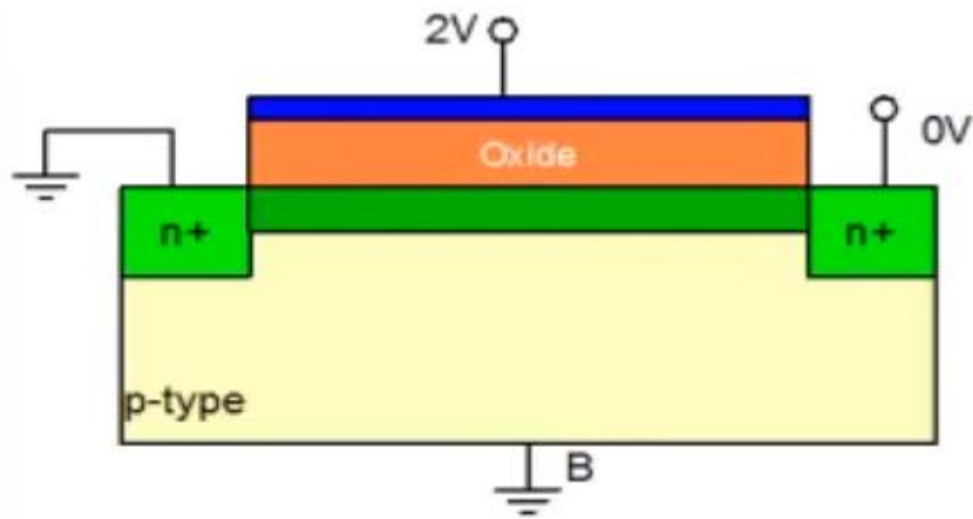
$$C'_{ox} = \frac{\epsilon_{ox}}{t_{ox}}$$

Inversion charge/area : $Q_{inv} = -C_{ox}(V_{GS} - V_{THN})$



When a positive drain voltage is applied, current flows from drain to source and inversion charge density decreases from source to drain end.

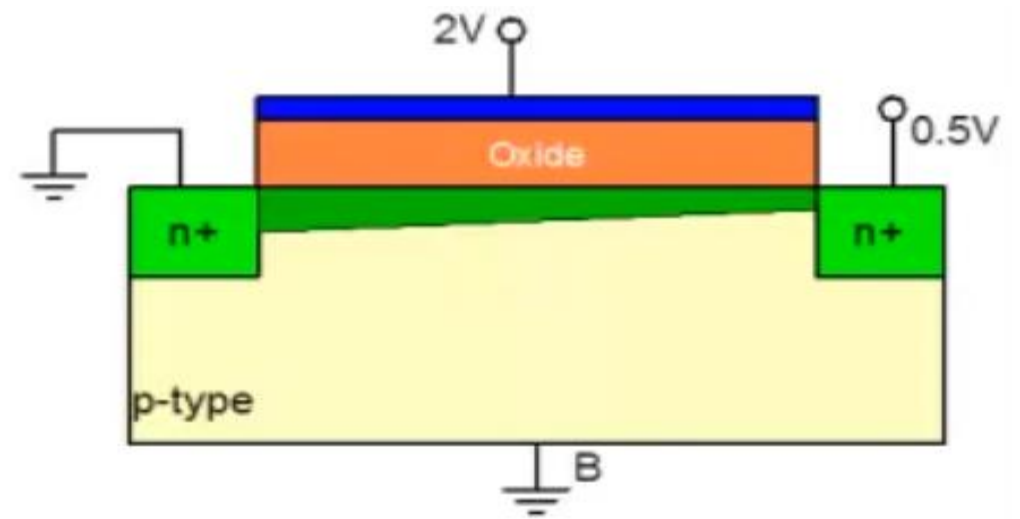




$$dR_{ch} = \frac{1}{q\mu n(x)} \frac{dx}{W \times t}$$

$$R_{ch} \propto \frac{1}{\int Q_{inv} dx}$$

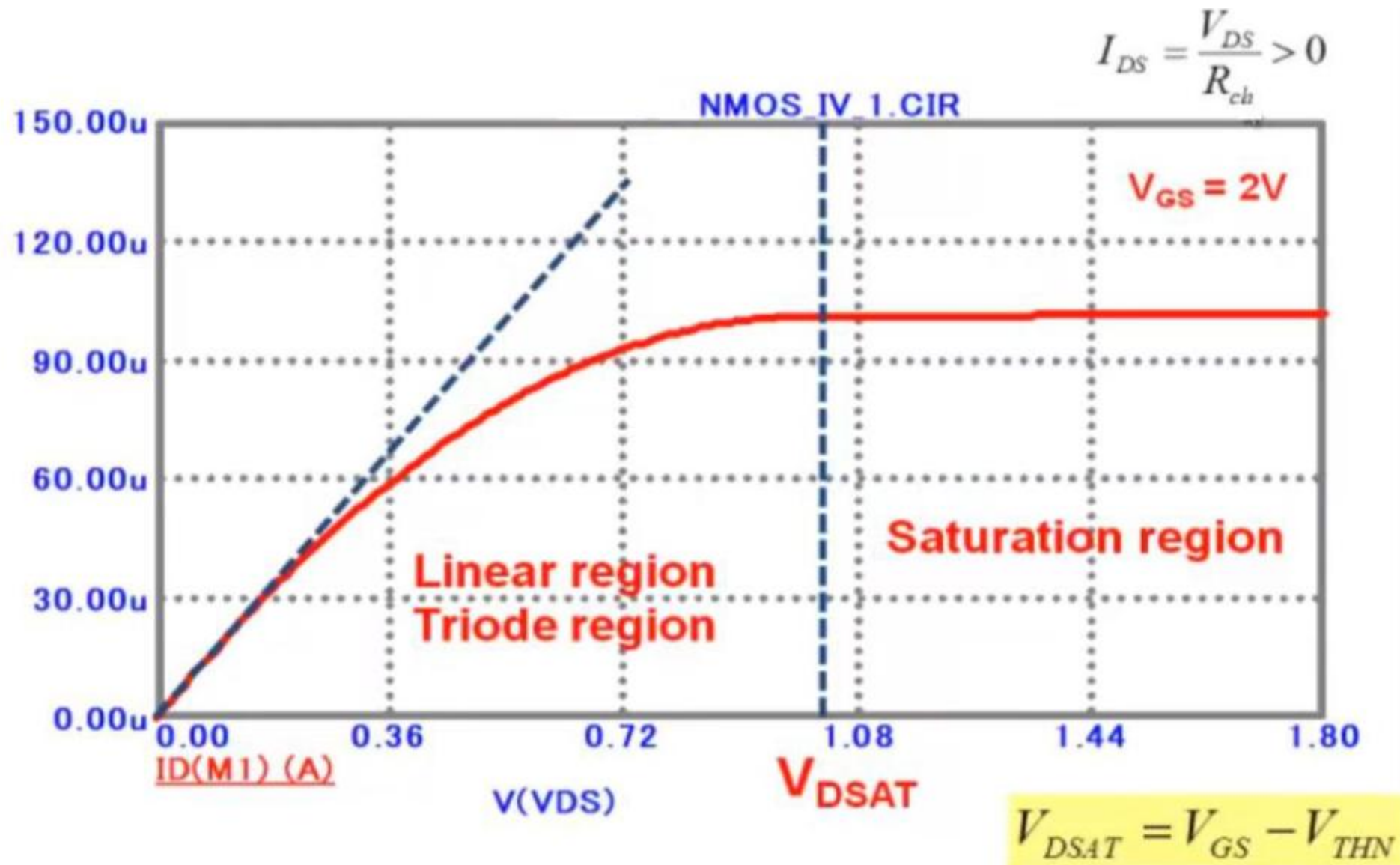
$$I_{DS} = \frac{V_{DS}}{R_{cho}} = 0$$



$$R_{ch} > R_{cho}$$

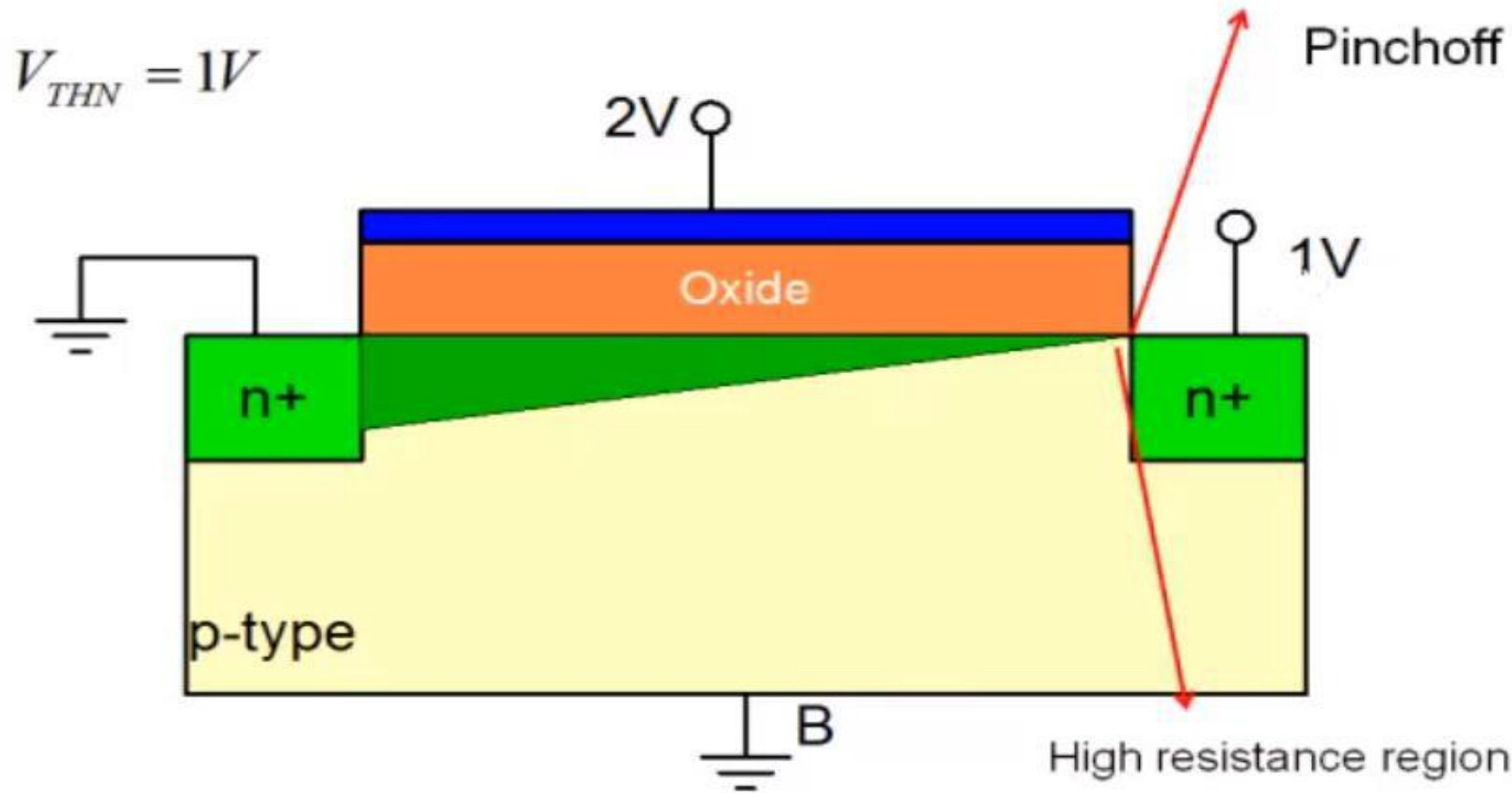
$$I_{DS} = \frac{V_{DS}}{R_{ch}} > 0$$

As drain voltage increases, channel resistance also increases causing drain current to depart from linear behavior

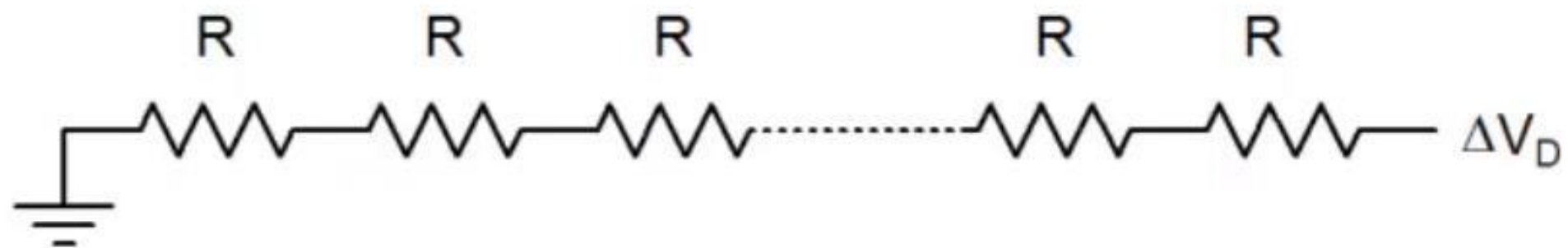


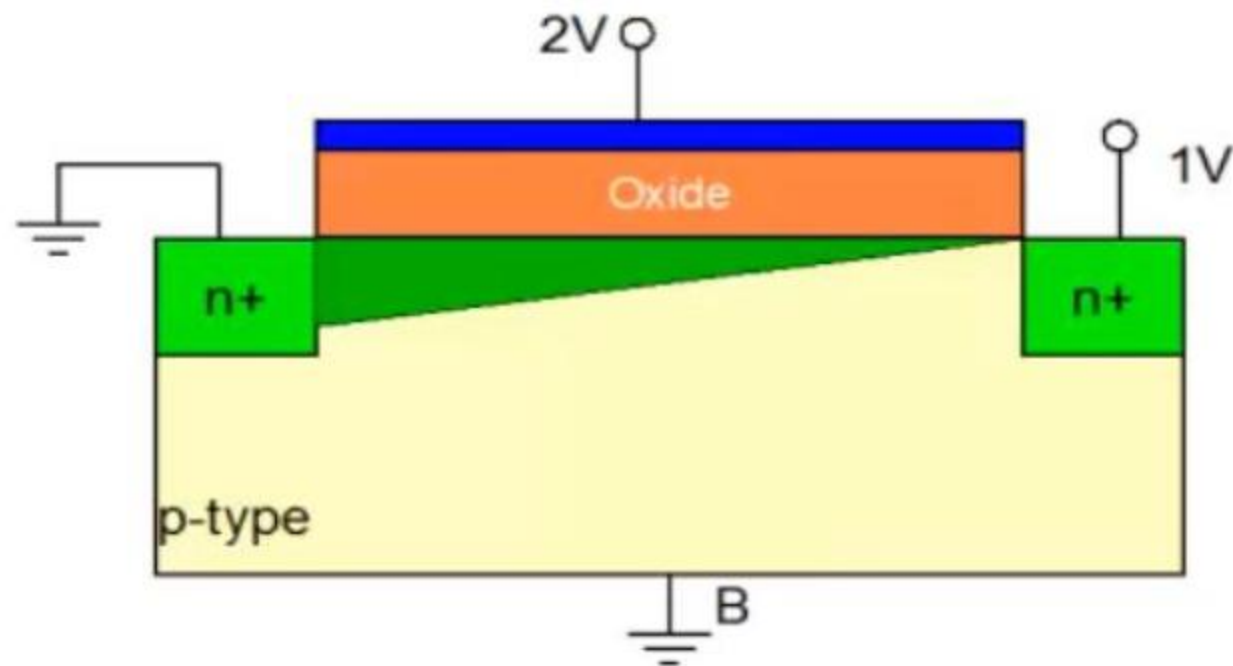
Note that **Saturation in MOSFET** is analogous to **forward active mode in BJT** and linear region is analogous to saturation in BJT.

$$Q_{inv}(x) = -C_{ox}(V_{GS} - V_{THN} - V(x)) \cong -C_{ox}(2 - 1 - 1) = 0$$

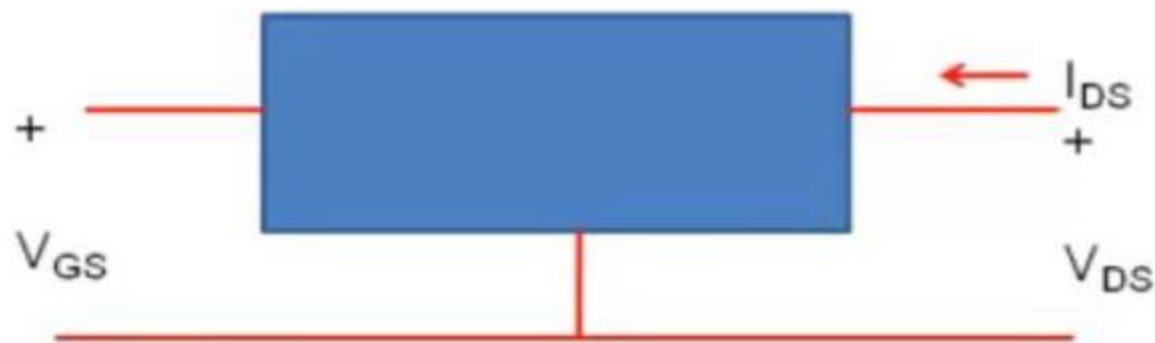


Any further increase in drain bias is absorbed in a small region next to the drain and rest of channel is not much affected and thus current becomes constant.



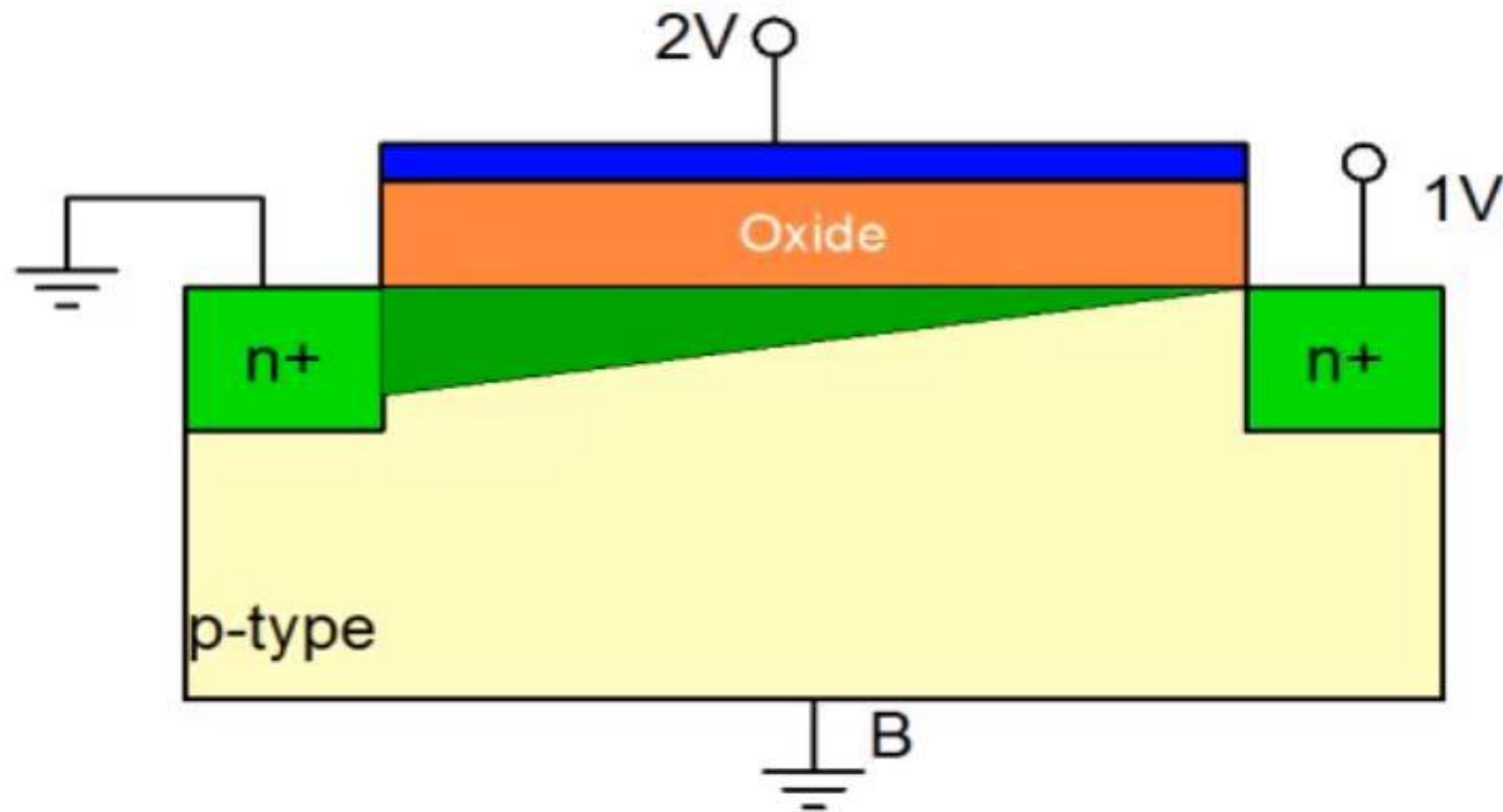


After pinchoff or saturation, drain current does not change much with drain voltage but is still very sensitive to gate voltage. MOSFET can now **AMPLIFY** signals



$$\frac{\partial I_{DS}}{\partial V_{GS}} \gg \frac{\partial I_{DS}}{\partial V_{DS}}$$

The voltage at which pinchoff occurs is the drain-saturation voltage V_{DSAT}



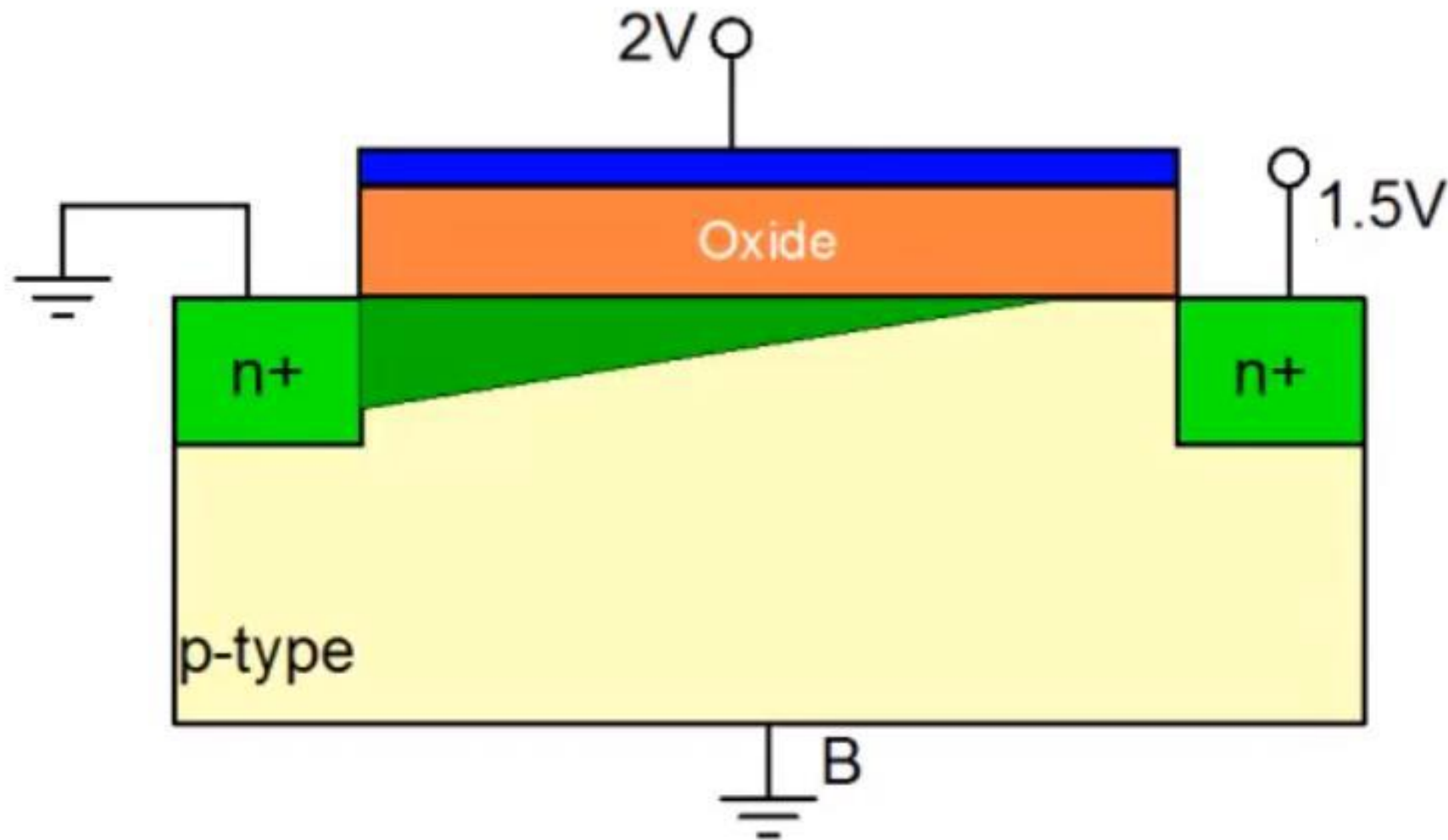
$$Q_{inv}(x) = -C_{ox}(V_{GS} - V_{THN} - V_{DSAT}) \cong 0$$

$$V_{DSAT} = V_{GS} - V_{THN}$$

This is a very simple picture. In short channel MOSFETs especially, saturation is a more complicated phenomenon.,

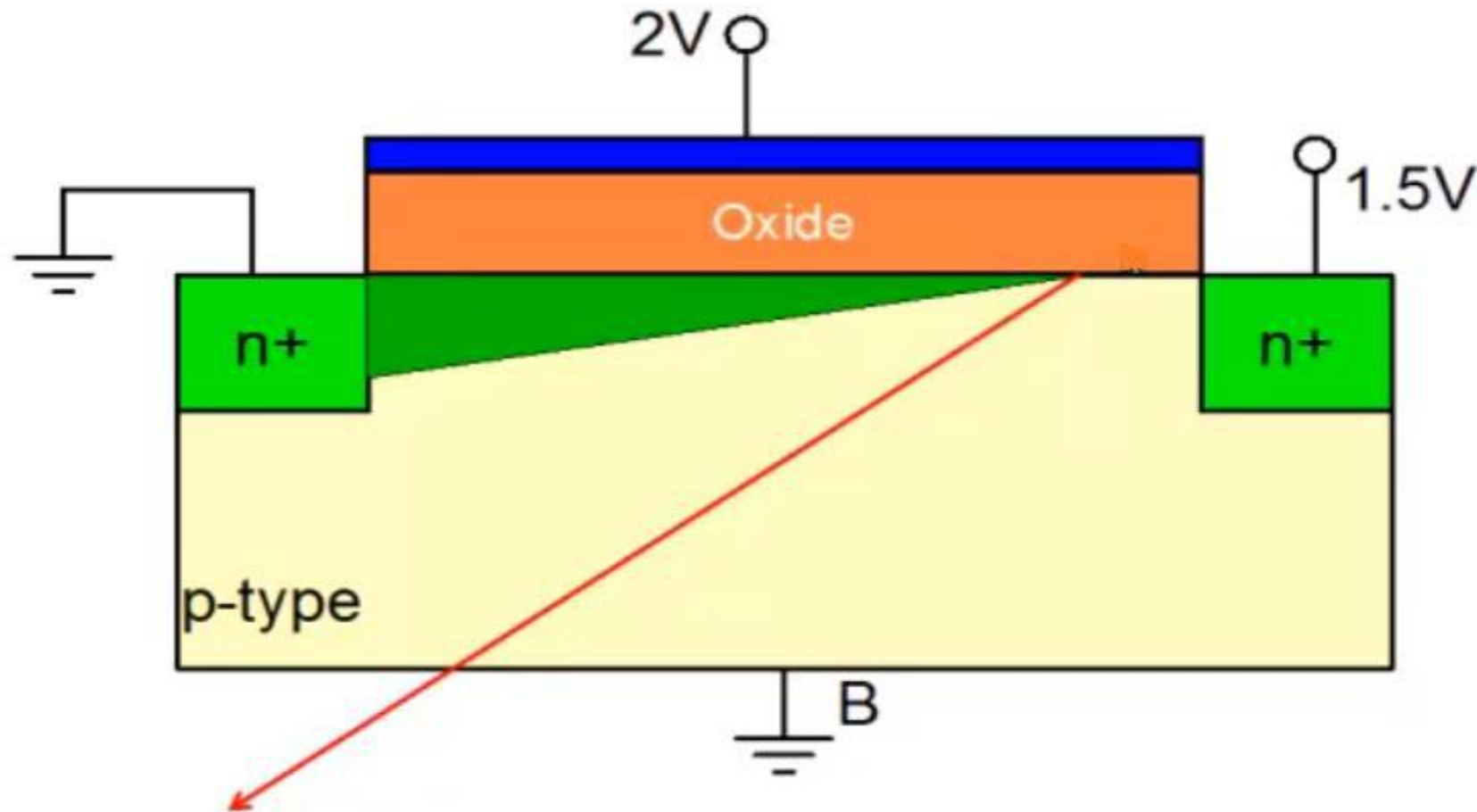
For voltages larger than saturation voltage

$$Q_{inv}(x) = -C_{ox}(V_{GS} - V_{THN} - V(x)) \cong -C_{ox}(2 - 1 - 1) = 0$$



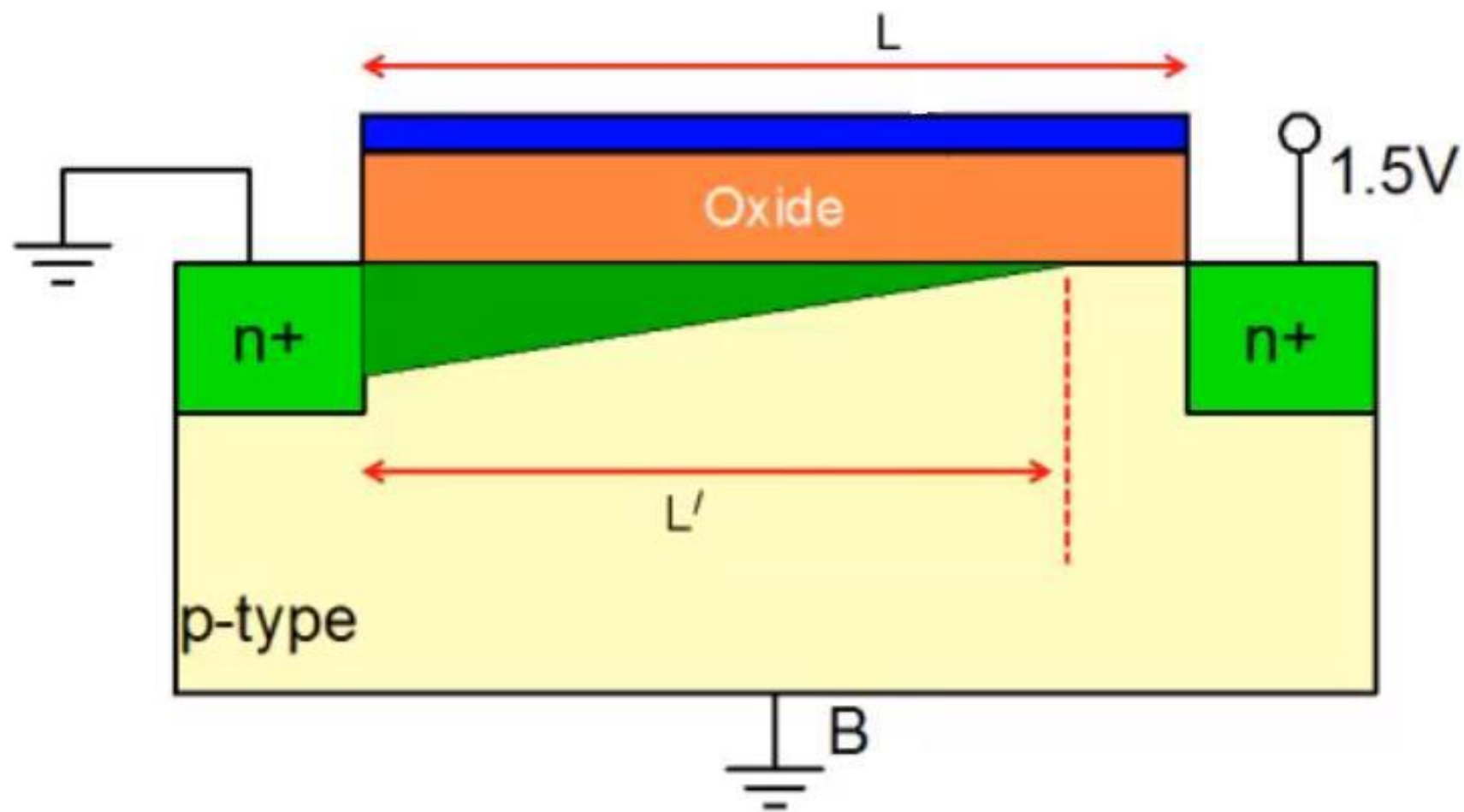
For voltages larger than saturation voltage

$$Q_{inv}(x) = -C_{ox}(V_{GS} - V_{THN} - V(x)) \cong -C_{ox}(2 - 1 - 1) = 0$$

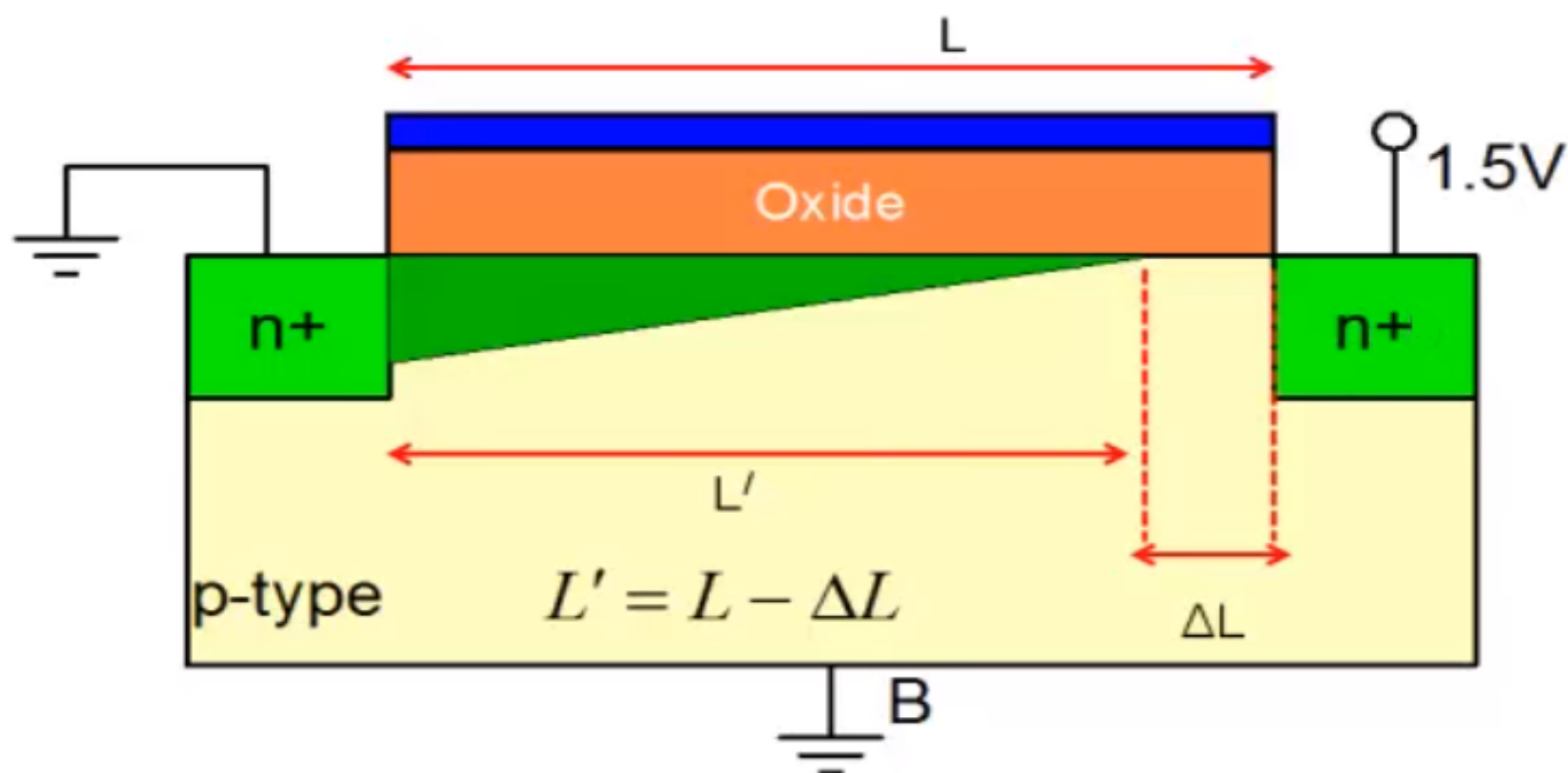


Pinchoff point moves left towards the source end. Voltage is $V_{DSAT} = 1V$

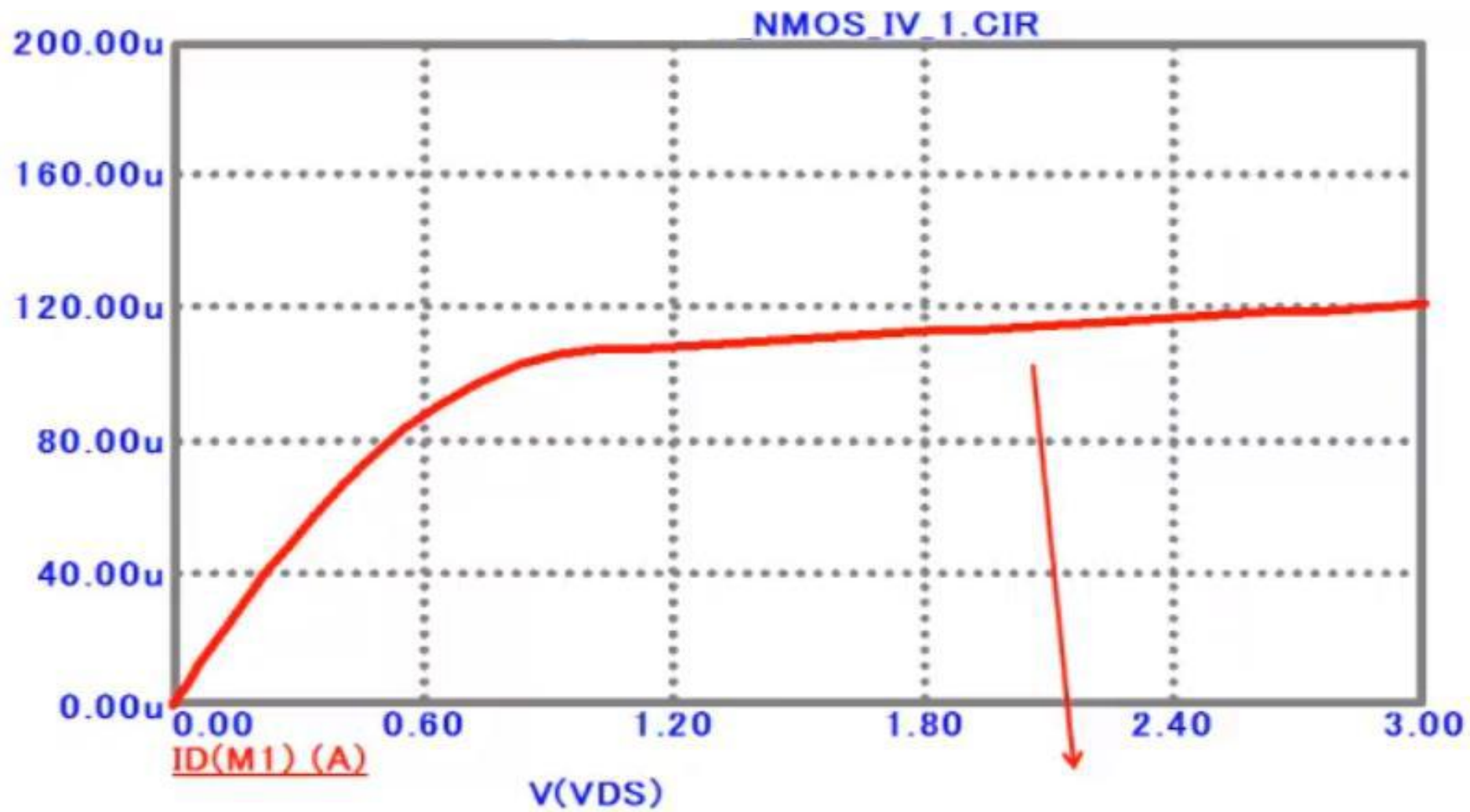
Channel Length Modulation



Channel Length Modulation

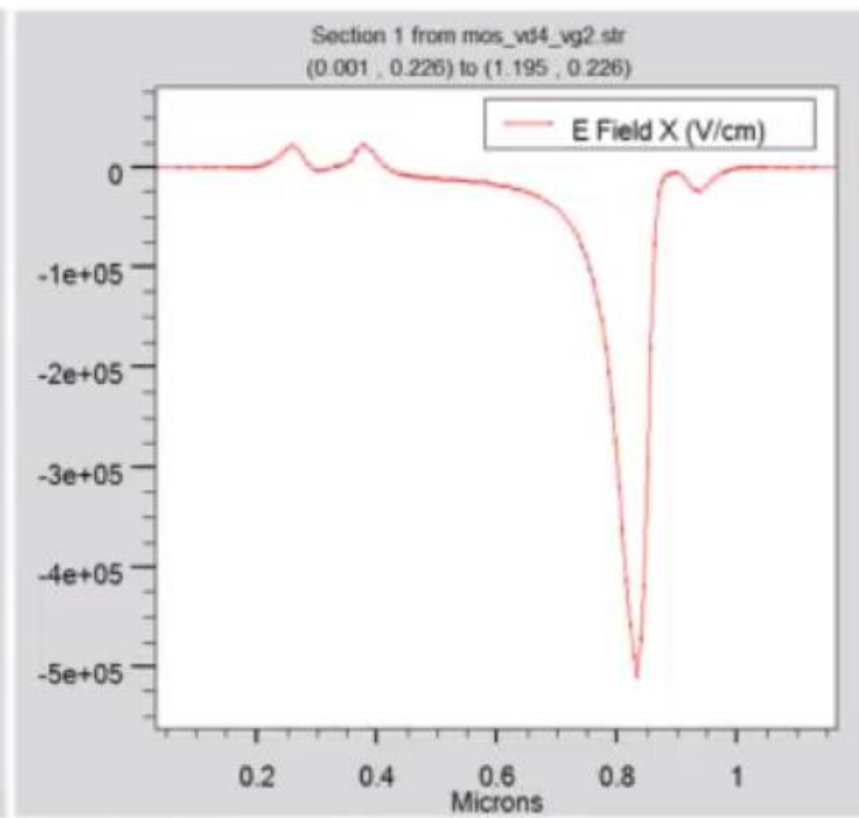
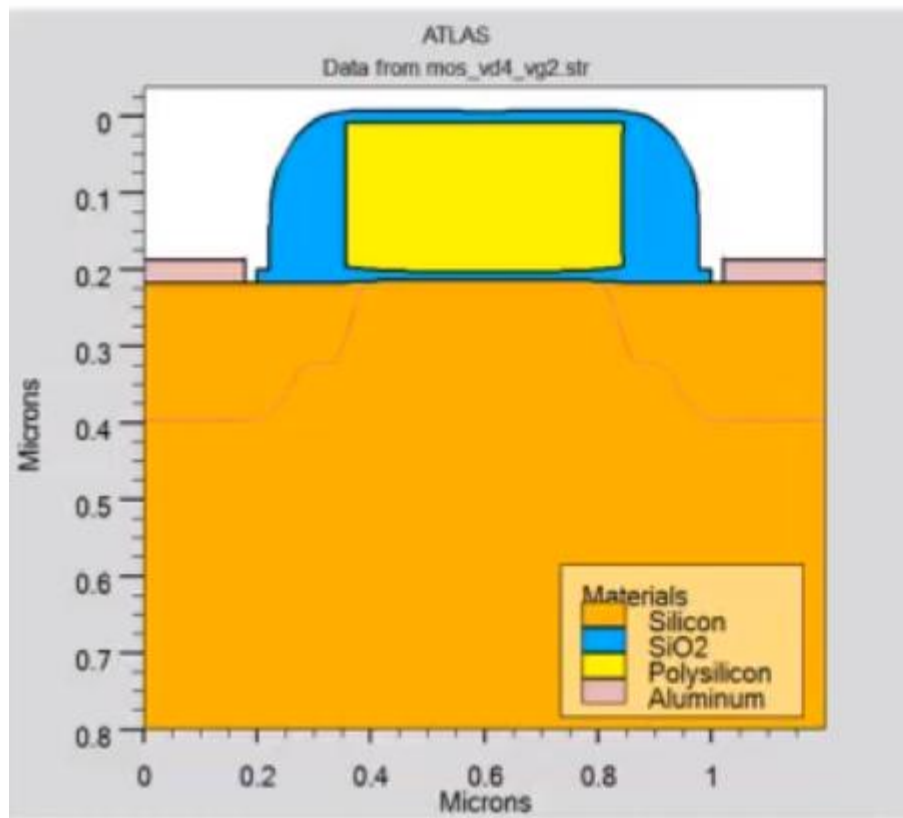


Effective channel length decreases as voltage increases beyond V_{DSAT} . As a result current increases a little with voltage

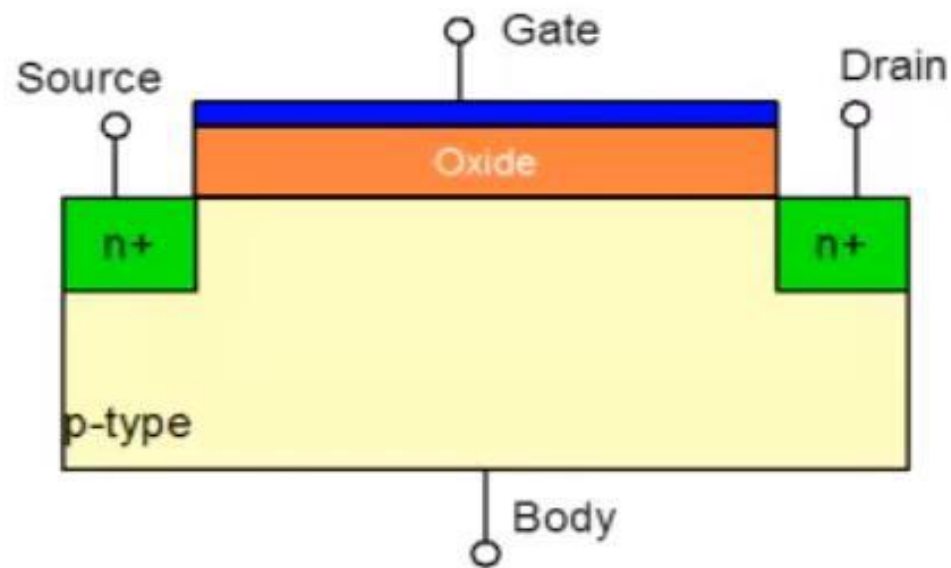


$$I_{DSAT} \times (1 + \lambda \times V_{DS})$$

λ : channel length modulation parameter



Threshold Voltage



$$V_{THN} = V_{THN0} + \gamma(\sqrt{2\phi_F + V_{SB}} - \sqrt{2\phi_F})$$

$$V_{THN0} = 1V$$

$$\gamma = \text{body parameter} \quad \text{Units : } \sqrt{V}$$

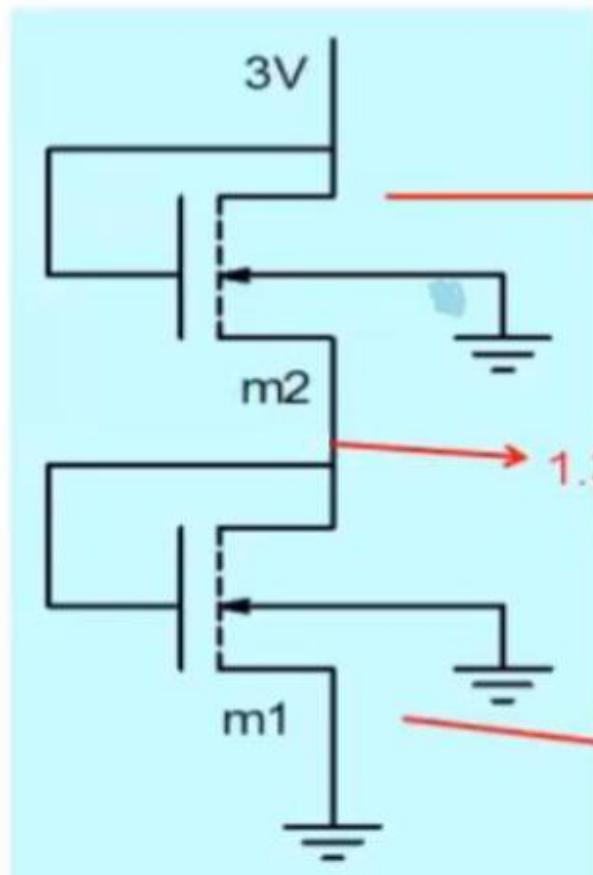
$$\gamma = 0.7 V^{1/2}$$

$$\text{Surface potential : } 2\phi_F$$

$$2\phi_F = 0.7V$$

$$V_{THN0} = 1V; \gamma = 0.7 V^{1/2}; 2\phi_F = 0.7V$$

$$V_{THN} = V_{THN0} + \gamma(\sqrt{2\phi_F + V_{SB}} - \sqrt{2\phi_F})$$



$$V_{SB2} = 1.3V; V_{THN2} = 1.4V$$

$$V_{THN1} = V_{THN0} = 1V$$

MOS Operating Regions

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graph TD; A[MOS Operating Regions] --> B[Above Threshold  
(VGS > VTN)]; A --> C[Subthreshold  
(VGS < VTN)]; B --> D[Saturation  
(VDS > VDSAT)]; B --> E[Triode  
(VDS < VDSAT)];
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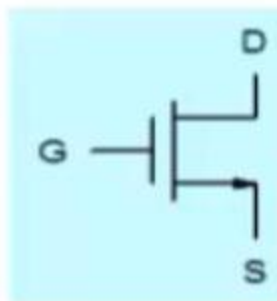
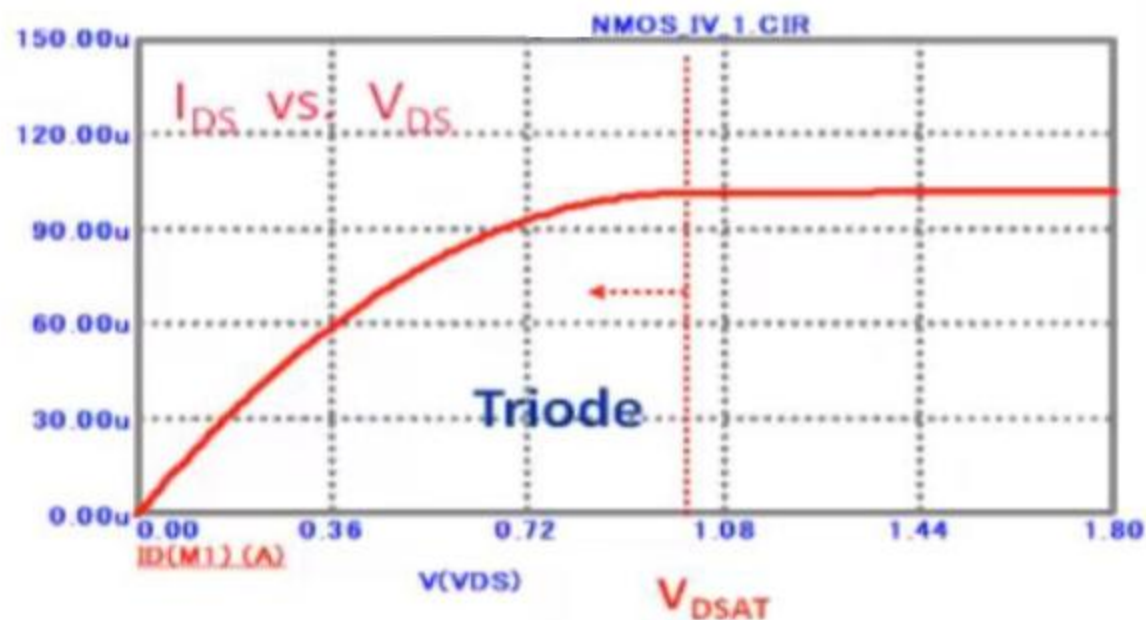
Above Threshold
 $(V_{GS} > V_{TN})$

Subthreshold
 $(V_{GS} < V_{TN})$

Saturation
 $(V_{DS} > V_{DSAT})$

Triode
 $(V_{DS} < V_{DSAT})$

dc Model: Triode (or Linear)



$$V_{GS} > V_{TN}$$

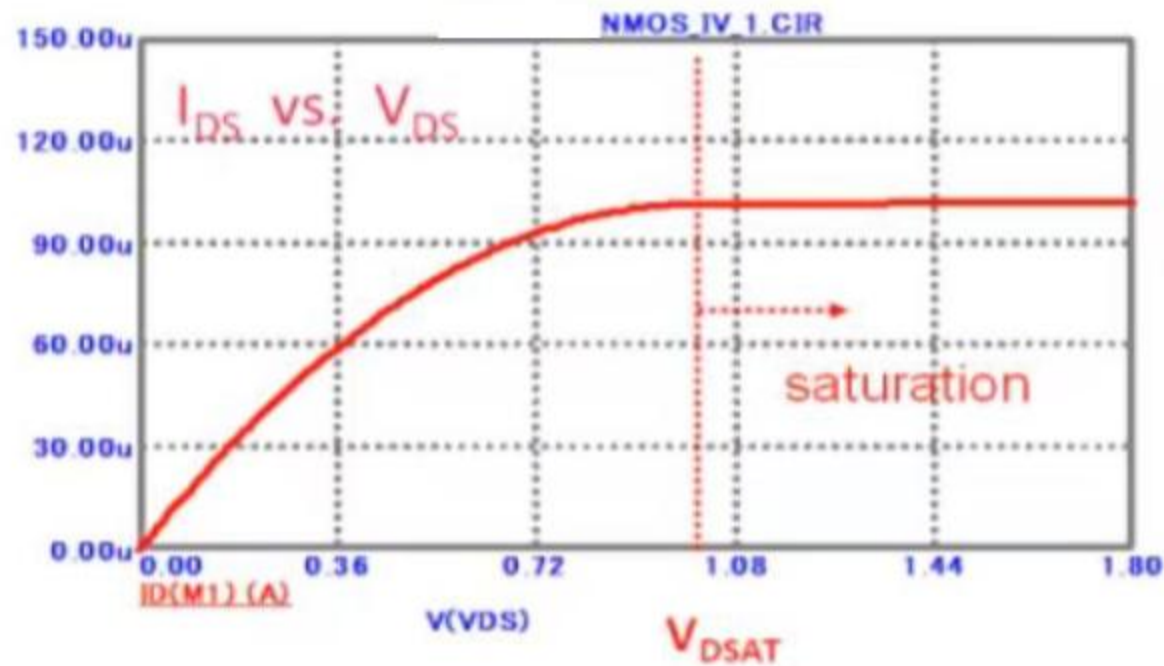
$$V_{DS} < V_{Dsat} = V_{GS} - V_{TN}$$

$$I_{DS} = \beta_N \left\{ (V_{GS} - V_{THN}) V_{DS} - \frac{V_{DS}^2}{2} \right\}$$

$$\beta_N = kP_N \cdot \frac{W}{L}$$

$$kP_N = \mu_n C_{ox'} : (\text{TransConductance parameter } \frac{\mu A}{V^2})$$

DC Model: Saturation Region



$$V_{GS} > V_{THN}$$

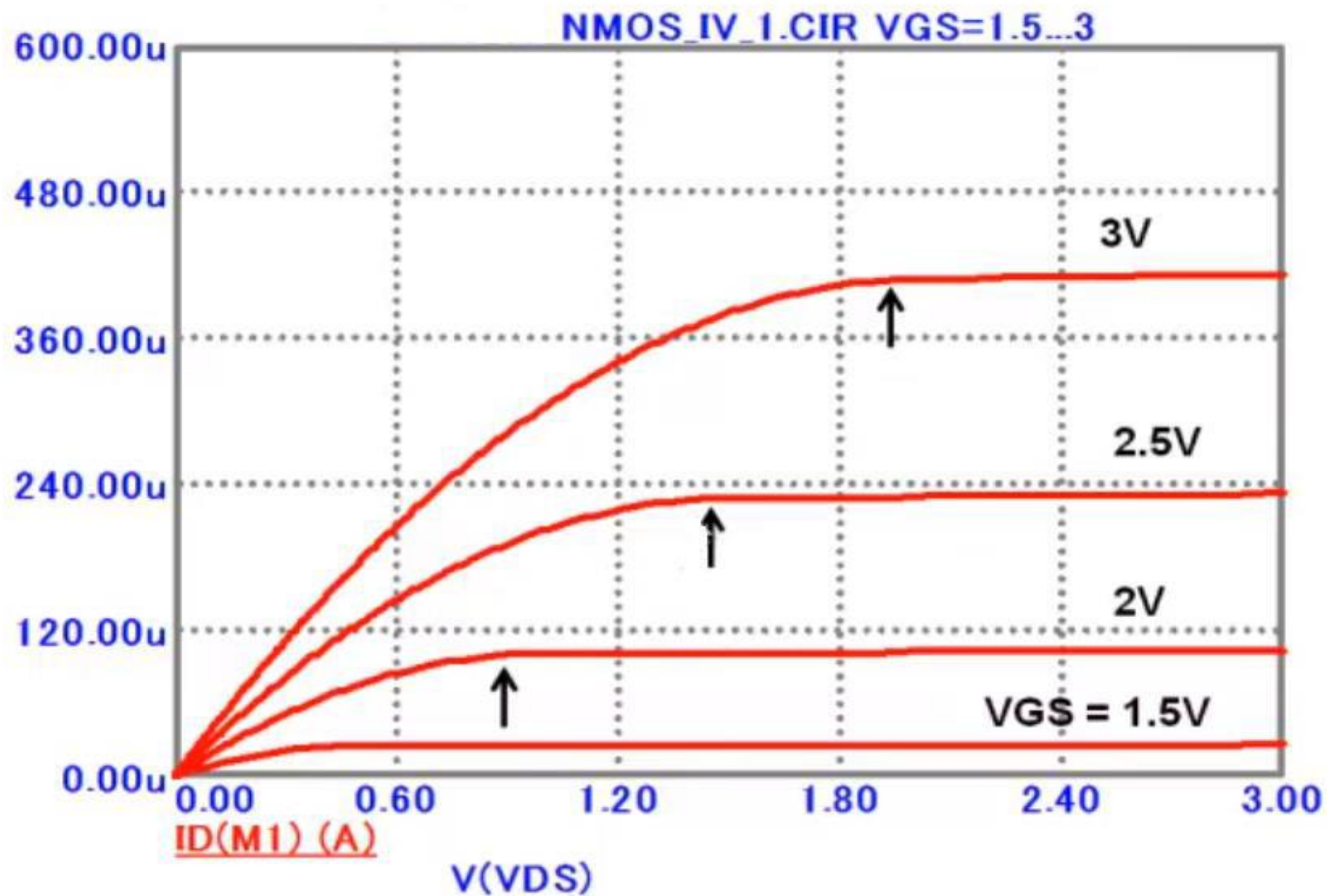
$$V_{DS} \geq V_{GS} - V_{THN}$$

$$I_{DS} = \frac{\beta_N}{2} (V_{GS} - V_{THN})^2 [1 + \lambda_n V_{DS}]$$

λ_N is the channel length modulation parameter

Note that unlike BJT, V_{DSAT} is not only larger but also dependent on applied gate-source voltage

Output Characteristics of MOSFET



dc model parameters

$$\text{Linear : } I_{DS} = \beta_N \left\{ (V_{GS} - V_{THN}) V_{DS} - \frac{V_{DS}^2}{2} \right\} \quad \beta_N = kP_N \cdot \frac{W}{L}$$

$$\text{Saturation : } I_{DS} = \frac{\beta_N}{2} (V_{GS} - V_{THN})^2 [1 + \lambda_n V_{DS}] \quad \lambda_N$$

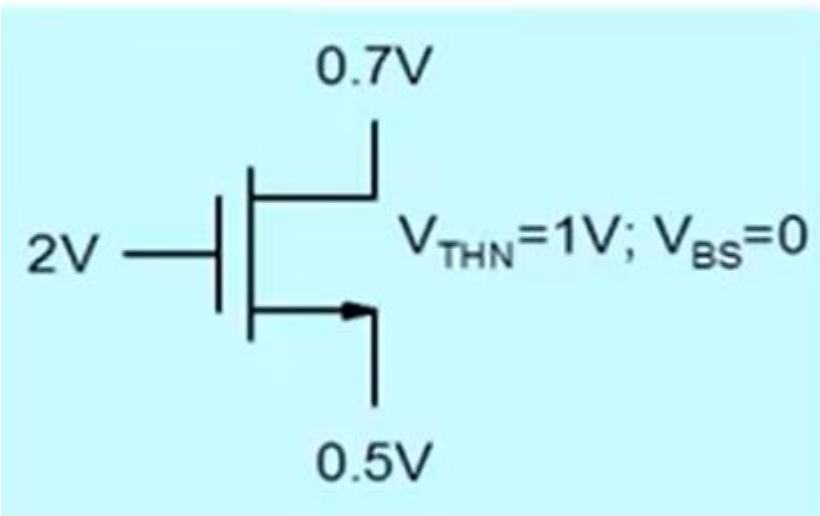
$$V_{THN} = V_{THN0} + \gamma (\sqrt{2\phi_F + V_{SB}} - \sqrt{2\phi_F})$$

$$V_{THN0} = 1V; \gamma = 0.7 V^{1/2}; 2\phi_F = 0.7V;$$

$$KP_N = 100 \mu A / V^2; L = 1 \mu m; \lambda = 0.01 V^{-1}$$

L is usually fixed, W is determined by designer

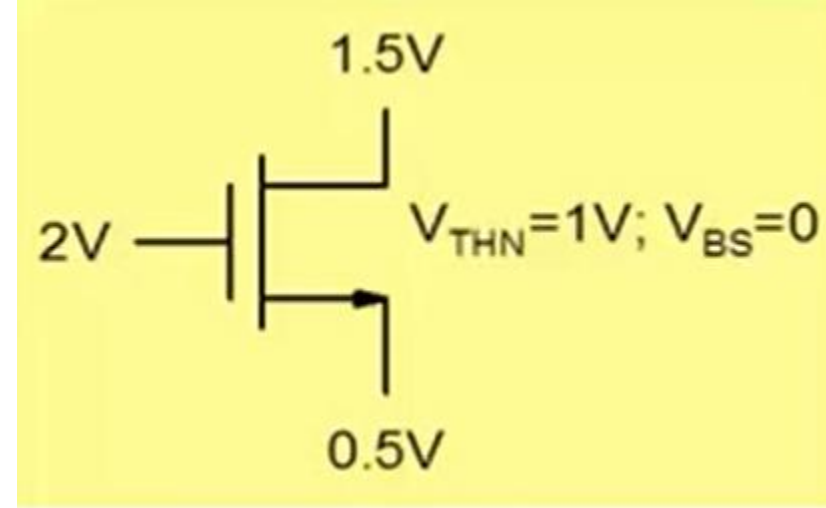
Which mode is the transistor operating in ?



$$V_{GS} = 1.5 ; V_{DS} = 0.2$$

$$V_{DSAT} = V_{GS} - V_{THN} = 0.5$$

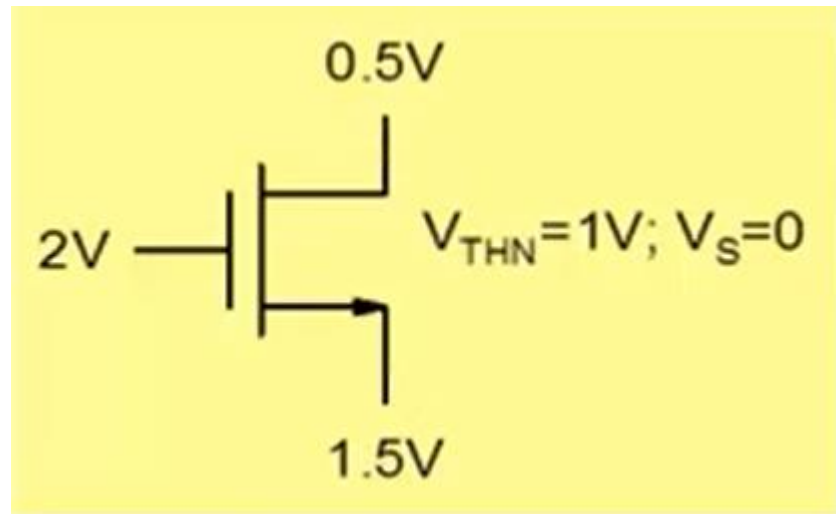
$$V_{DS} < V_{DSAT} \Rightarrow \text{Linear}$$



$$V_{DSAT} = 0.5 ; V_{DS} = 1V$$

Saturation

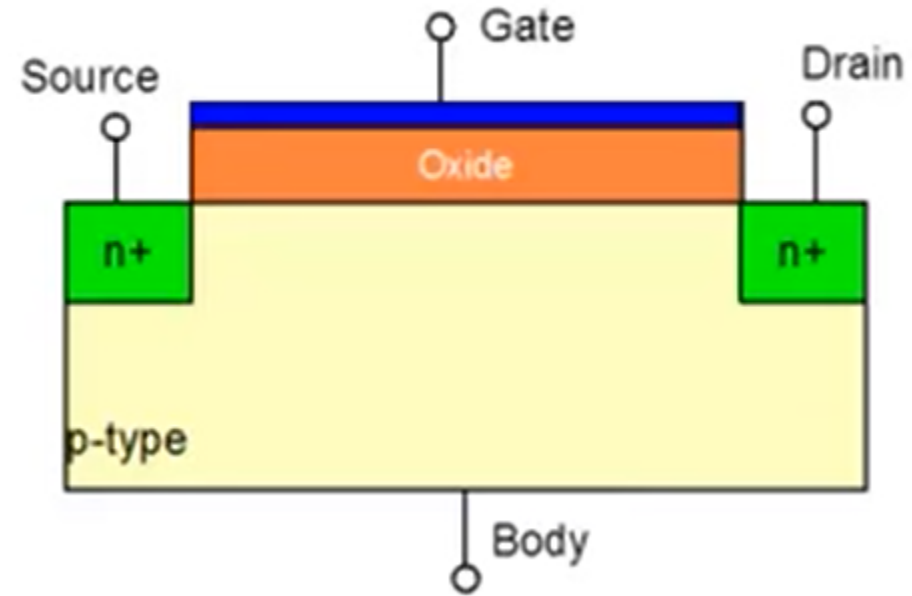
Which mode is the transistor operating in ?



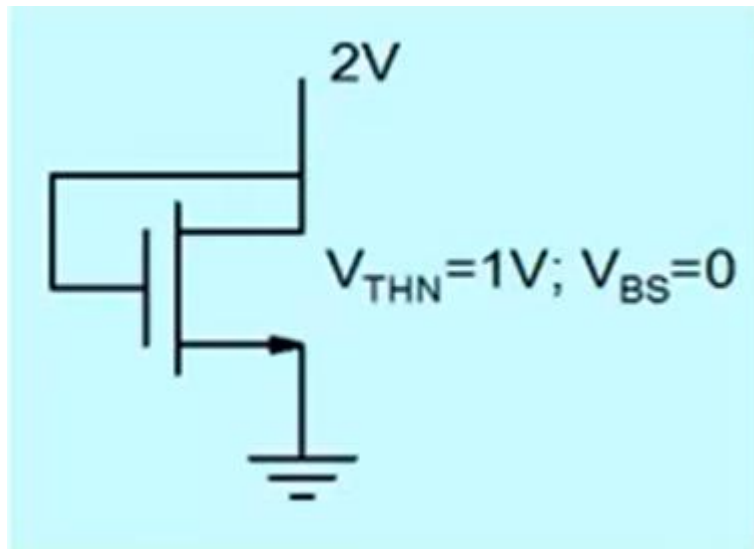
$$V_{GS} = 0.5 ; V_{DS} = -1V$$

$$V_{DSAT} = V_{GS} - V_{THN} = 0.5$$

$$V_{DS} > V_{DSAT} \Rightarrow \text{saturation}$$



$$V_{GS} = 1.5 ; V_{DS} = 1V$$



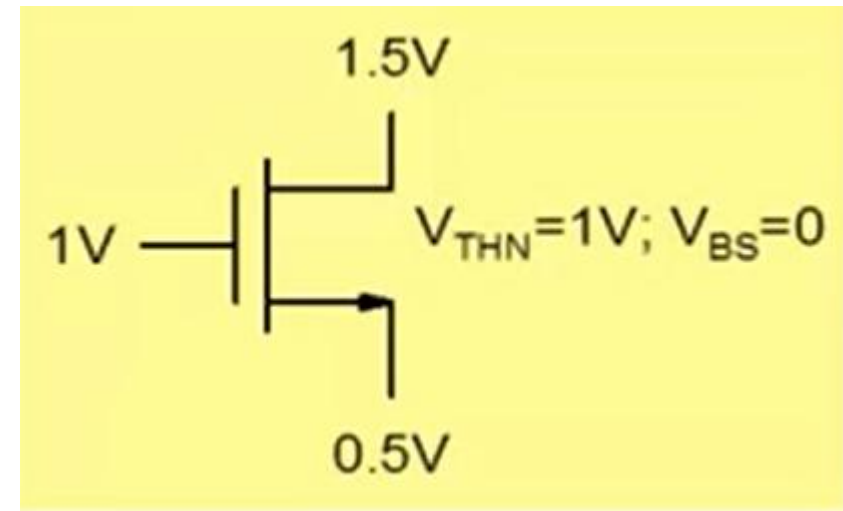
$$V_{GS} = 2; V_{DS} = 2$$

Saturation

$$I_{DS} = \frac{\beta_N}{2} (V_{GS} - V_{THN})^2 [1 + \lambda_n V_{DS}]$$

$$I_X \cong \frac{\beta_N}{2} (V_X - V_{THN})^2$$

Diode with a turn-on voltage of V_{THN}



$$V_{GS} = 0.5V < V_{THN}$$

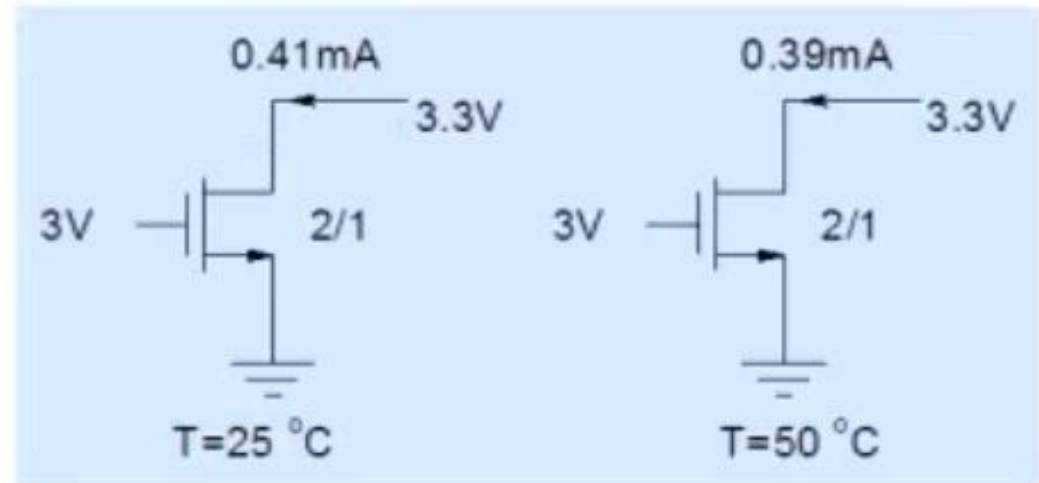
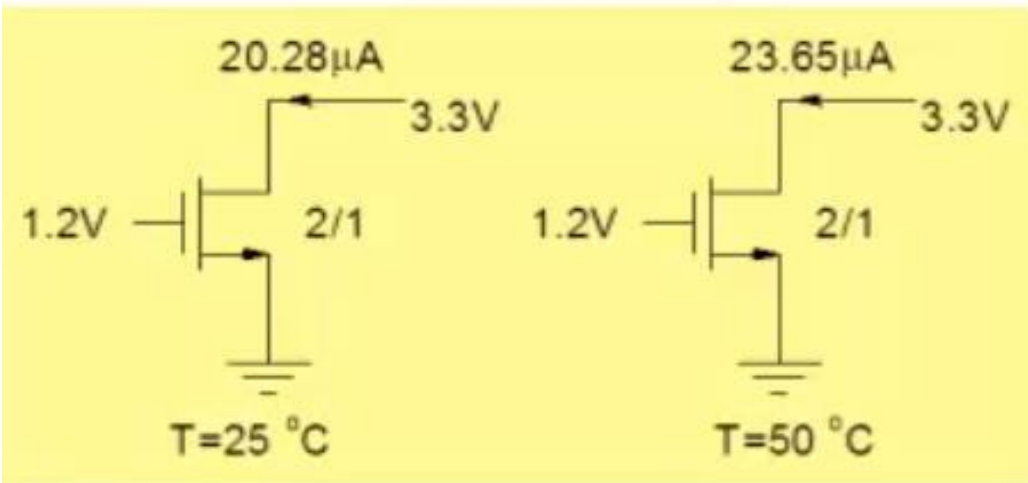
Transistor is in sub-threshold mode of operation

Temperature dependence

$$I_{DS} = \frac{KP_N}{2} \times \frac{W}{L} \times (V_{GS} - V_{THN})^2$$

Increase in temperature causes both transconductance parameter KP_N and threshold voltage V_{THN} to decrease

Although both K_{PN} and V_{THN} decrease with temperature, the former causes a decrease in current while the latter causes an increase in current.



MOS Operating Regions

```
graph TD; A[MOS Operating Regions] --> B[Above Threshold<br/>(V<sub>GS</sub> > V<sub>TN</sub>)]; A --> C[Subthreshold<br/>(V<sub>GS</sub> < V<sub>TN</sub>)]; B --> D[Saturation<br/>(V<sub>DS</sub> > V<sub>DSAT</sub>)]; B --> E[Triode<br/>(V<sub>DS</sub> < V<sub>DSAT</sub>)];
```

Above Threshold
 $(V_{GS} > V_{TN})$

Subthreshold
 $(V_{GS} < V_{TN})$

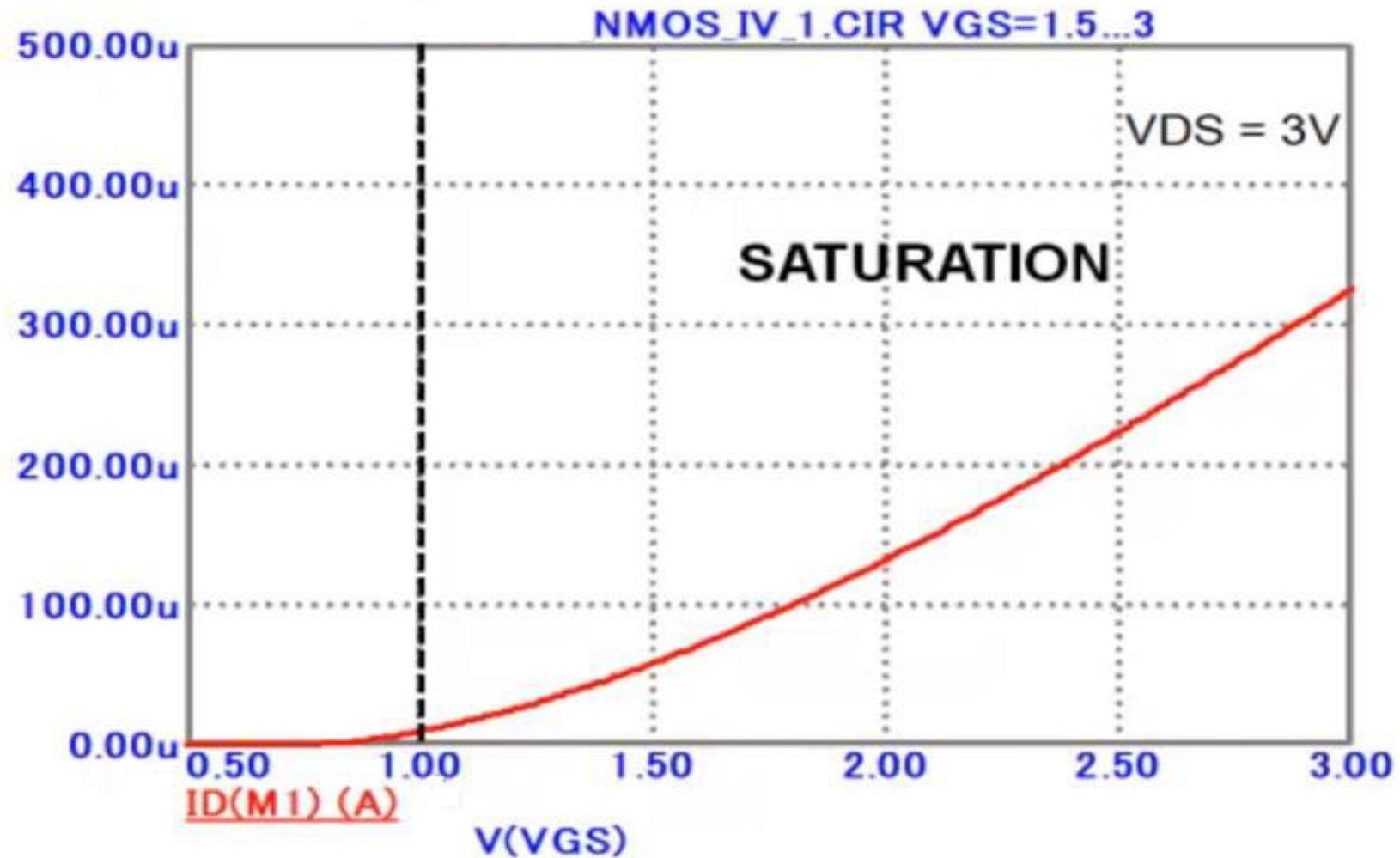
Saturation

$(V_{DS} > V_{DSAT})$

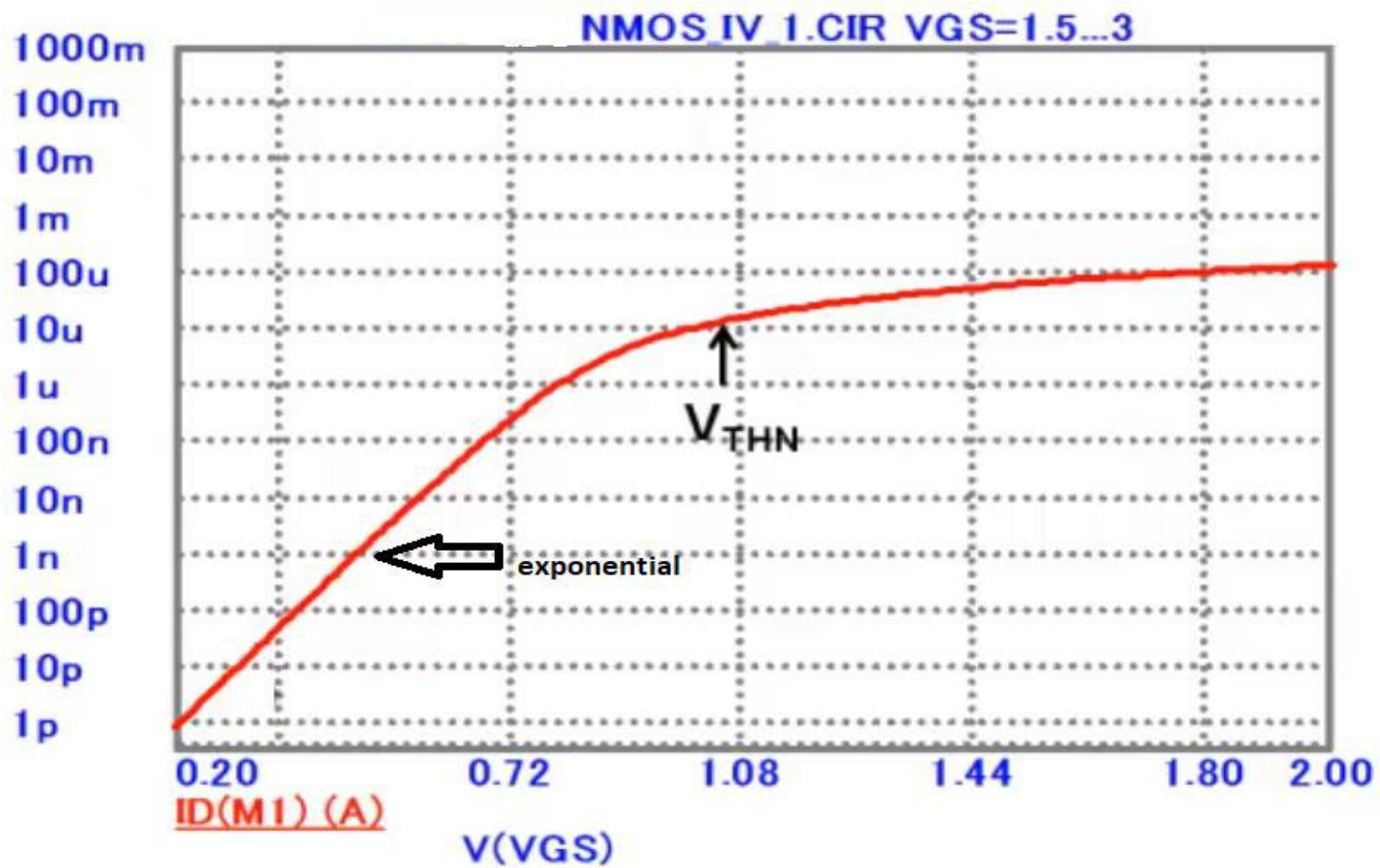
Triode

$(V_{DS} < V_{DSAT})$

Transfer characteristics of NMOS



Current is very small until gate-source voltage **exceeds threshold voltage**
 V_{THN}



- Logscale
- Ignoring leakages

$$I_D \propto e^{\frac{qV_{GS}}{N'kT}} \qquad g_m = \frac{\partial I_{DS}}{\partial V_{gs}} = \frac{I_{DS}}{nV_T}$$

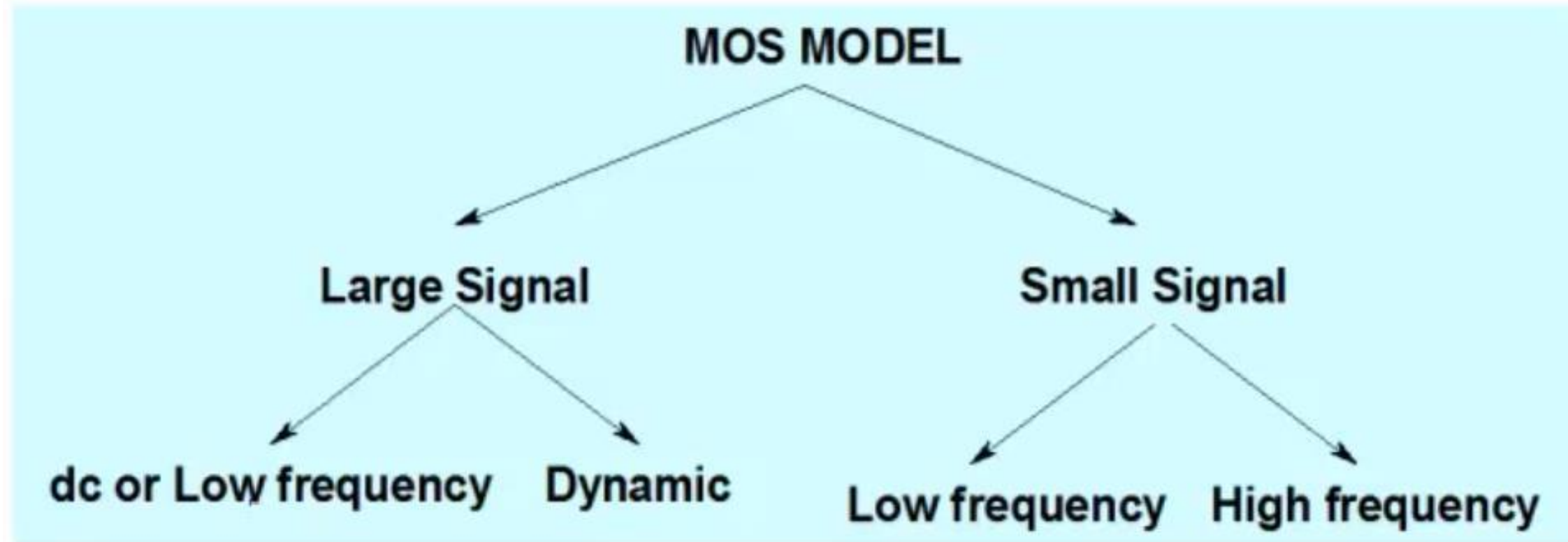
- In subthreshold region, MOS acts like a BJT

$$BJT: \qquad I_C = I_S e^{V_{BE}/V_T}$$

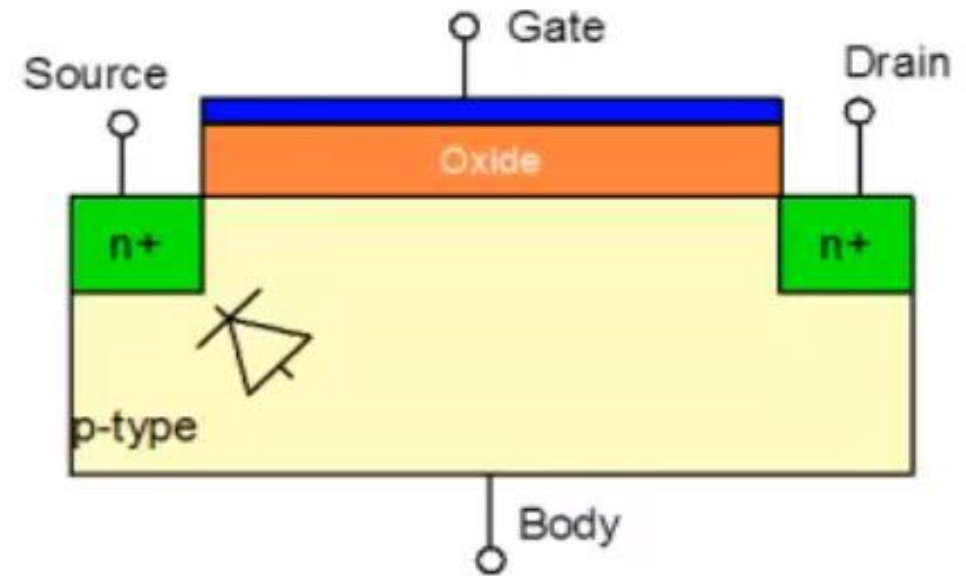
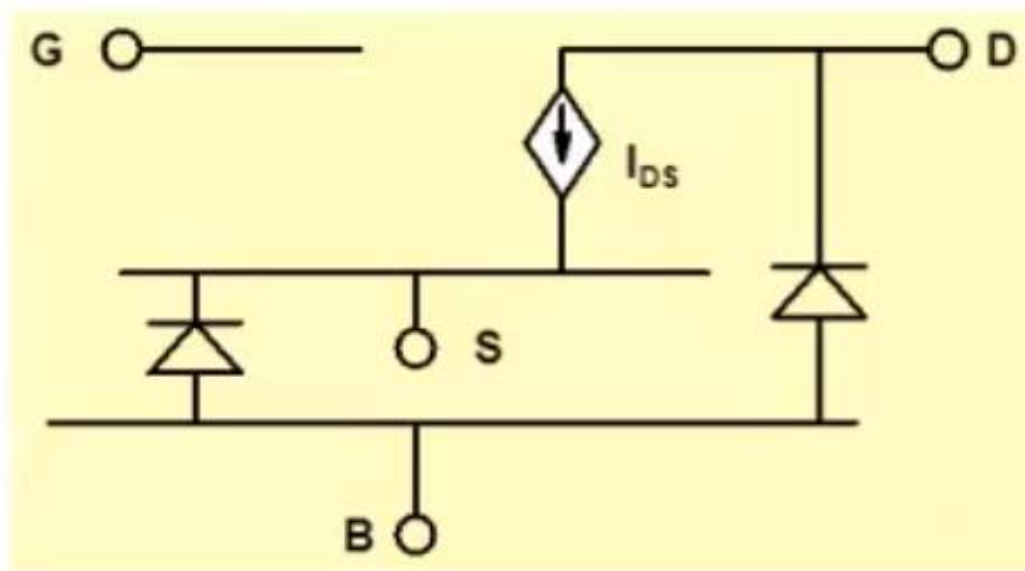
$$MOS: \quad I_{DS} = I_{S_1} e^{V_{GS}/\eta V_T}$$

- The advantage of MOS is that it offers almost infinite input impedance. Its disadvantage is that current levels are low.

MOS models : The classification of models can be done on the basis of magnitude and frequency of applied voltages

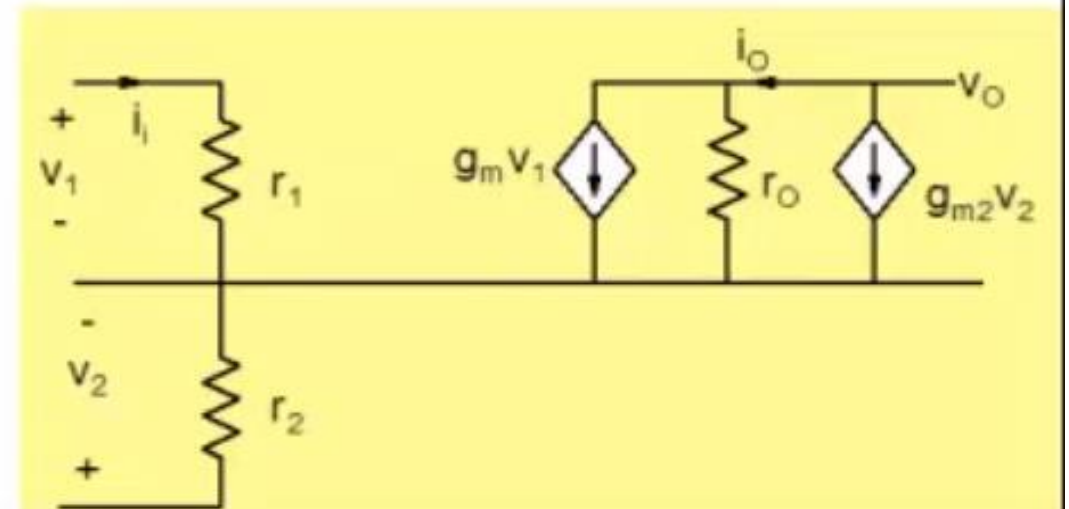
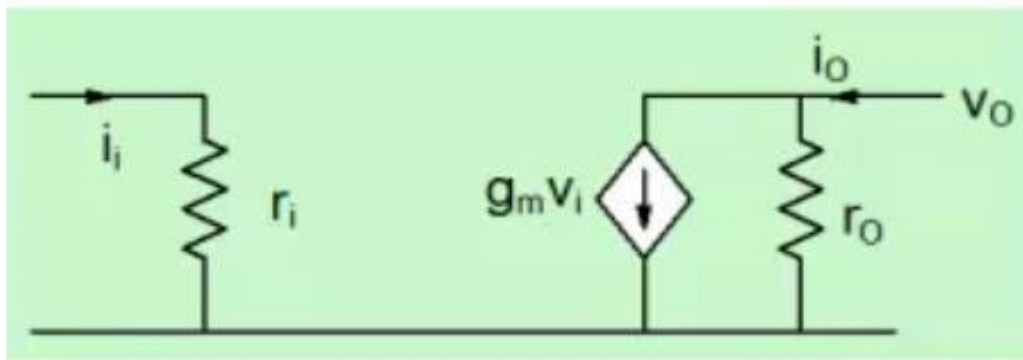
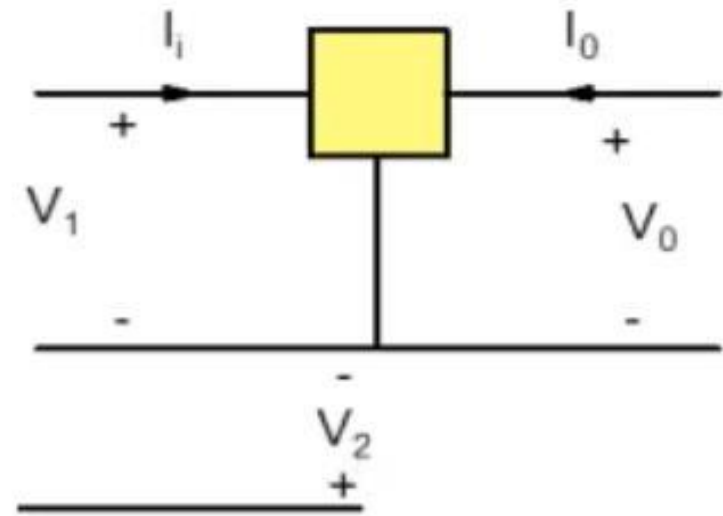
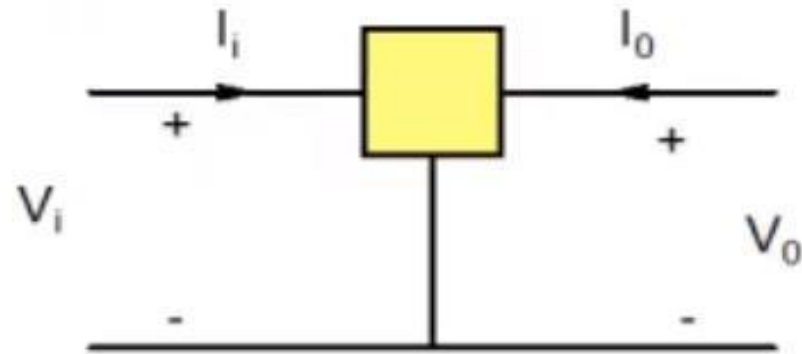


The dc model of the transistor in triode and saturation region can be represented in the form of an equivalent circuit:

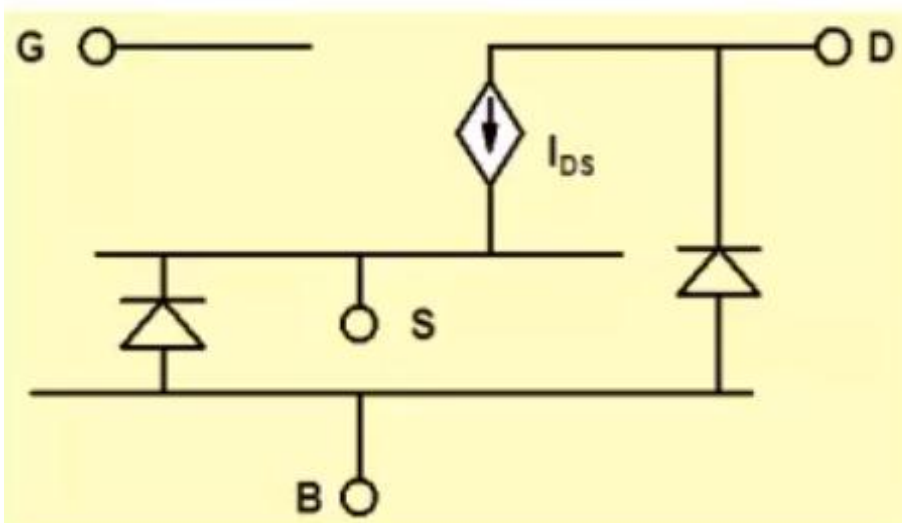


Series resistance associated with gate, source, drain and body terminals is not shown but can play an important role.

Complete **small signal model** (dc) for a 3-terminal unilateral device.



Small Signal Model (dc/low frequency)



$$I_{ds} = \frac{\beta_N}{2} (V_{gs} - V_{THN})^2 (1 + \lambda_n V_{ds})$$

$$V_{gs} = V_{GSQ} + v_{gs}$$

$$V_{ds} = V_{DSQ} + v_{ds}$$

$$V_{sb} = V_{SBQ} + v_{sb}$$

$$I_{ds} = I_{DSQ} + i_{ds}$$

$$I_{DSQ} + i_{ds} = \frac{\beta_N}{2} \left[V_{GSQ} + v_{gs} - V_{THN} (V_{BSQ} + v_{bs}) \right]^2 (1 + \lambda_n V_{DSQ} + \lambda_n v_{ds})$$

↑ not really multiplication, but function

$$i_{ds} \cong I_{DSQ} \left\{ \left(\lambda_n v_{ds} + \frac{2v_{gs}}{V_{GSQ} - V_{THN}} + \frac{\gamma \times v_{bs}}{(V_{GSQ} - V_{THN}) \times \sqrt{2\phi_F - V_{BSQ}}} \right) \right\}$$

$$i_{ds} = \frac{v_{ds}}{r_o} + g_m v_{gs} + g_{mb} v_{bs}$$

$$r_o = \frac{1}{\lambda_n I_{DSQ}}$$

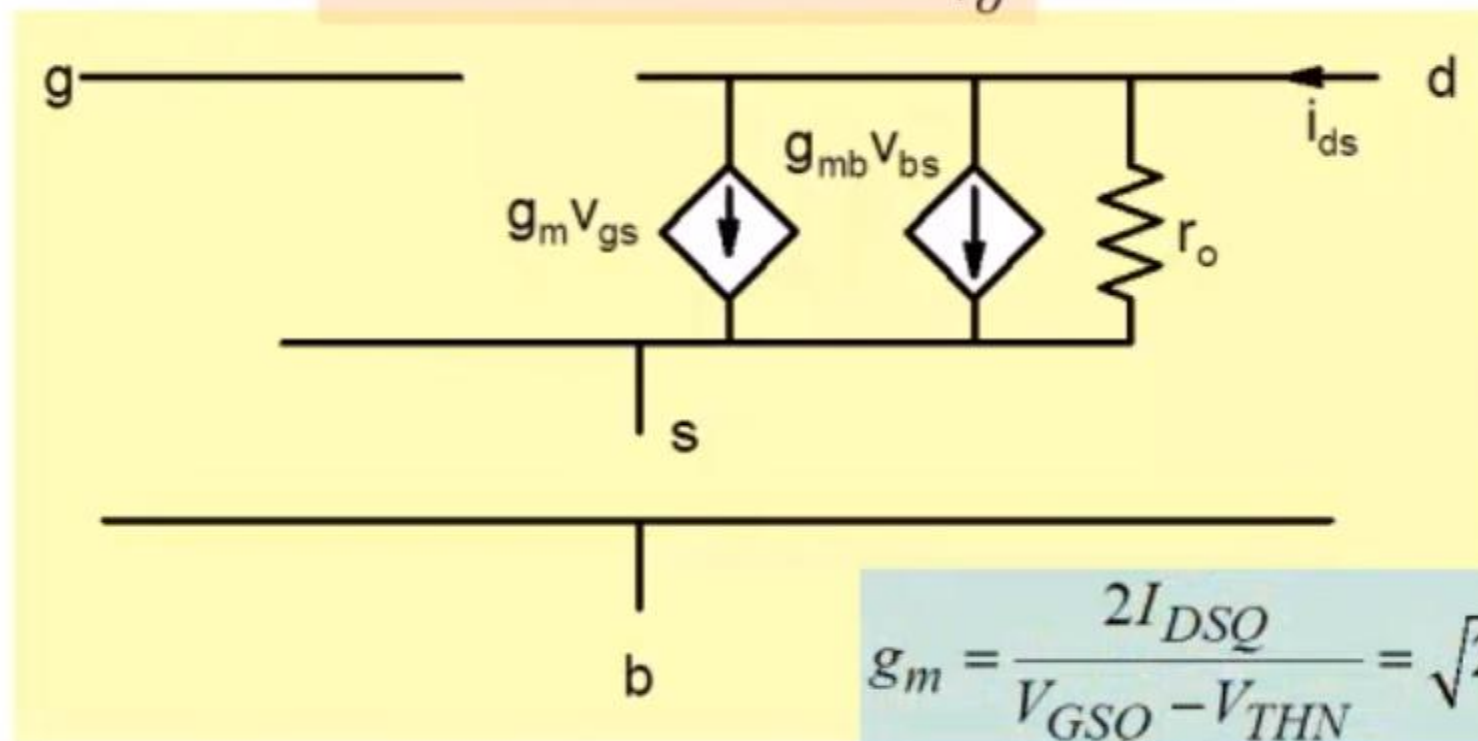
$$g_m = \frac{2I_{DSQ}}{V_{GSQ} - V_{THN}} = \sqrt{2I_{DSQ}\beta}$$

$$g_{mb} = g_m \cdot \eta$$

$$\eta = \frac{\gamma}{2\sqrt{2\phi_F + V_{SBQ}}}$$

Low frequency Small Signal model

$$i_{ds} = g_m v_{gs} + g_{mb} v_{bs} + \frac{v_{ds}}{r_o}$$



$$g_m = \frac{2I_{DSQ}}{V_{GSQ} - V_{THN}} = \sqrt{2I_{DSQ}\beta}$$

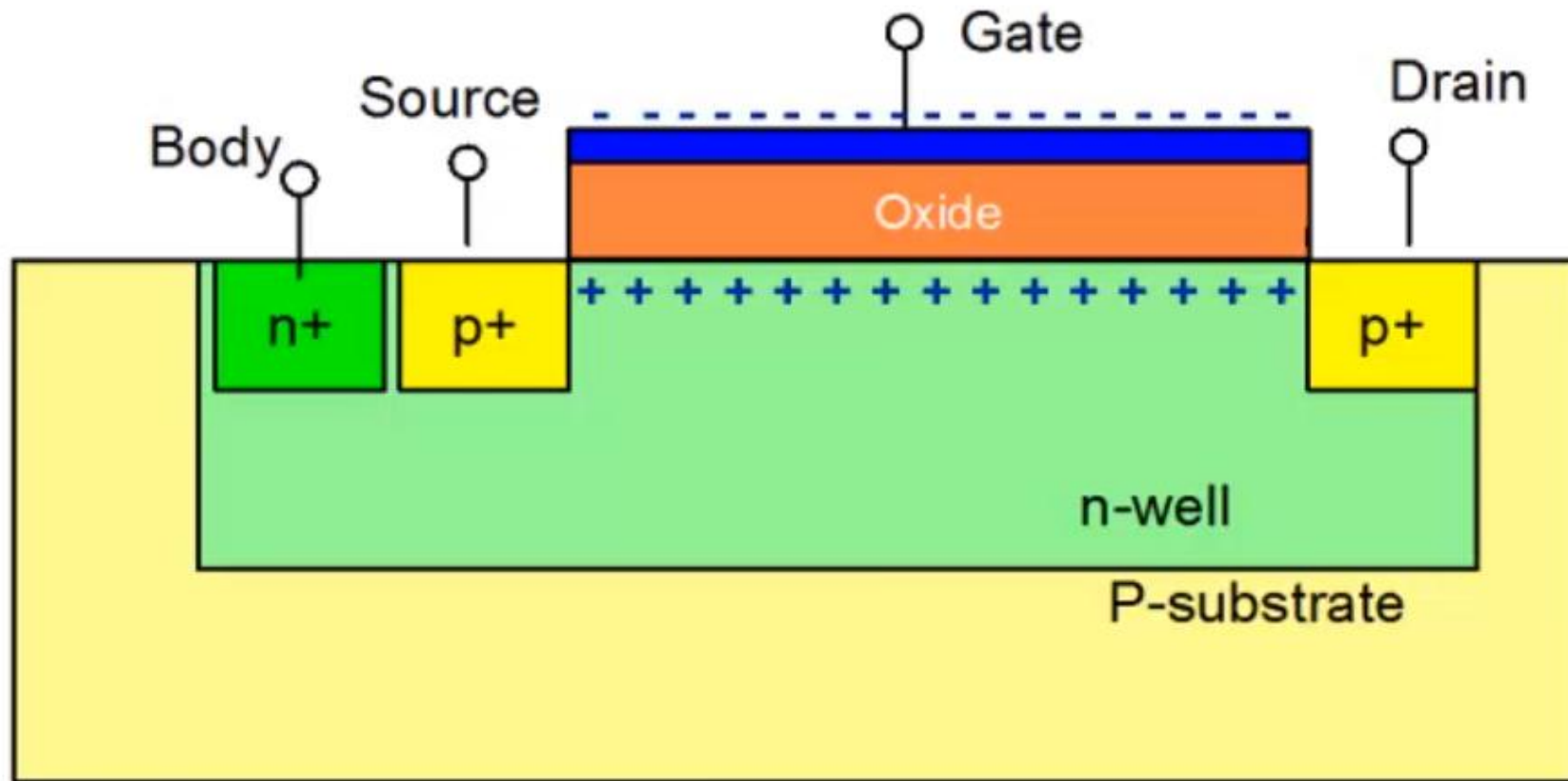
$$r_o = \frac{1}{\lambda_n I_{DSQ}}$$

$$g_{mb} = g_m \cdot \eta$$

$$\eta = \frac{\gamma}{2\sqrt{2\Phi_F + V_{SBQ}}}$$

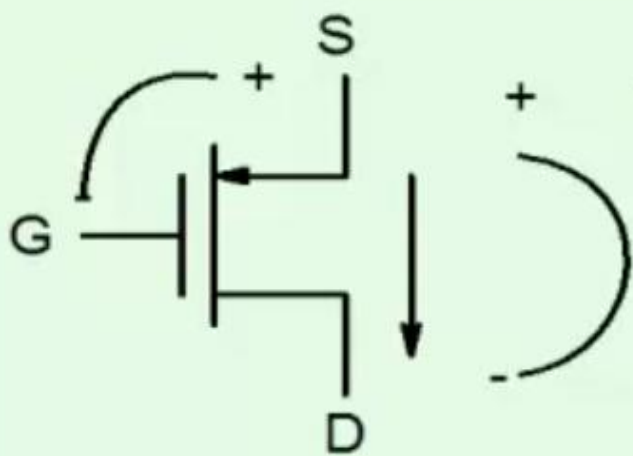
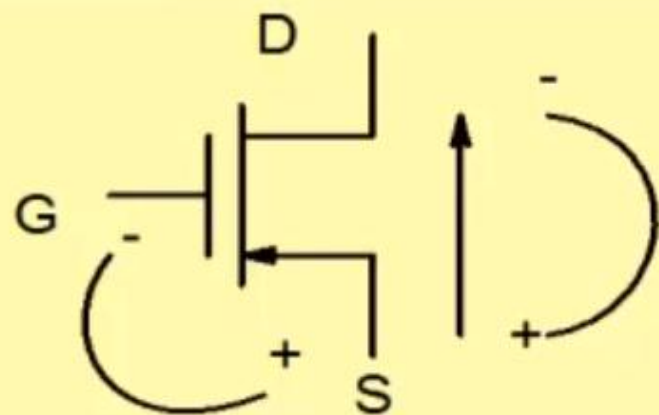
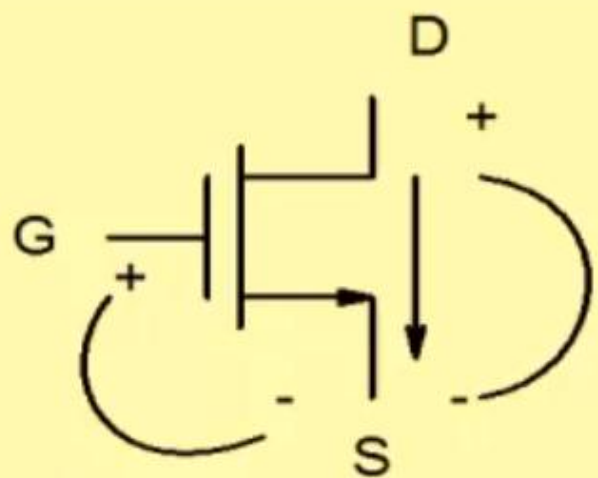
The small signal approximation $i_{ds} = g_m v_{gs}$ is accurate when $v_{gs} \ll 2(V_{GS} - V_{THN})$

PMOS



V_{GS} is negative ; Threshold voltage V_{TP} is negative

V_{DS} is negative ; I_{DS} is negative



$$V_{GSN} \rightarrow V_{SGP}$$

$$I_{DSN} \rightarrow I_{SDP}$$

$$V_{DSN} \rightarrow V_{SDP}$$

Transformations

$$V_{GSN} \rightarrow V_{SGP}$$

$$V_{DSN} \rightarrow V_{SDP}$$

$$V_{BSN} \rightarrow V_{SBP}$$

$$V_{THN} \rightarrow -V_{THP}$$

$$I_{DSN} \rightarrow I_{SDP}$$

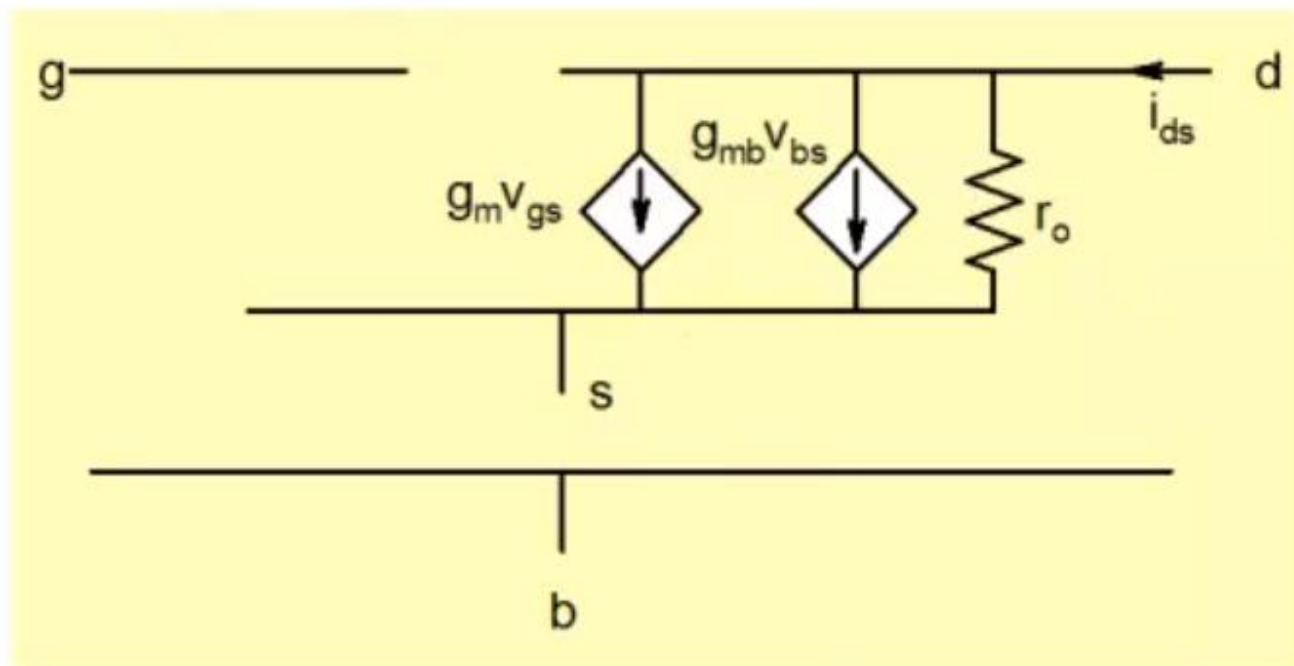
$$I_{DS} = \frac{\beta_N}{2} (V_{GS} - V_{THN})^2 [1 + \lambda_n V_{DS}] \rightarrow$$

$$I_{SD} = \frac{\beta_P}{2} (V_{SG} + V_{THP})^2 [1 + \lambda_p V_{SD}]$$

$$i_{sd} = g_m v_{sg} + g_{mb} v_{sb} + \frac{v_{sd}}{r_o}$$

$$i_{ds} = g_m v_{gs} + g_{mb} v_{bs} + \frac{v_{ds}}{r_o} \text{ same as NMOS}$$

Small Signal Model



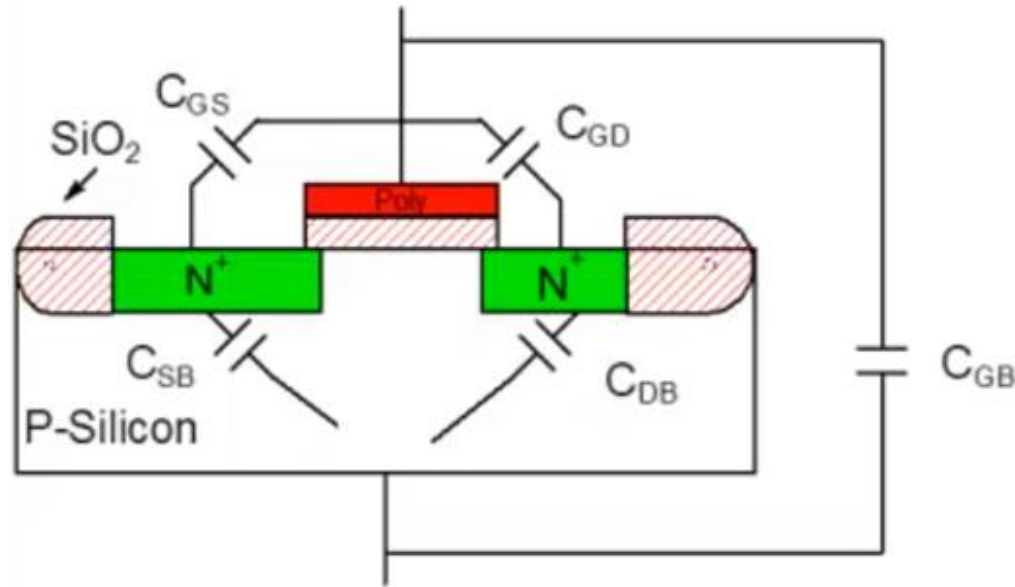
$$g_m = \frac{2I_{SDQ}}{V_{SGQ} + V_{THP}}$$

$$r_o = \frac{1}{\lambda_p I_{SDQ}}$$

Capacitance Model

Saturation

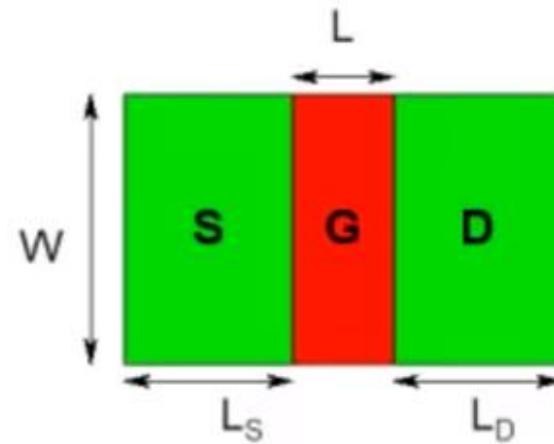
- There are five distinct components of capacitance as illustrated below



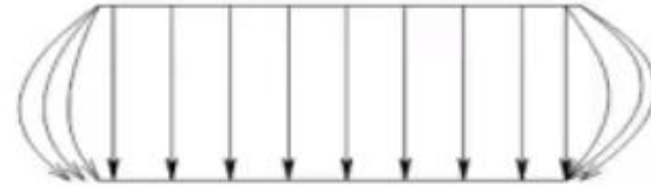
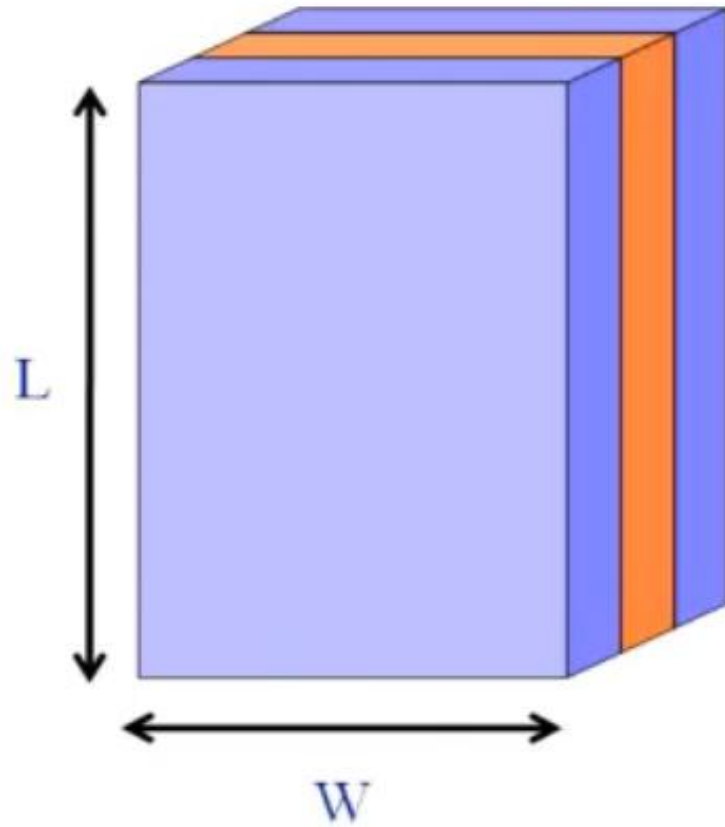
$$C_{gs} \cong \frac{2}{3} C_{ox'} \cdot W \cdot L + C_{gso} \cdot W$$

(Channel) (overlap)

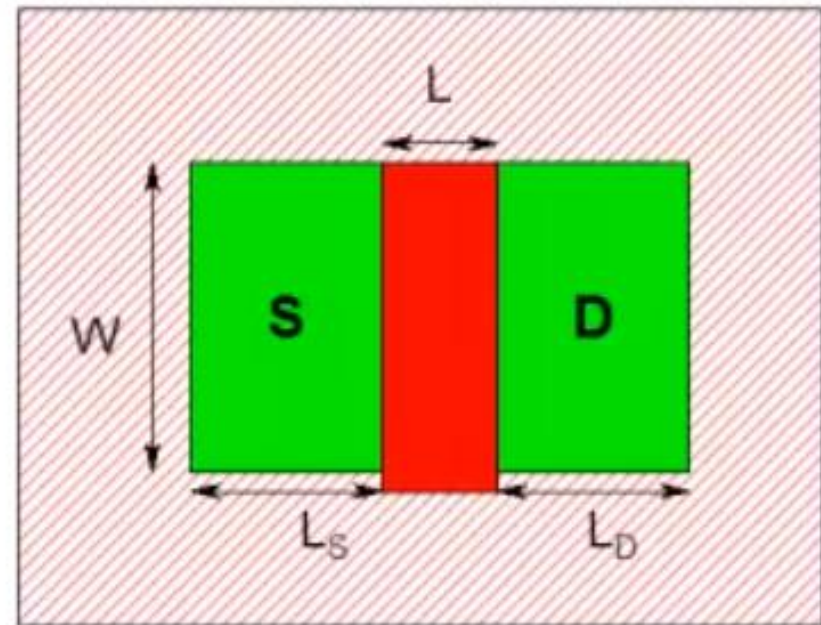
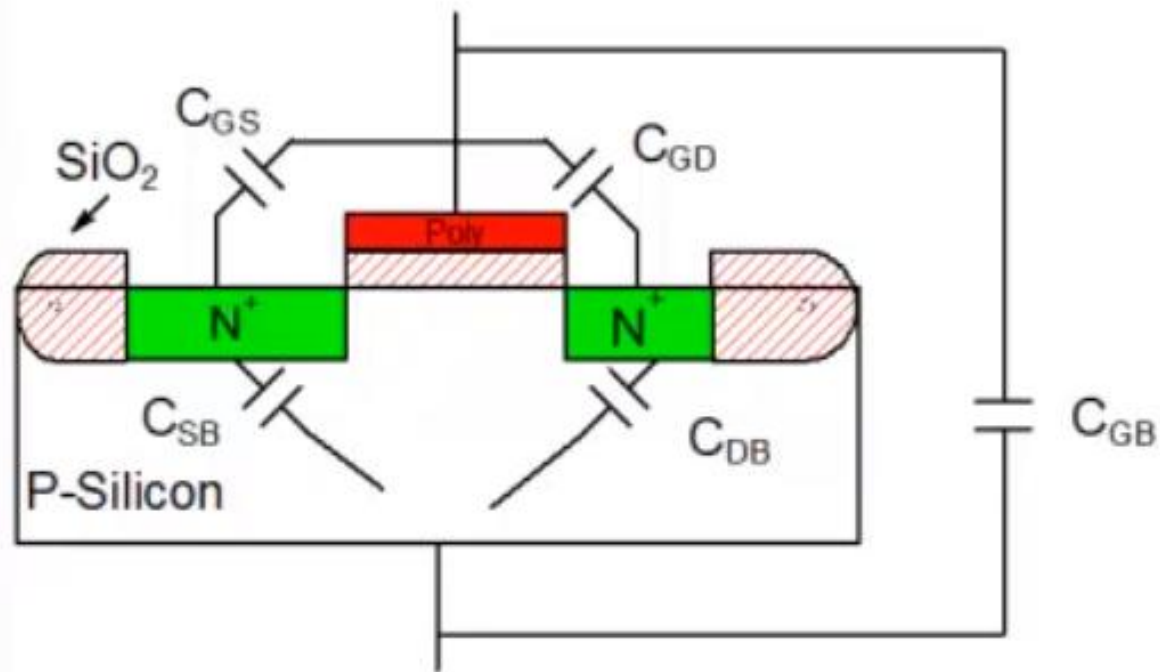
$$C_{gd} = C_{GDO} \cdot W$$



Capacitances: Area and Perimeter Components



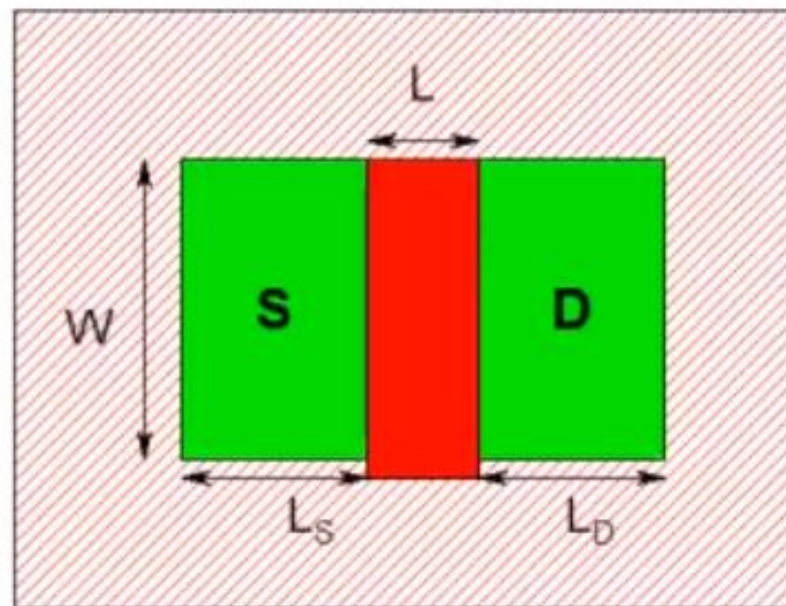
$$C = \frac{\epsilon}{d} \times W \times L + C_p \times (2L + 2W)$$



$$C_{sb} = \frac{C_j \cdot A_s}{\left(1 + \frac{V_{SB}}{P_B}\right)^{M_j}} + \frac{C_{jsw} \cdot P_s}{\left(1 + \frac{V_{SB}}{P_{BSW}}\right)^{M_{jsw}}}, \quad P_s = 2L_s + W, \quad A_s = W \cdot L_s$$

$$C_{db} = \frac{C_{jsw} \cdot P_D}{\left(1 + \frac{V_{DB}}{P_{BSW}}\right)^{M_{jsw}}} + \frac{C_j \cdot A_D}{\left(1 + \frac{V_{DB}}{P_B}\right)^{M_j}} \quad P_D = 2L_D + W$$

$$C_{gb} = C_{GBO} \cdot L \sim \text{often negligible}$$



Triode/Linear Region

$$C_{gs} = \frac{1}{2} C_{ox'} . W . L + C_{GSO} . W$$

$$C_{gd} = \frac{1}{2} C_{ox'} . W . L + C_{GDO} . W$$

$$C_{sb} = \text{same as before}$$

$$C_{db} = \text{same as before}$$

$$C_{gb} = \text{same as before}$$

Assuming $V_{DS} \sim 0$

Cutoff Region

$$C_{gs} = C_{GSO} . W$$

$$C_{gd} = C_{GDO} . W$$

$$C_{sb} = \text{same as before}$$

$$C_{db} = \text{same as before}$$

$$C_{gb} = C_{GBO} . L + C_{ox'} . W . L$$

Assuming Tr. is in accumulation

Summary

$$C_{gs} \cong \frac{2}{3} C_{ox'} \cdot W \cdot L + C_{gso} \cdot W \quad C_{gd} = C_{GDO} \cdot W$$

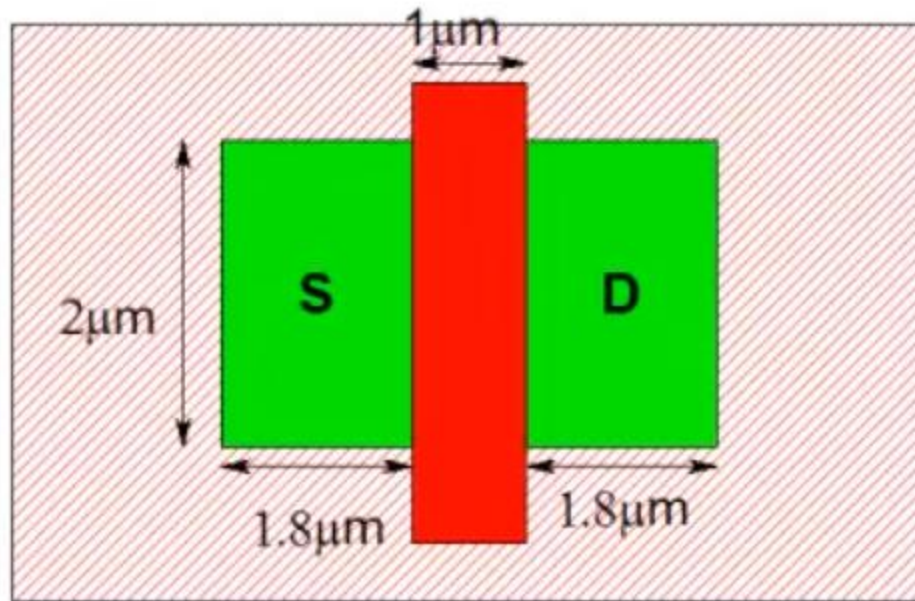
$$C_{sb} = \frac{C_j \cdot A_s}{\left(1 + \frac{V_{SB}}{P_B}\right)^{M_j}} + \frac{C_{jsw} \cdot P_s}{\left(1 + \frac{V_{SB}}{P_{BSW}}\right)^{M_{jsw}}}, \quad P_s = 2L_s + W, \quad A_s = W \cdot L_s$$

$$C_{db} = \frac{C_{jsw} \cdot P_D}{\left(1 + \frac{V_{DB}}{P_{BSW}}\right)^{M_{jsw}}} + \frac{C_j \cdot A_D}{\left(1 + \frac{V_{DB}}{P_B}\right)^{M_j}} \quad P_D = 2L_D + W$$

The capacitance model presented herein requires 10 parameters:

$$C_{GSO}, C_{GDO}, C_{GBO}, C'_{OX}, C_J, P_B, M_J, C_{JSW}, P_{BSW}, M_{JSW}$$

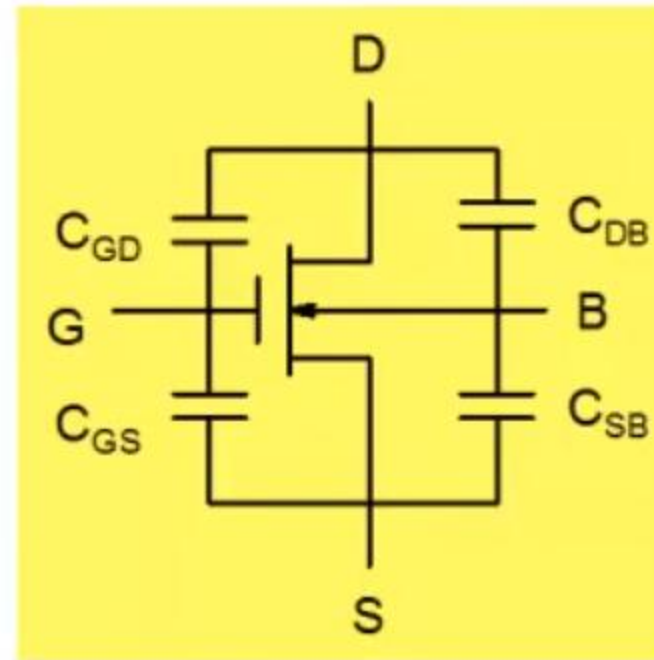
Typical Values of Capacitances



$$C_{gs} = 4.1 \text{ fF}; C_{gd} = 0.43 \text{ fF}$$

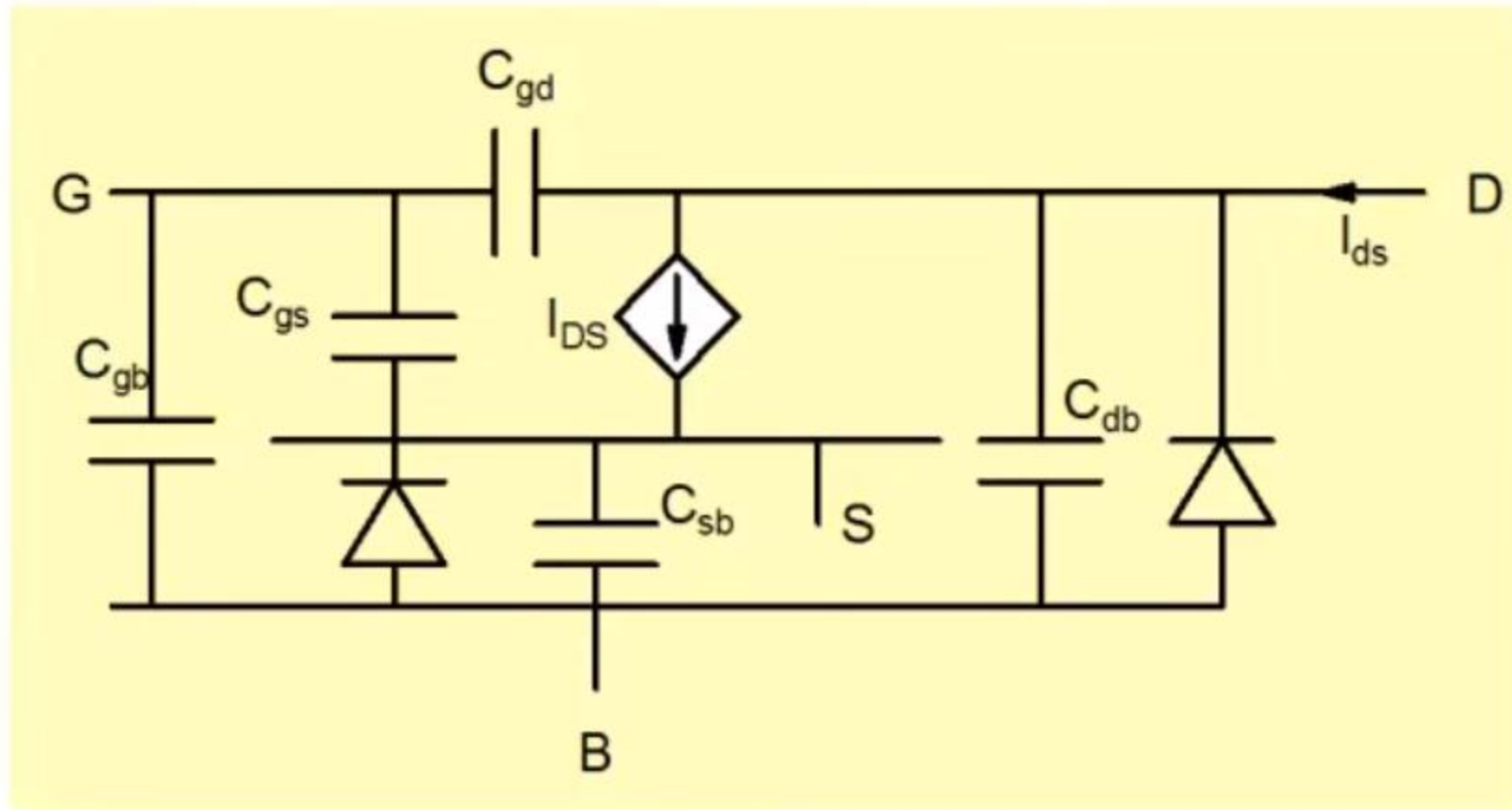
$$C_{sb} = 4.47 \text{ fF}$$

$$C_{db} = 2.75 \text{ fF}$$

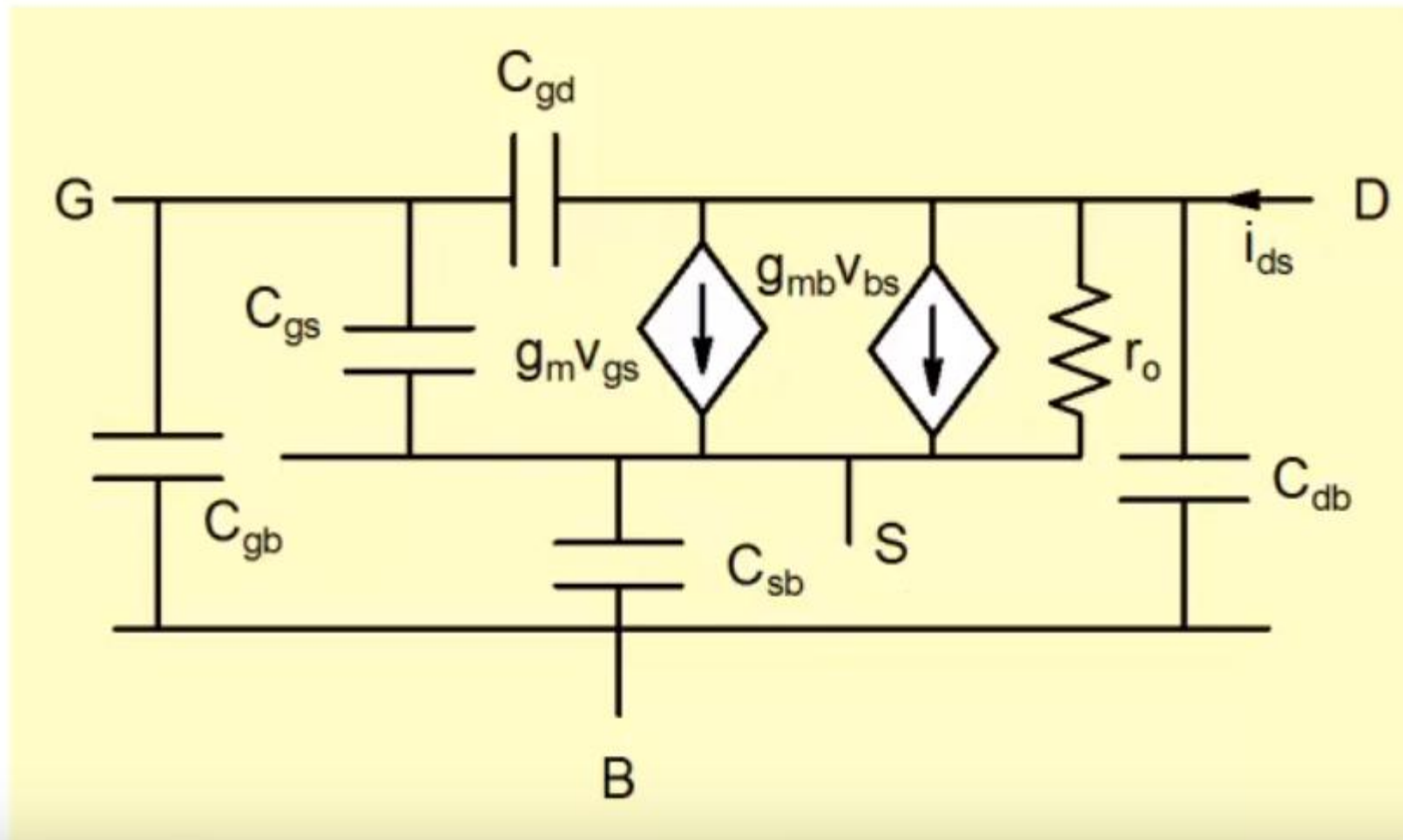


$$V_{SB} = 0; V_{DS} = 2 \text{ V}$$

Complete Large Signal Model



High Frequency Small Signal Model



- We have so far discussed simple MOS models which are suitable for 'hand-analysis' of circuits. For more accurate prediction of circuit characteristics using circuit simulation more accurate MOS models are required.

- SPICE and its various variants are the most popular circuit simulation tool. In SPICE, there are a number of MOS models that are available including Level-1, level-2, Level-3, BSIM1, BSIM2, BSIM3, BSIM4 etc.

- Level-1 model is the simplest and is basically similar to the large signal model that we have described earlier. A popular model for submicron devices is BSIM3 model.

* **SPICE- Simulation Program with Integrated Circuit Emphasis**

* **BSIM- Berkeley Short-channel IGFET Model**

BSIM3 : Berkeley Short Channel IGFET (Insulated gate Field Effect) Model

$$I_{ds} = \frac{I_{dso}(V_{dseff})}{1 + \frac{R_{ds}I_{dso}(V_{dseff})}{V_{dseff}}} \left(1 + \frac{V_{ds} - V_{dseff}}{V_A} \right) \left(1 + \frac{V_{ds} - V_{dseff}}{V_{ASCBE}} \right)$$

$$I_{dso} = \frac{W_{eff}\mu_{eff}C_{ox}V_{gsteff}(1 - A_{bulk}\frac{V_{dseff}}{2(V_{gsteff} + 2\psi_t)})V_{dseff}}{L_{eff}[1 + V_{dseff} / (E_{sat}L_{eff})]}$$

$$V_A = V_{Asat} + (1 + \frac{P_{vag}V_{gsteff}}{E_{sat}L_{eff}})(\frac{1}{V_{ACLM}} + \frac{1}{V_{ADIBLC}})^{-1}$$

$$V_{ACLM} = \frac{A_{bulk}E_{sat}L_{eff} + V_{gsteff}}{P_{CLM}A_{bulk}E_{sat}l_{itl}}(V_{ds} - V_{dseff})$$

LATEST MODELS

Typical SPICE model files for each future generation are available here.

Attention: By using a **PTM** file, you agree to acknowledge both the URL of **PTM**: <http://ptm.asu.edu/> and the related publications in all documents and publications involving its usage.

New!

June 01, 2012:

PTM releases a new set of models for multi-gate transistors (**PTM-MG**), for both HP and LSTP applications. It is based on [BSIM-CMG](#), a dedicated model for multi-gate devices.

Acknowledgement: PTM-MG is developed in collaboration with ARM.

Please start from [models](#) and [param.inc](#).

- 7nm PTM-MG [HP NMOS](#), [HP PMOS](#), [LSTP NMOS](#), [LSTP PMOS](#)
- 10nm PTM-MG [HP NMOS](#), [HP PMOS](#), [LSTP NMOS](#), [LSTP PMOS](#)
- 14nm PTM-MG [HP NMOS](#), [HP PMOS](#), [LSTP NMOS](#), [LSTP PMOS](#)
- 16nm PTM-MG [HP NMOS](#), [HP PMOS](#), [LSTP NMOS](#), [LSTP PMOS](#)
- 20nm PTM-MG [HP NMOS](#), [HP PMOS](#), [LSTP NMOS](#), [LSTP PMOS](#)

The entire package is also available here: [PTM-MG](#)

November 15, 2008:

PTM releases a new set of models for low-power applications (PTM LP), incorporating high-k/metal gate and stress effect.

- 16nm PTM LP model: [V2.1](#)
- 22nm PTM LP model: [V2.1](#)
- 32nm PTM LP model: [V2.1](#)
- 45nm PTM LP model: [V2.1](#)

September 30, 2008:

PTM releases a new set of models for high-performance applications (PTM HP), incorporating high-k/metal gate and stress effect.

- 16nm PTM HP model: [V2.1](#)
- 22nm PTM HP model: [V2.1](#)



https://bsim.berkeley.edu/models/bsim4/



BSIM Group

[About](#)[News](#)[Models](#)[Publications](#)[Members](#)[Links](#)[BSIM-CMG](#)[BSIM-IMG](#)[BSIM-SOI](#)[BSIM-BULK](#)[BSIM4](#)

BSIM4, as the extension of **BSIM3** model, introduces physical effects into sub-100nm regime. It is a physics-based, accurate, scalable, robust SPICE model for circuit simulation and CMOS technology development. It is developed by the BSIM Research Group in the Department of Electrical Engineering and Computer Sciences (EECS) at the University of California, Berkeley. All suggestions for model improvements are charted by the Compact Model Coalition (CMC).

BSIM4 has been used for the 0.13 um, 90 nm, 65 nm, 45/40 nm, 23/28 nm, and 22/20nm technology nodes.

See **BSIM3**, a predecessor of **BSIM4**, [here](#).

Latest Release

BSIM4 4.8.1 was released on Feb. 15, 2017.

We would like to thank CMC members for testing beta models and providing valuable feedbacks during model development.

Download **BSIM4 4.8.1** model package, including

- Model code in C

* PTM Low Power 16nm Metal Gate / High-K / Strained-Si

* nominal Vdd = 0.9V

.model nmos nmos level = 54

+version = 4.0	binunit = 1	paramchk= 1	mobmod = 0
+capmod = 2	igcmmod = 1	igbmod = 1	geomod = 1
+diomod = 1	rdsmod = 0	rbodymod= 1	rgatemod= 1
+permod = 1	acnqsmode= 0	trnqsmode= 0	
+tnom = 27	toxe = 1.2e-009	toxp = 9e-010	toxm = 1.2e-009
+dtox = 3e-010	epsrox = 3.9	wint = 5e-009	lint = 0
+ll = 0	wl = 0	lln = 1	wln = 1
+lw = 0	ww = 0	lwn = 1	wwn = 1
+lwl = 0	wwl = 0	xpart = 0	toxref = 1.2e-009
+vth0 = 0.68191	k1 = 0.4	k2 = 0	k3 = 0
+k3b = 0	w0 = 2.5e-006	dvt0 = 1	dvt1 = 2
+dvt2 = 0	dvt0w = 0	dvt1w = 0	dvt2w = 0
+dsb = 0.1	minv = 0.05	voff1 = 0	dvt0 = 1e-011
+dvtp1 = 0.1	lpe0 = 0	lpeb = 0	xj = 5e-009
+ngate = 1e+023	ndep = 7e+018	nsd = 2e+020	phin = 0
+cdsc = 0	cdscb = 0	cdscd = 0	cit = 0
+voff = -0.1014	nfactor = 1.6	eta0 = 0.0095	etab = 0
+vfb = -0.55	u0 = 0.028	ua = 6e-010	ub = 1.2e-018
+uc = 0	vsat = 200000	a0 = 1	ags = 0
+a1 = 0	a2 = 1	b0 = 0	b1 = 0
+keta = 0.04	dwg = 0	dwb = 0	pclm = 0.02
+pdiblc1 = 0.001	pdiblc2 = 0.001	pdiblc3 = -0.005	drout = 0.5
+pvag = 1e-020	delta = 0.01	pscbe1 = 8.14e+008	pscbe2 = 1e-007
+fprout = 0.2	pdits = 0.01	pditsd = 0.23	pditsl = 2300000
+rsh = 5	rds = 170	rsd = 75	rdw = 75
+rdsmin = 0	rdwmin = 0	rsmin = 0	prwg = 0
+prwb = 0	wr = 1	alpha0 = 0.074	alpha1 = 0.005
+beta0 = 30	agidl = 0.0002	bgidl = 2.1e+009	cgidl = 0.0002
+egidl = 0.8	aigbacc = 0.012	bigbacc = 0.0028	cigbacc = 0.002